Python Text Mining and Analytics: Transforming Text into Insights

TGS-2026064715 | WSQ | v15 | 14 September 2026

Tertiary Infotech Academy Pte Ltd · UEN 201200696W



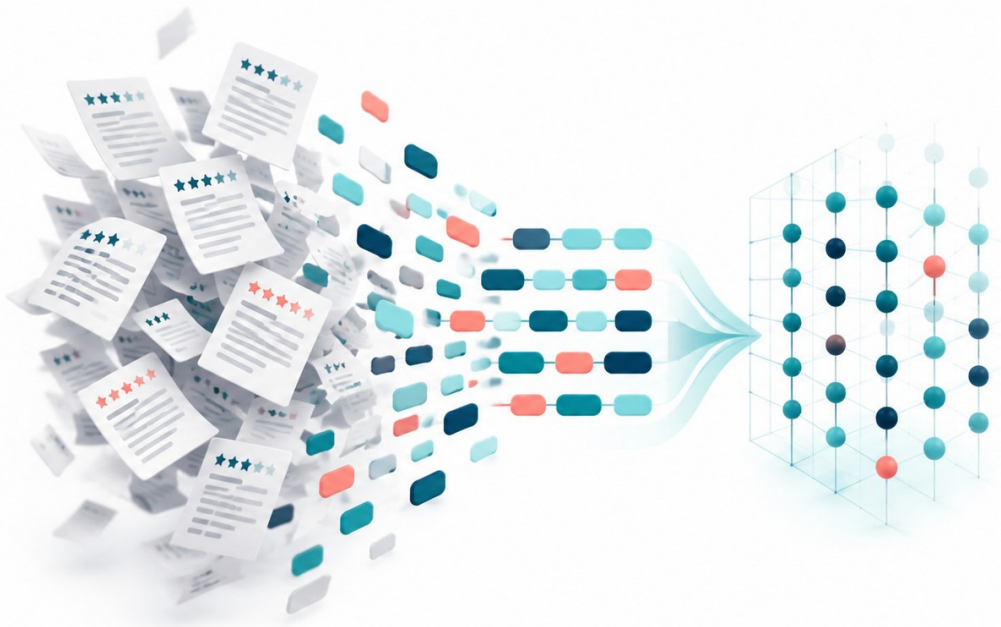
Digital attendance

Complete AM, PM and assessment digital attendance through the provider QR process. Minimum attendance requirements apply.

Topic 1: CRISP-DM and corpus design

Technical evidence from a synthetic Singapore hotel-review corpus

DATA → MODEL → INSIGHT



Business question

INPUT / ARTIFACT

negative share by hotel

ANALYTICAL DECISION

define target, unit and decision

OBSERVE

negative share by hotel

TRANSFORM

define target, unit and decision

CHECK

sampling bias

Business question: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

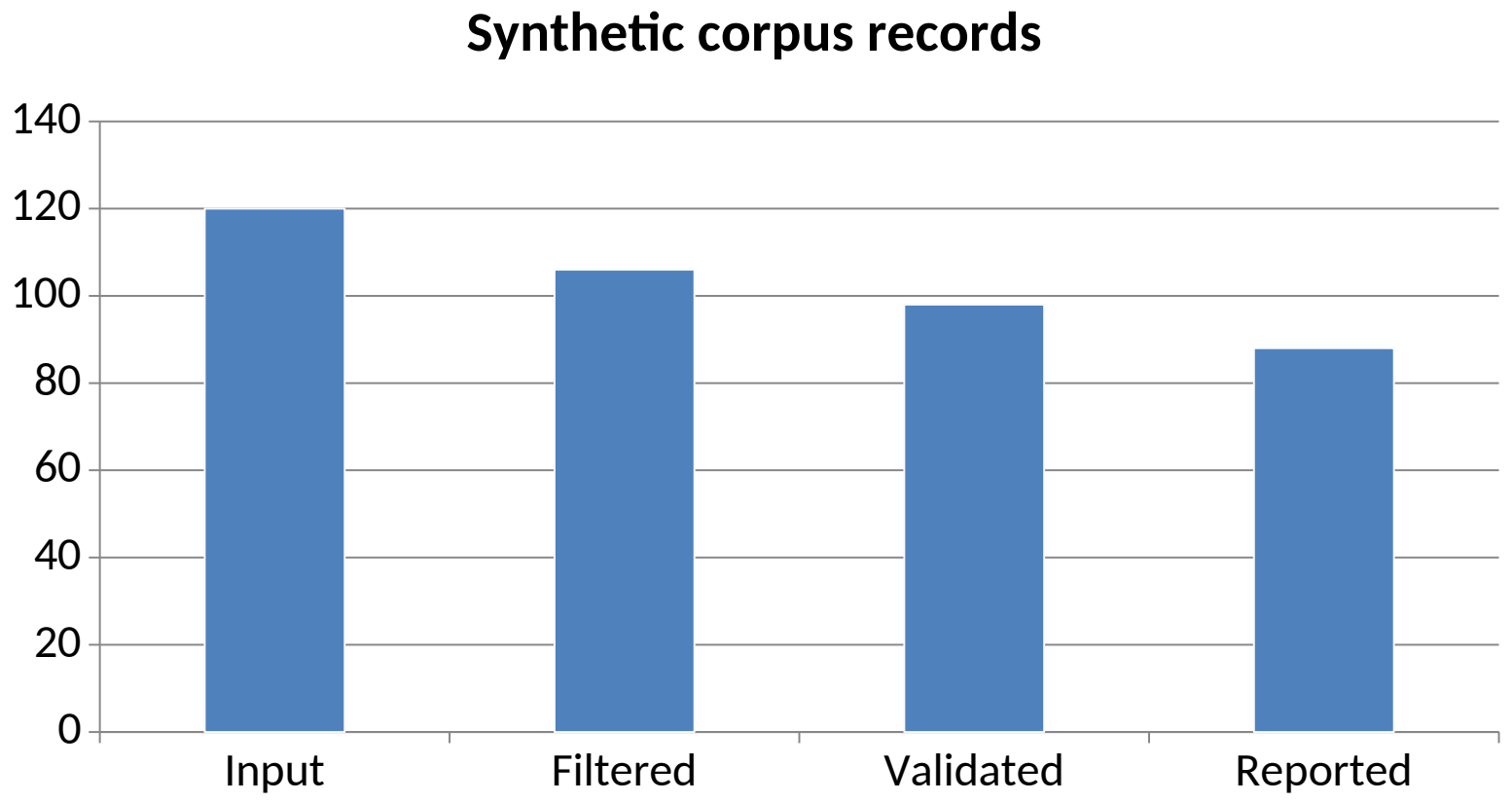
negative share by hotel

PYTHON EXPRESSION OR ANALYTICAL RULE

review_id + sentiment

Evidence to inspect: define target, unit and decision.

Business question: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: define target, unit and decision.

Editable PowerPoint chart

Business question: failure and control

FAILURE MODE

sampling bias

CONTROL / EVIDENCE

Check negative share by hotel; confirm define target, unit and decision.

CRISP-DM phases

INPUT / ARTIFACT

six linked phases

ANALYTICAL DECISION

iterate evaluation into preparation

OBSERVE

six linked phases

TRANSFORM

iterate evaluation into preparation

CHECK

linear waterfall assumption

CRISP-DM phases: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

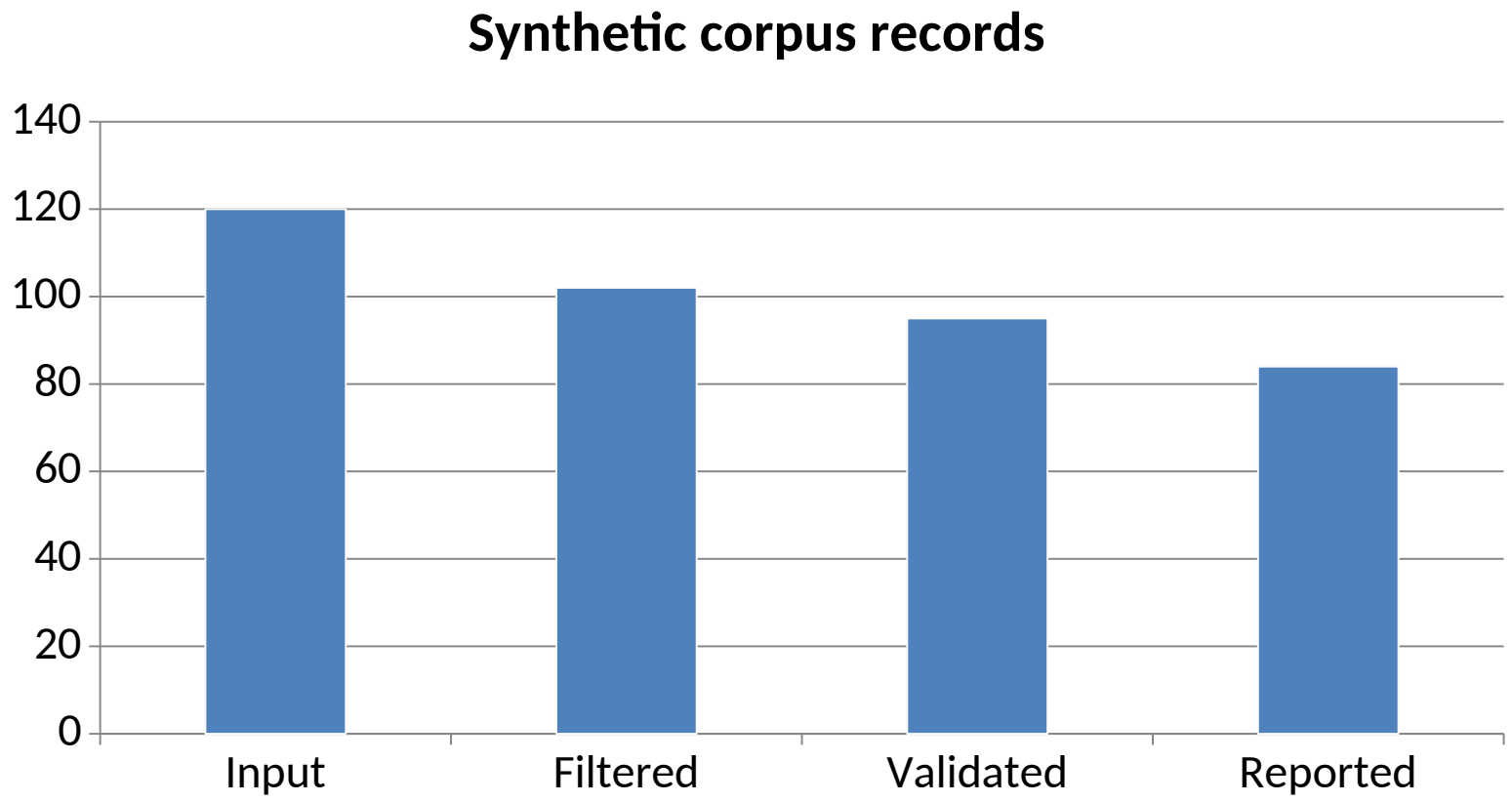
six linked phases

PYTHON EXPRESSION OR ANALYTICAL RULE

business → data → prepare → model → evaluate → deploy

Evidence to inspect: iterate evaluation into preparation.

CRISP-DM phases: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: iterate evaluation into preparation.

Editable PowerPoint chart

CRISP-DM phases: failure and control

FAILURE MODE

linear waterfall assumption

CONTROL / EVIDENCE

Check six linked phases; confirm iterate evaluation into preparation.

Data inventory

INPUT / ARTIFACT

review_id, date, hotel, text

ANALYTICAL DECISION

profile rows and nulls

OBSERVE

review_id, date, hotel, text

TRANSFORM

profile rows and nulls

CHECK

duplicate IDs

Data inventory: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

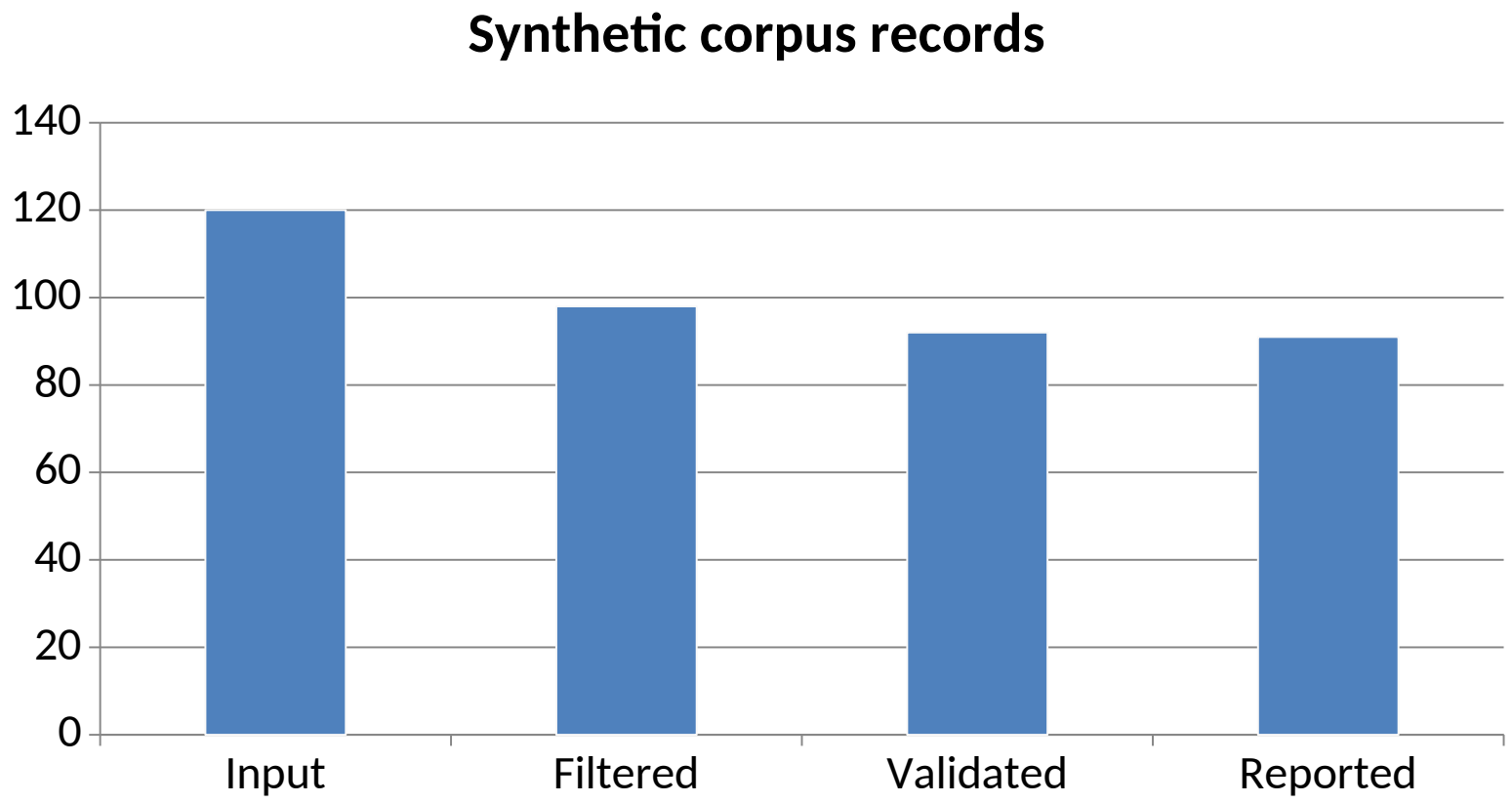
review_id, date, hotel, text

PYTHON EXPRESSION OR ANALYTICAL RULE

df.info(); df.isna().sum()

Evidence to inspect: profile rows and nulls.

Data inventory: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: profile rows and nulls.

Editable PowerPoint chart

Data inventory: failure and control

FAILURE MODE

duplicate IDs

CONTROL / EVIDENCE

Check review_id, date, hotel, text; confirm profile rows and nulls.

Label schema

INPUT / ARTIFACT

positive / neutral / negative

ANALYTICAL DECISION

define annotation rules

OBSERVE

positive / neutral / negative

TRANSFORM

define annotation rules

CHECK

rating-label disagreement

Label schema: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

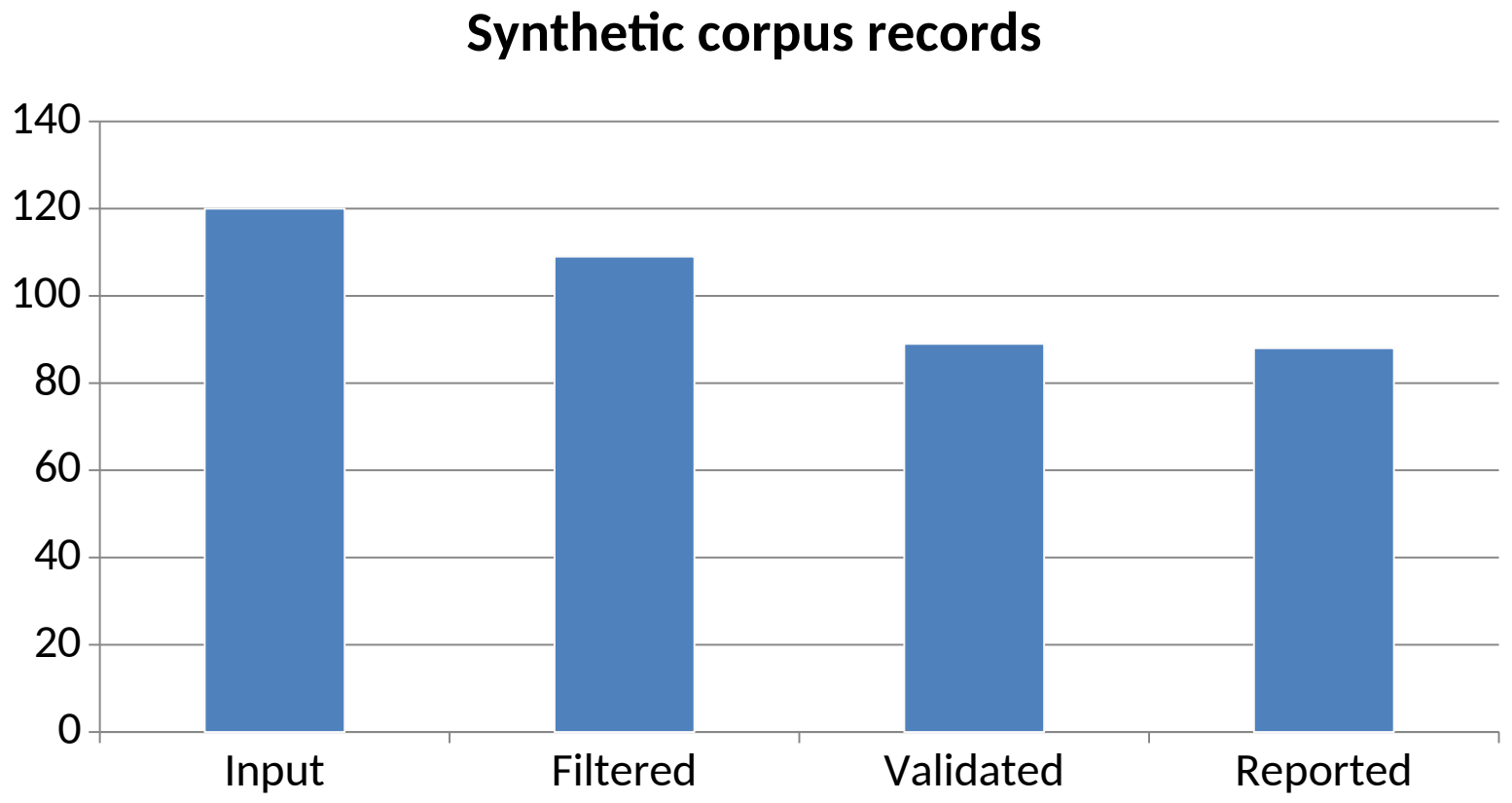
positive / neutral / negative

PYTHON EXPRESSION OR ANALYTICAL RULE

```
df.sentiment.value_counts()
```

Evidence to inspect: define annotation rules.

Label schema: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: define annotation rules.

Editable PowerPoint chart

Label schema: failure and control

FAILURE MODE

rating-label disagreement

CONTROL / EVIDENCE

Check positive / neutral / negative; confirm define annotation rules.

Corpus provenance

INPUT / ARTIFACT

synthetic hotel reviews

ANALYTICAL DECISION

retain source and date fields

OBSERVE

synthetic hotel reviews

TRANSFORM

retain source and date fields

CHECK

unlicensed scraped text

Corpus provenance: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

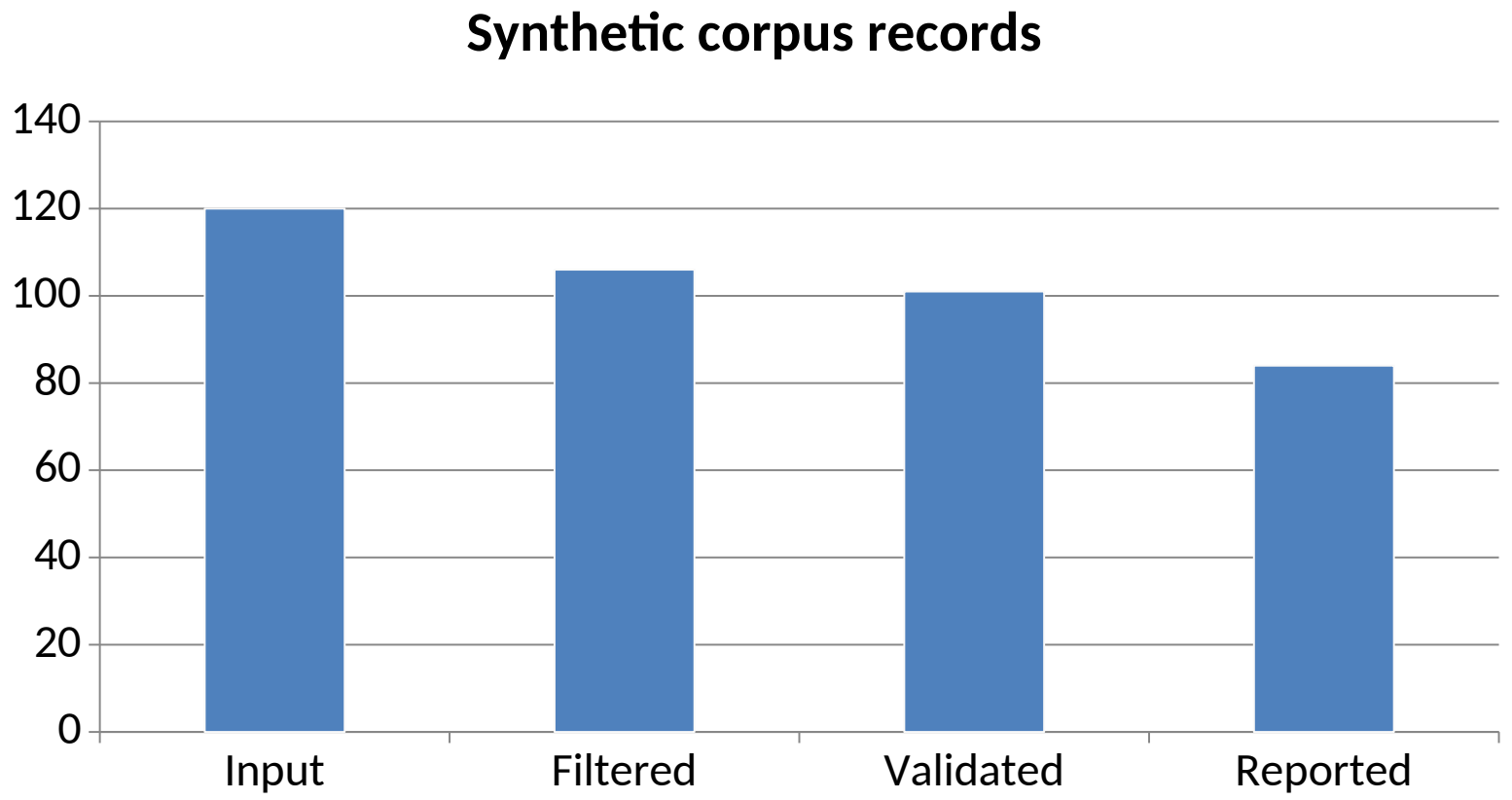
synthetic hotel reviews

PYTHON EXPRESSION OR ANALYTICAL RULE

review_id, channel, date

Evidence to inspect: retain source and date fields.

Corpus provenance: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: retain source and date fields.

Editable PowerPoint chart

Corpus provenance: failure and control

FAILURE MODE

unlicensed scraped text

CONTROL / EVIDENCE

Check synthetic hotel reviews; confirm retain source and date fields.

Sampling frame

INPUT / ARTIFACT

hotel × channel × month

ANALYTICAL DECISION

compare strata proportions

OBSERVE

hotel × channel × month

TRANSFORM

compare strata proportions

CHECK

channel overrepresentation

Sampling frame: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

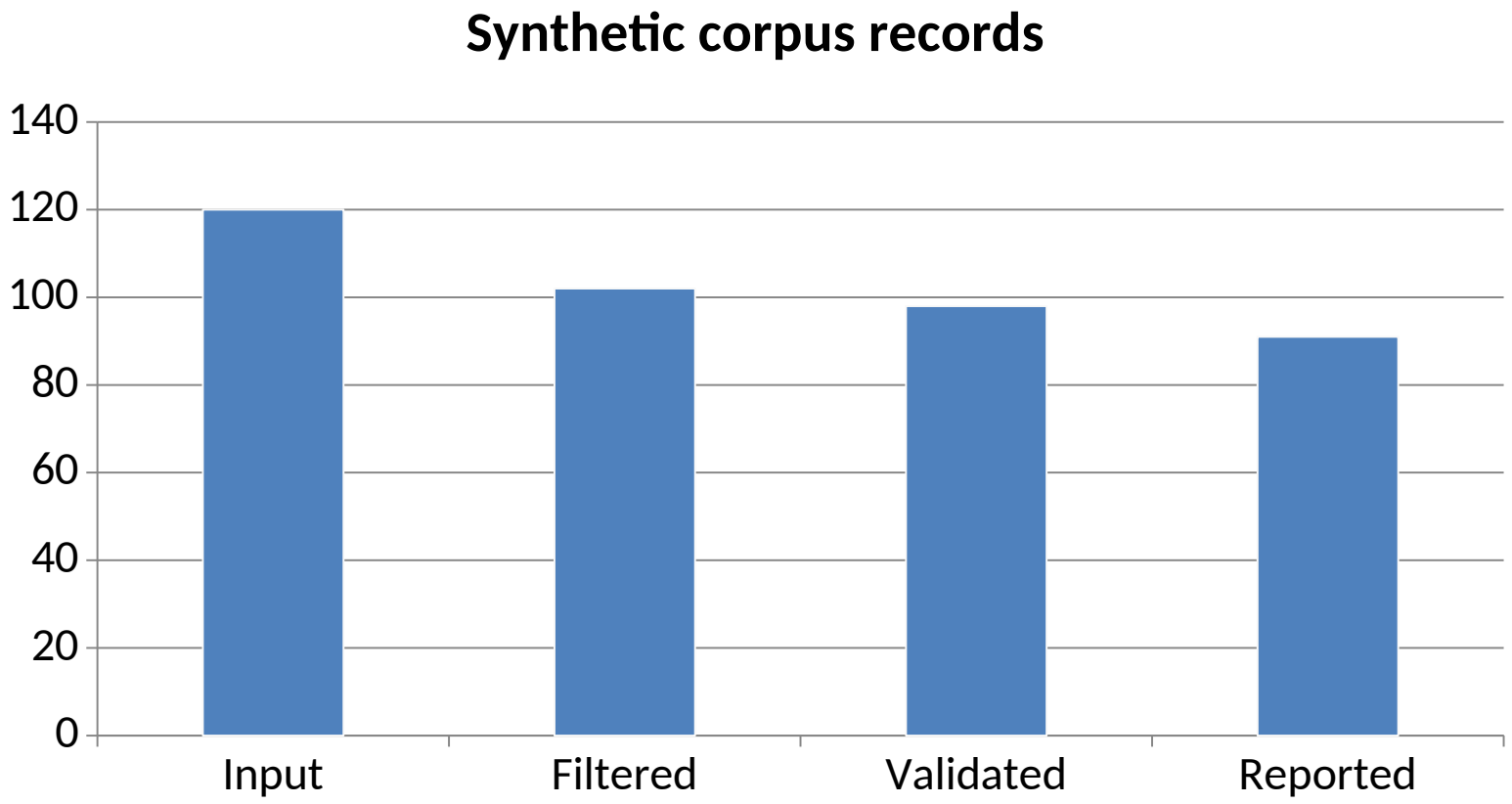
hotel × channel × month

PYTHON EXPRESSION OR ANALYTICAL RULE

```
pd.crosstab(df.hotel, df.channel)
```

Evidence to inspect: compare strata proportions.

Sampling frame: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: compare strata proportions.

Editable PowerPoint chart

Sampling frame: failure and control

FAILURE MODE

channel overrepresentation

CONTROL / EVIDENCE

Check hotel × channel × month; confirm compare strata proportions.

Success metric

INPUT / ARTIFACT

macro F1 and negative recall

ANALYTICAL DECISION

tie metric to response triage

OBSERVE

macro F1 and negative recall

TRANSFORM

tie metric to response triage

CHECK

accuracy hides minority failure

Success metric: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

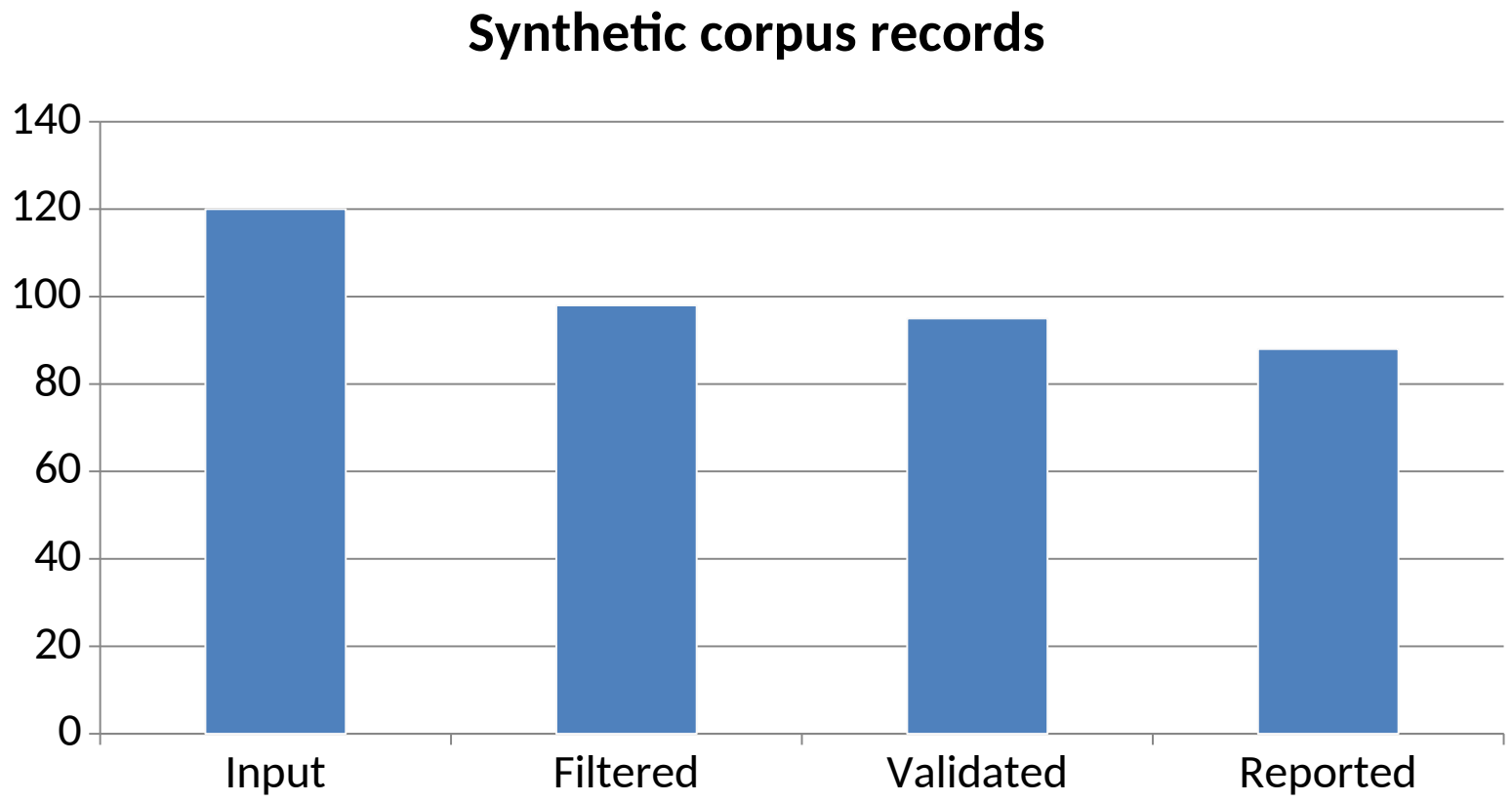
macro F1 and negative recall

PYTHON EXPRESSION OR ANALYTICAL RULE

```
classification_report(y_true, y_pred)
```

Evidence to inspect: tie metric to response triage.

Success metric: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: tie metric to response triage.

Editable PowerPoint chart

Success metric: failure and control

FAILURE MODE

accuracy hides minority failure

CONTROL / EVIDENCE

Check macro F1 and negative recall; confirm tie metric to response triage.

Train-test boundary

INPUT / ARTIFACT

stratified holdout

ANALYTICAL DECISION

fit vocabulary on train only

OBSERVE

stratified holdout

TRANSFORM

fit vocabulary on train only

CHECK

TF-IDF leakage

Train-test boundary: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

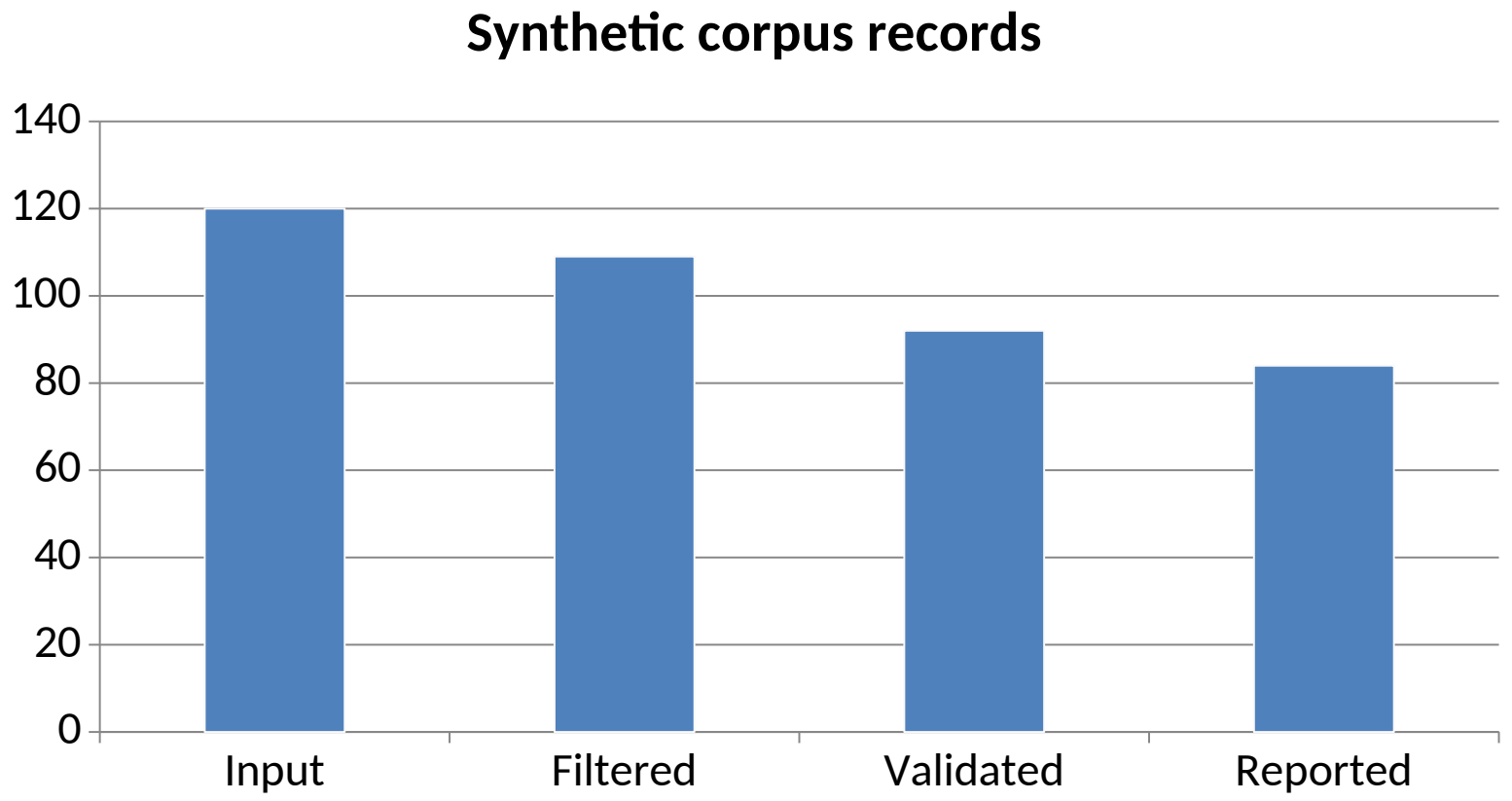
stratified holdout

PYTHON EXPRESSION OR ANALYTICAL RULE

```
train_test_split(..., stratify=y)
```

Evidence to inspect: fit vocabulary on train only.

Train-test boundary: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: fit vocabulary on train only.

Editable PowerPoint chart

Train-test boundary: failure and control

FAILURE MODE

TF-IDF leakage

CONTROL / EVIDENCE

Check stratified holdout; confirm fit vocabulary on train only.

Baseline analysis

INPUT / ARTIFACT

majority and keyword rules

ANALYTICAL DECISION

compare value above baseline

OBSERVE

majority and keyword rules

TRANSFORM

compare value above baseline

CHECK

overclaiming uplift

Baseline analysis: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

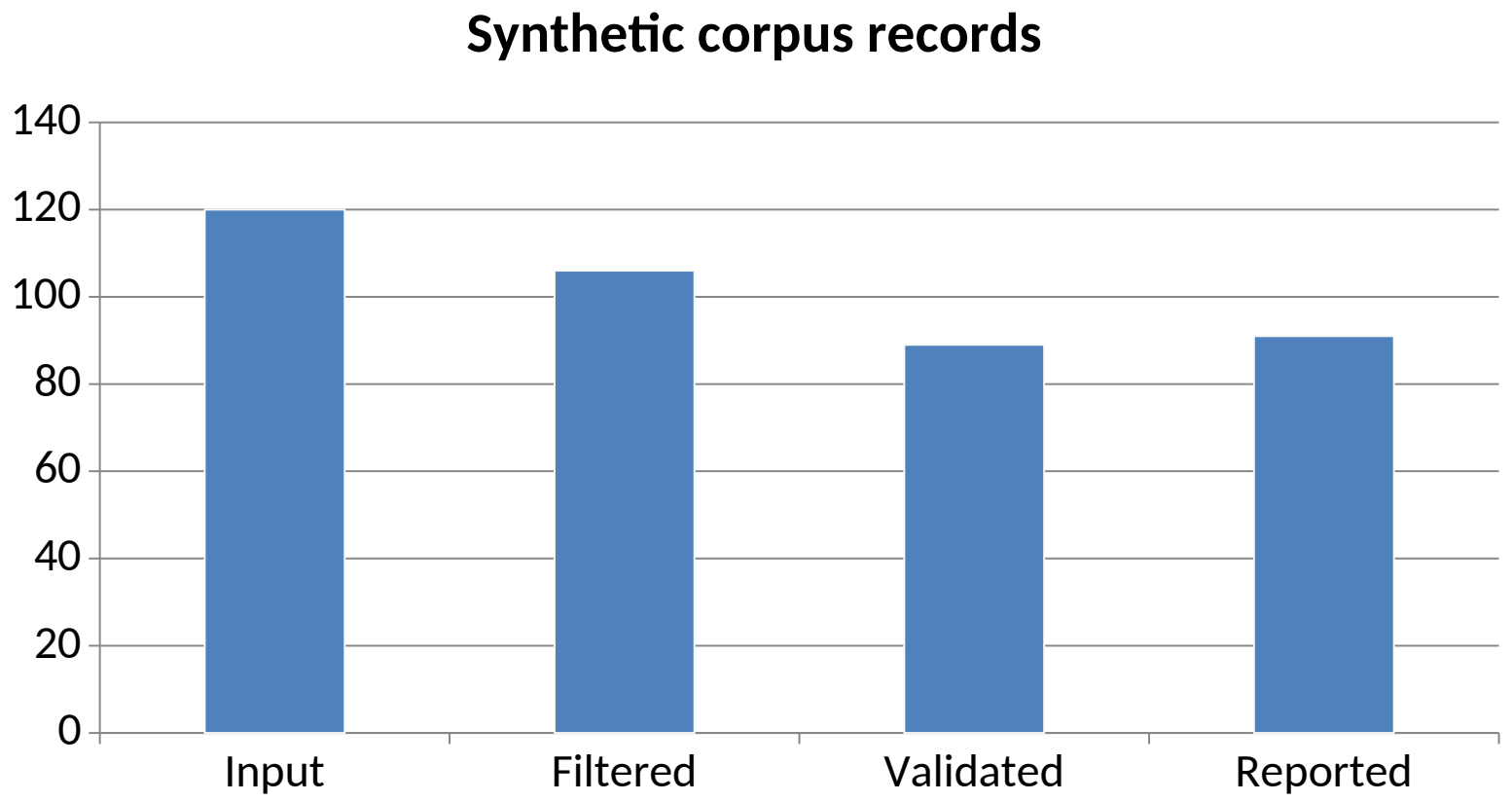
majority and keyword rules

PYTHON EXPRESSION OR ANALYTICAL RULE

```
DummyClassifier(strategy="most_frequent")
```

Evidence to inspect: compare value above baseline.

Baseline analysis: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: compare value above baseline.

Editable PowerPoint chart

Baseline analysis: failure and control

FAILURE MODE

overclaiming uplift

CONTROL / EVIDENCE

Check majority and keyword rules; confirm compare value above baseline.

Deployment evidence

INPUT / ARTIFACT

weekly insight dashboard

ANALYTICAL DECISION

specify owner and drift monitor

OBSERVE

weekly insight dashboard

TRANSFORM

specify owner and drift monitor

CHECK

model decay

Deployment evidence: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

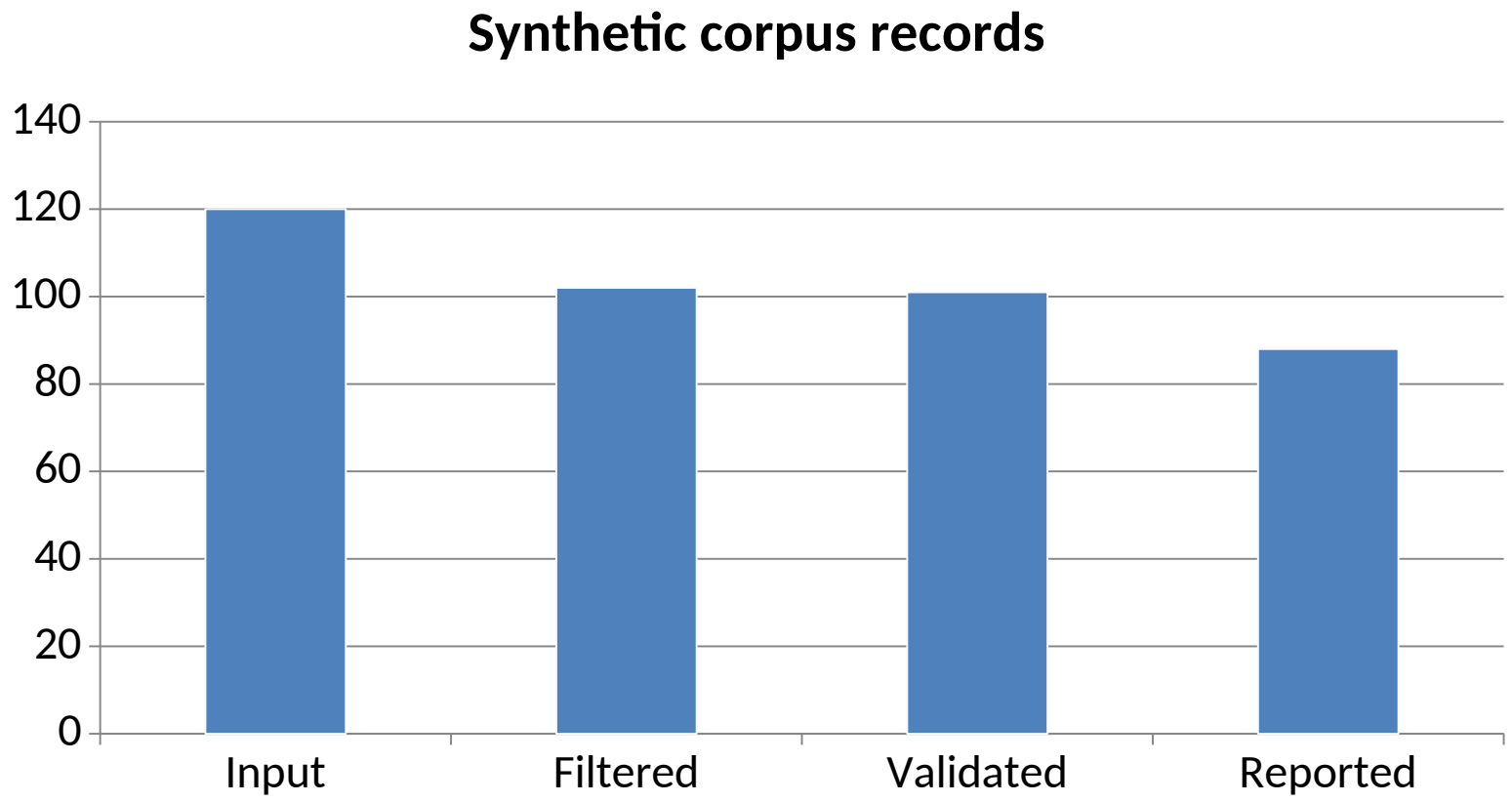
weekly insight dashboard

PYTHON EXPRESSION OR ANALYTICAL RULE

review volume + macro F1 trend

Evidence to inspect: specify owner and drift monitor.

Deployment evidence: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: specify owner and drift monitor.

Editable PowerPoint chart

Deployment evidence: failure and control

FAILURE MODE

model decay

CONTROL / EVIDENCE

Check weekly insight dashboard; confirm specify owner and drift monitor.

NLU and NLG boundary

INPUT / ARTIFACT

intent extraction vs text generation

ANALYTICAL DECISION

choose task-specific output

OBSERVE

intent extraction vs text generation

TRANSFORM

choose task-specific output

CHECK

generated text mistaken for evidence

NLU and NLG boundary: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

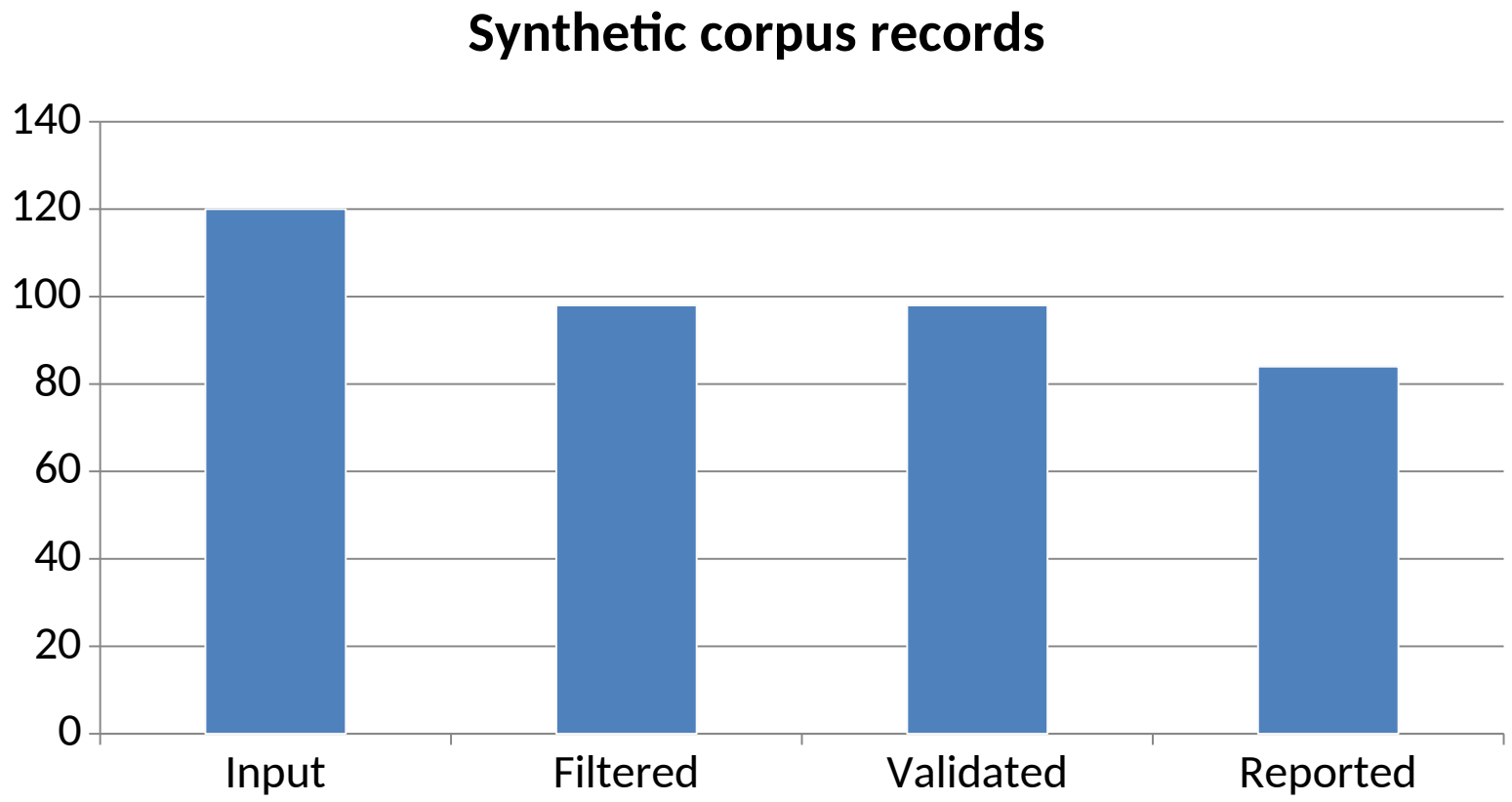
intent extraction vs text generation

PYTHON EXPRESSION OR ANALYTICAL RULE

NLU: classify; NLG: compose

Evidence to inspect: choose task-specific output.

NLU and NLG boundary: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: choose task-specific output.

Editable PowerPoint chart

NLU and NLG boundary: failure and control

FAILURE MODE

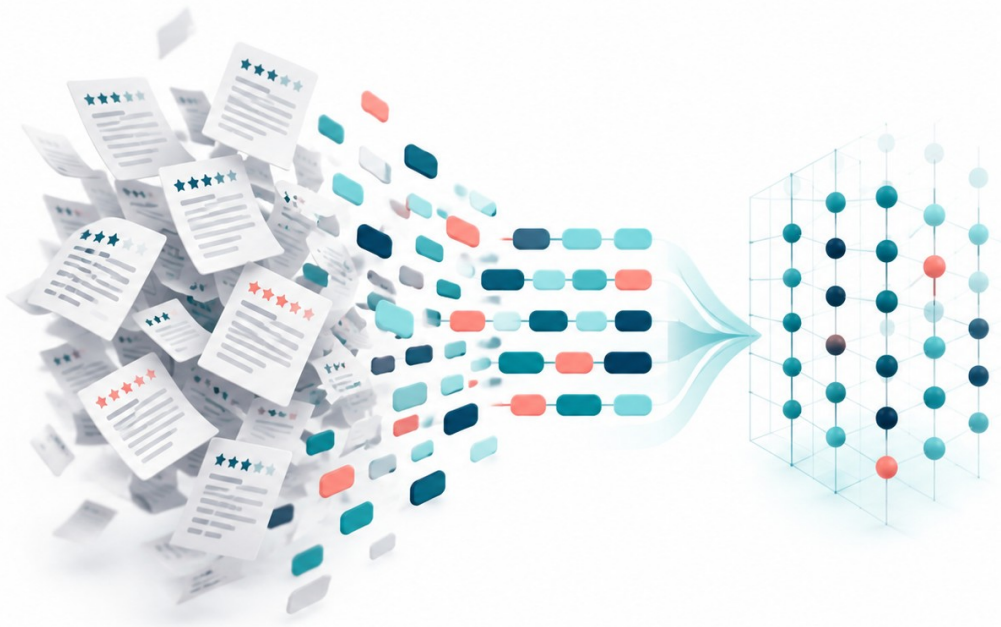
generated text mistaken for evidence

CONTROL / EVIDENCE

Check intent extraction vs text generation; confirm choose task-specific output.

Topic 2: Text cleaning and vectorization

Technical evidence from a synthetic Singapore hotel-review corpus



DATA → MODEL → INSIGHT

Corpus ingestion

INPUT / ARTIFACT

CSV to DataFrame

ANALYTICAL DECISION

validate encoding and schema

OBSERVE

CSV to DataFrame

TRANSFORM

validate encoding and schema

CHECK

silent dtype coercion

Corpus ingestion: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

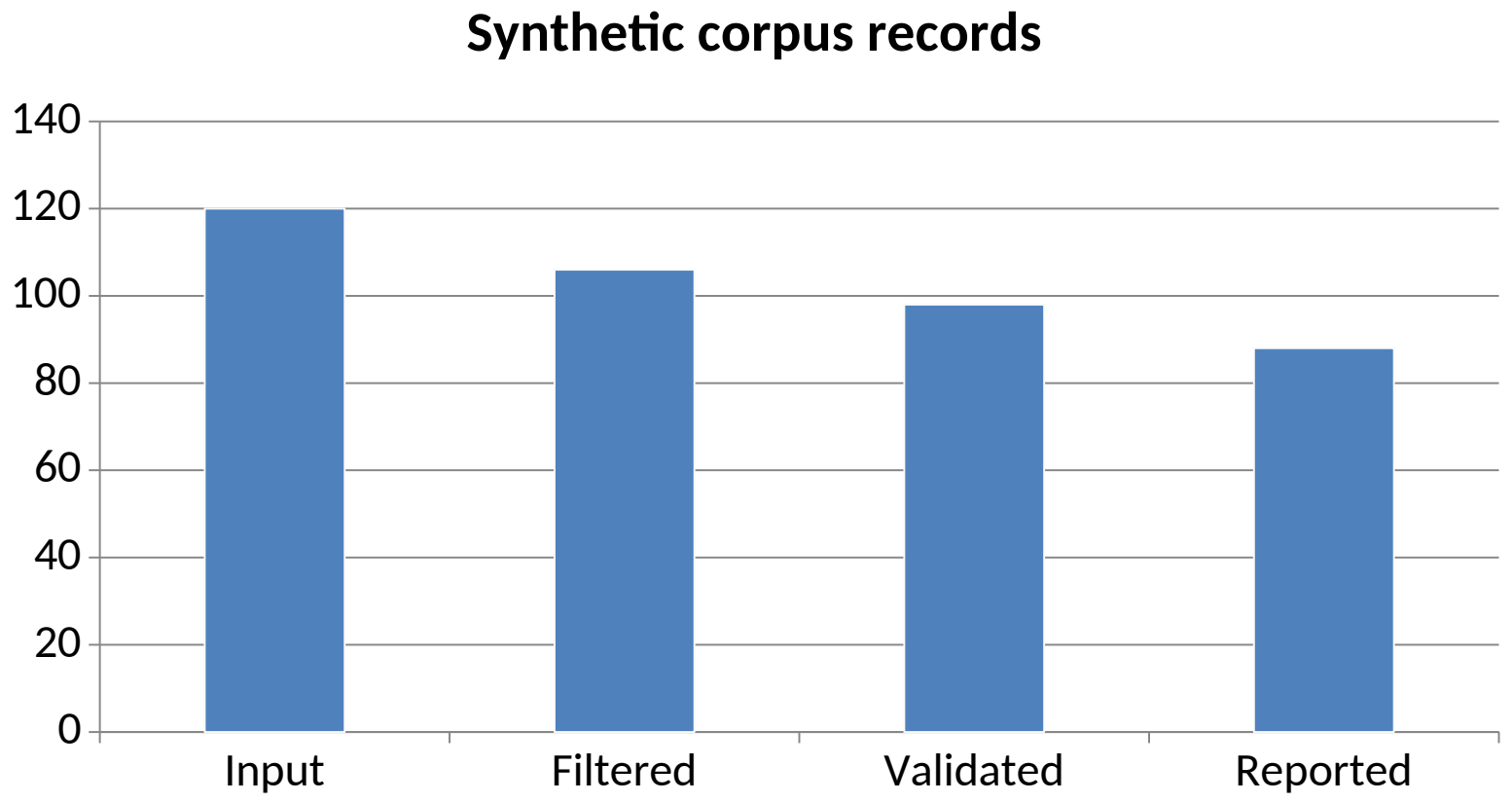
CSV to DataFrame

PYTHON EXPRESSION OR ANALYTICAL RULE

```
pd.read_csv("hotel_reviews.csv")
```

Evidence to inspect: validate encoding and schema.

Corpus ingestion: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: validate encoding and schema.

Editable PowerPoint chart

Corpus ingestion: failure and control

FAILURE MODE

silent dtype coercion

CONTROL / EVIDENCE

Check CSV to DataFrame; confirm validate encoding and schema.

Unicode normalization

INPUT / ARTIFACT

NFKC form

ANALYTICAL DECISION

compare before/after tokens

OBSERVE

NFKC form

TRANSFORM

compare before/after tokens

CHECK

meaningful symbols lost

Unicode normalization: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

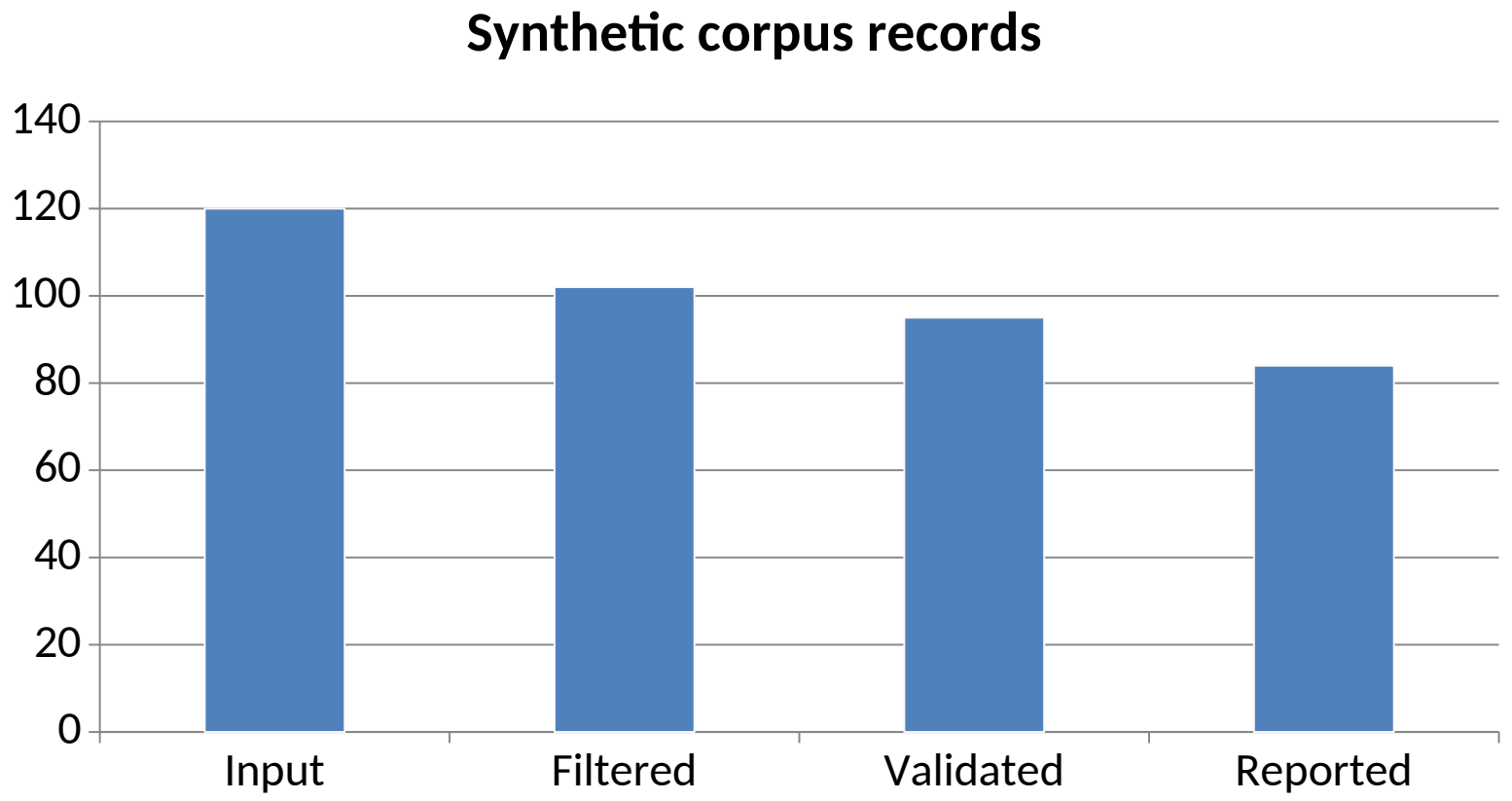
NFKC form

PYTHON EXPRESSION OR ANALYTICAL RULE

```
unicodedata.normalize("NFKC", text)
```

Evidence to inspect: compare before/after tokens.

Unicode normalization: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: compare before/after tokens.

Editable PowerPoint chart

Unicode normalization: failure and control

FAILURE MODE

meaningful symbols lost

CONTROL / EVIDENCE

Check NFKC form; confirm compare before/after tokens.

Tokenization

INPUT / ARTIFACT

regex word tokens

ANALYTICAL DECISION

measure token count

OBSERVE

regex word tokens

TRANSFORM

measure token count

CHECK

contractions split

Tokenization: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

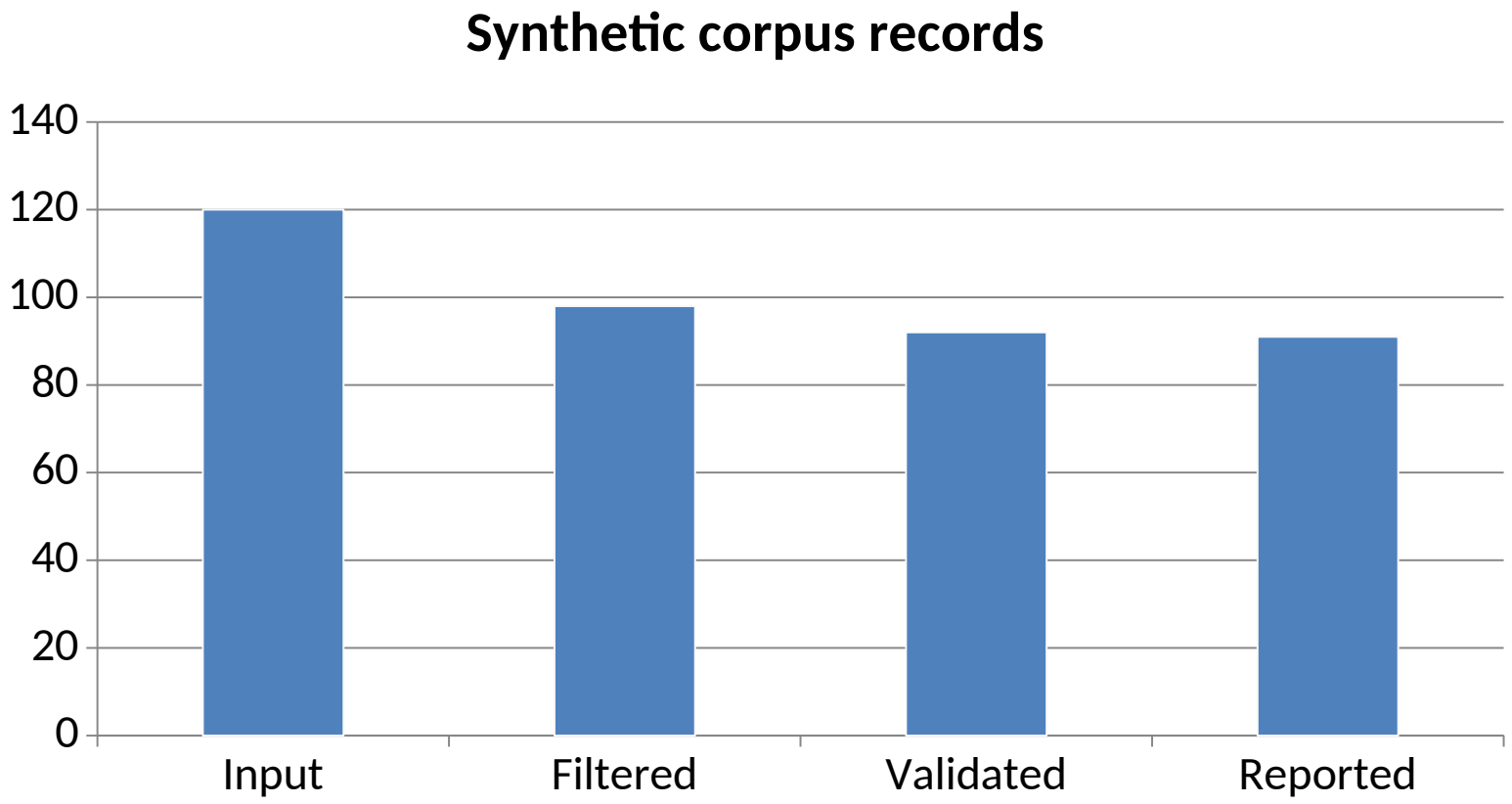
regex word tokens

PYTHON EXPRESSION OR ANALYTICAL RULE

```
re.findall(r"[A-Za-z]+", text)
```

Evidence to inspect: measure token count.

Tokenization: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: measure token count.

Editable PowerPoint chart

Tokenization: failure and control

FAILURE MODE

contractions split

CONTROL / EVIDENCE

Check regex word tokens; confirm measure token count.

Case folding

INPUT / ARTIFACT

lowercase vocabulary

ANALYTICAL DECISION

collapse variant terms

OBSERVE

lowercase vocabulary

TRANSFORM

collapse variant terms

CHECK

acronym loss

Case folding: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

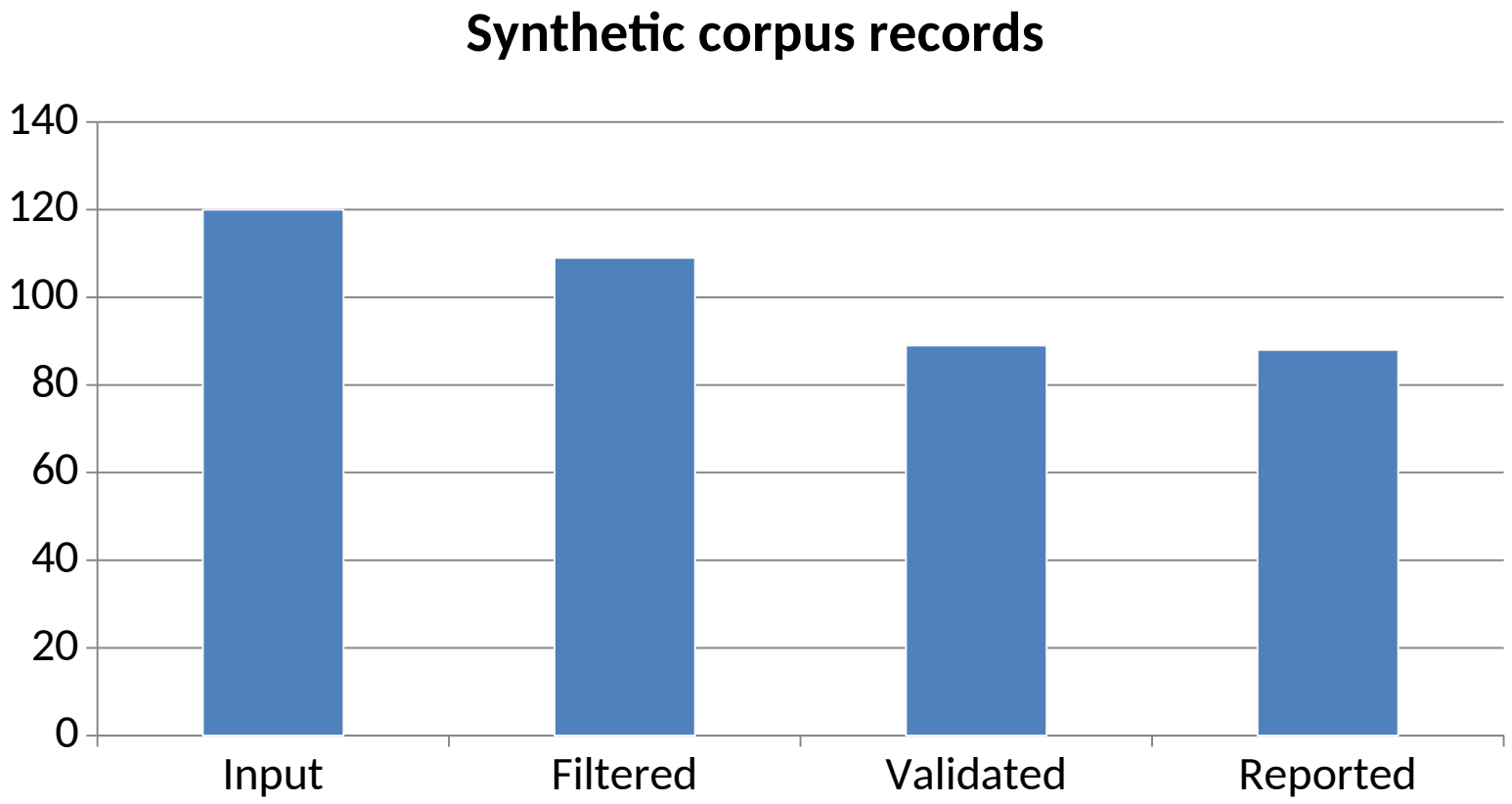
lowercase vocabulary

PYTHON EXPRESSION OR ANALYTICAL RULE

```
text.lower()
```

Evidence to inspect: collapse variant terms.

Case folding: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: collapse variant terms.

Editable PowerPoint chart

Case folding: failure and control

FAILURE MODE

acronym loss

CONTROL / EVIDENCE

Check lowercase vocabulary; confirm collapse variant terms.

Punctuation policy

INPUT / ARTIFACT

preserve useful punctuation

ANALYTICAL DECISION

compare sentiment cue retention

OBSERVE

preserve useful punctuation

TRANSFORM

compare sentiment cue retention

CHECK

negation lost

Punctuation policy: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

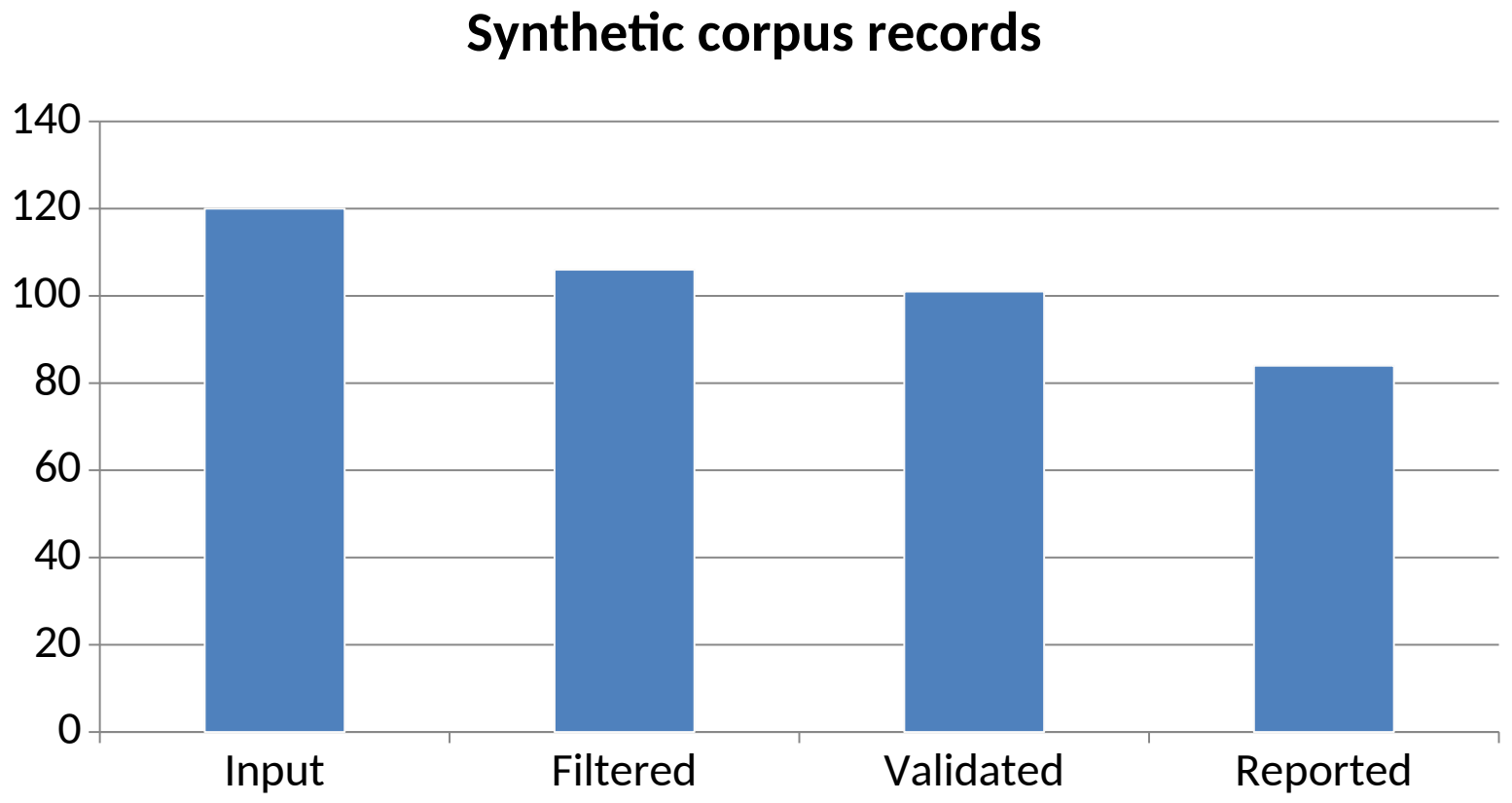
preserve useful punctuation

PYTHON EXPRESSION OR ANALYTICAL RULE

```
re.sub(r"[^\w\s]", " ", text)
```

Evidence to inspect: compare sentiment cue retention.

Punctuation policy: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: compare sentiment cue retention.

Editable PowerPoint chart

Punctuation policy: failure and control

FAILURE MODE

negation lost

CONTROL / EVIDENCE

Check preserve useful punctuation; confirm compare sentiment cue retention.

Stop-word policy

INPUT / ARTIFACT

domain stop list

ANALYTICAL DECISION

retain not/no for sentiment

OBSERVE

domain stop list

TRANSFORM

retain not/no for sentiment

CHECK

negation removal

Stop-word policy: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

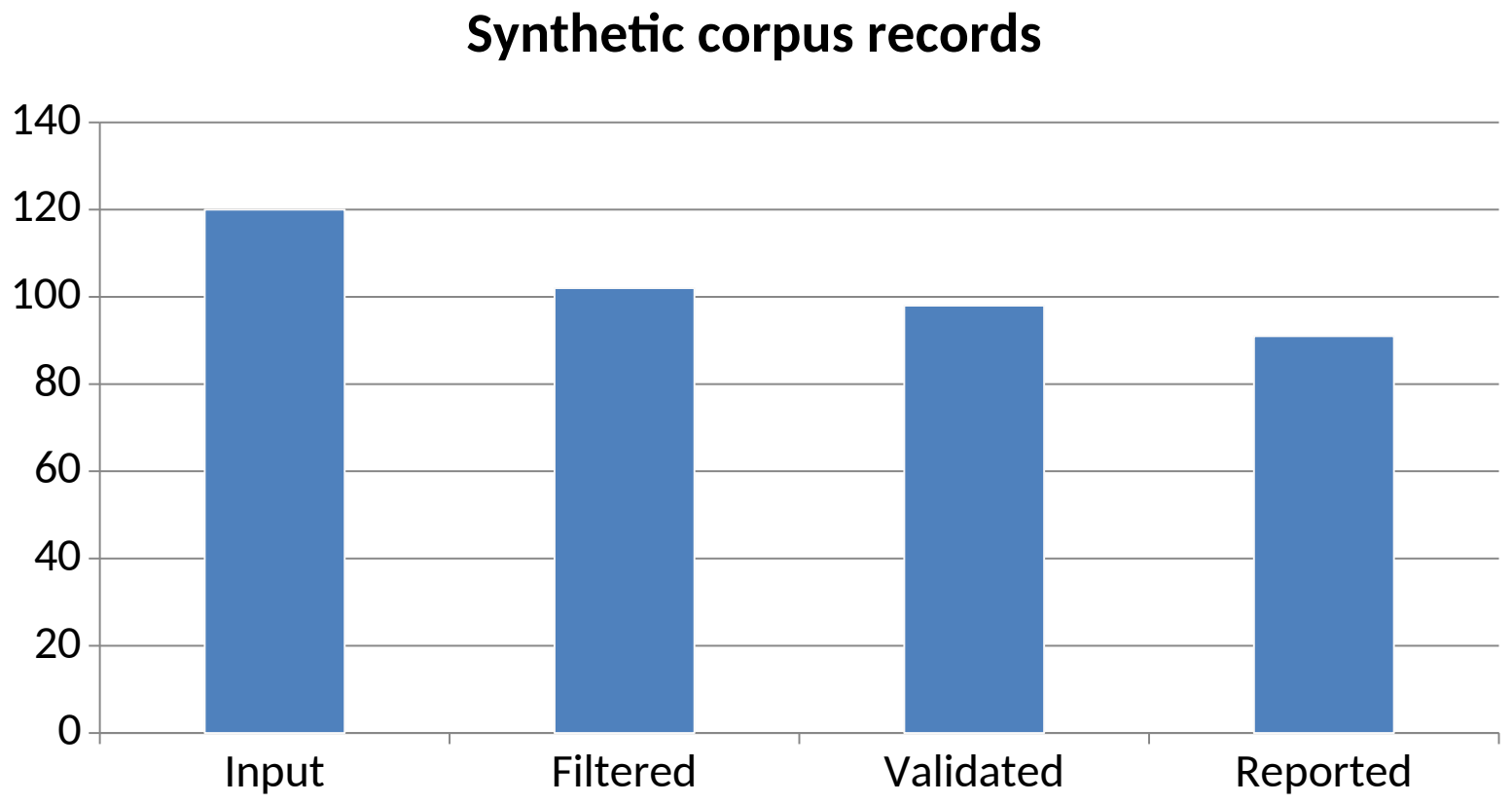
domain stop list

PYTHON EXPRESSION OR ANALYTICAL RULE

```
CountVectorizer(stop_words=None)
```

Evidence to inspect: retain not/no for sentiment.

Stop-word policy: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: retain not/no for sentiment.

Editable PowerPoint chart

Stop-word policy: failure and control

FAILURE MODE

negation removal

CONTROL / EVIDENCE

Check domain stop list; confirm retain not/no for sentiment.

Stemming

INPUT / ARTIFACT

Porter stems

ANALYTICAL DECISION

compare stem collisions

OBSERVE

Porter stems

TRANSFORM

compare stem collisions

CHECK

interpretability loss

Stemming: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

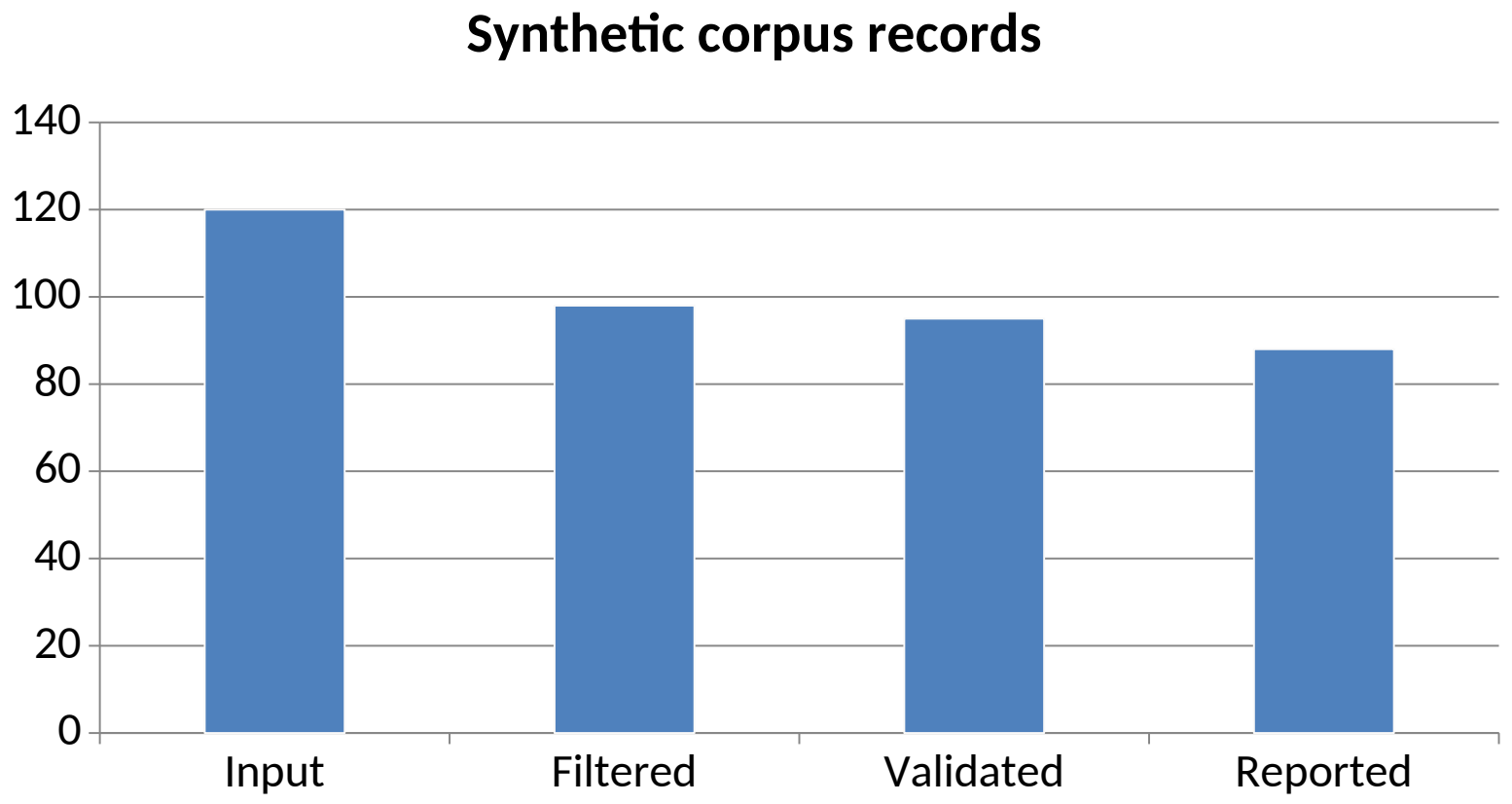
Porter stems

PYTHON EXPRESSION OR ANALYTICAL RULE

```
PorterStemmer().stem("booking")
```

Evidence to inspect: compare stem collisions.

Stemming: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: compare stem collisions.

Editable PowerPoint chart

Stemming: failure and control

FAILURE MODE

interpretability loss

CONTROL / EVIDENCE

Check Porter stems; confirm compare stem collisions.

Lemmatization

INPUT / ARTIFACT

WordNet lemmas

ANALYTICAL DECISION

inspect part-of-speech dependence

OBSERVE

WordNet lemmas

TRANSFORM

inspect part-of-speech dependence

CHECK

missing WordNet corpus

Lemmatization: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

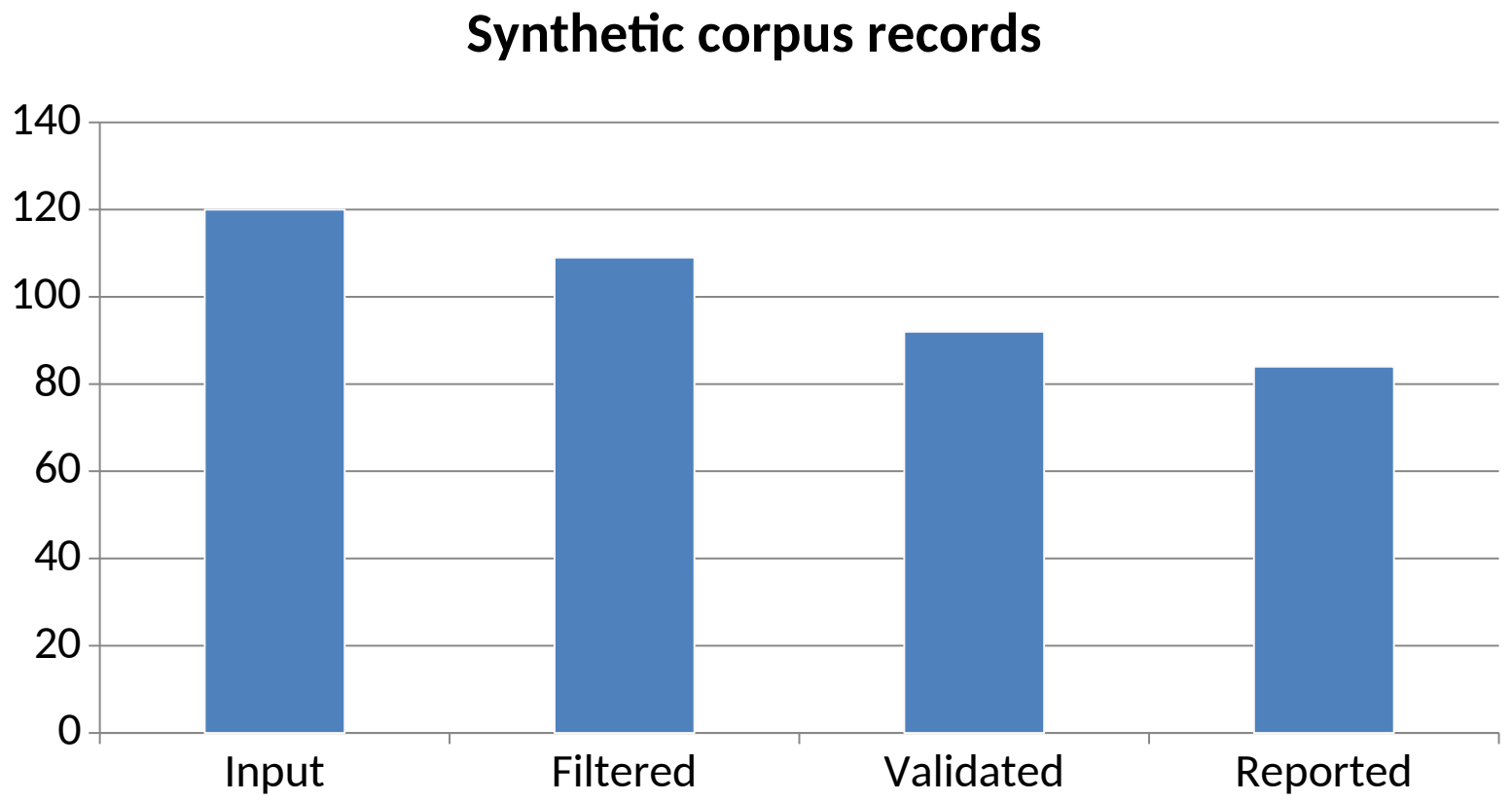
WordNet lemmas

PYTHON EXPRESSION OR ANALYTICAL RULE

```
WordNetLemmatizer().lemmatize("rooms")
```

Evidence to inspect: inspect part-of-speech dependence.

Lemmatization: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: inspect part-of-speech dependence.

Editable PowerPoint chart

Lemmatization: failure and control

FAILURE MODE

missing WordNet corpus

CONTROL / EVIDENCE

Check WordNet lemmas; confirm inspect part-of-speech dependence.

Term frequency

INPUT / ARTIFACT

document-term matrix

ANALYTICAL DECISION

inspect vocabulary dimensions

OBSERVE

document-term matrix

TRANSFORM

inspect vocabulary dimensions

CHECK

rare-term noise

Term frequency: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

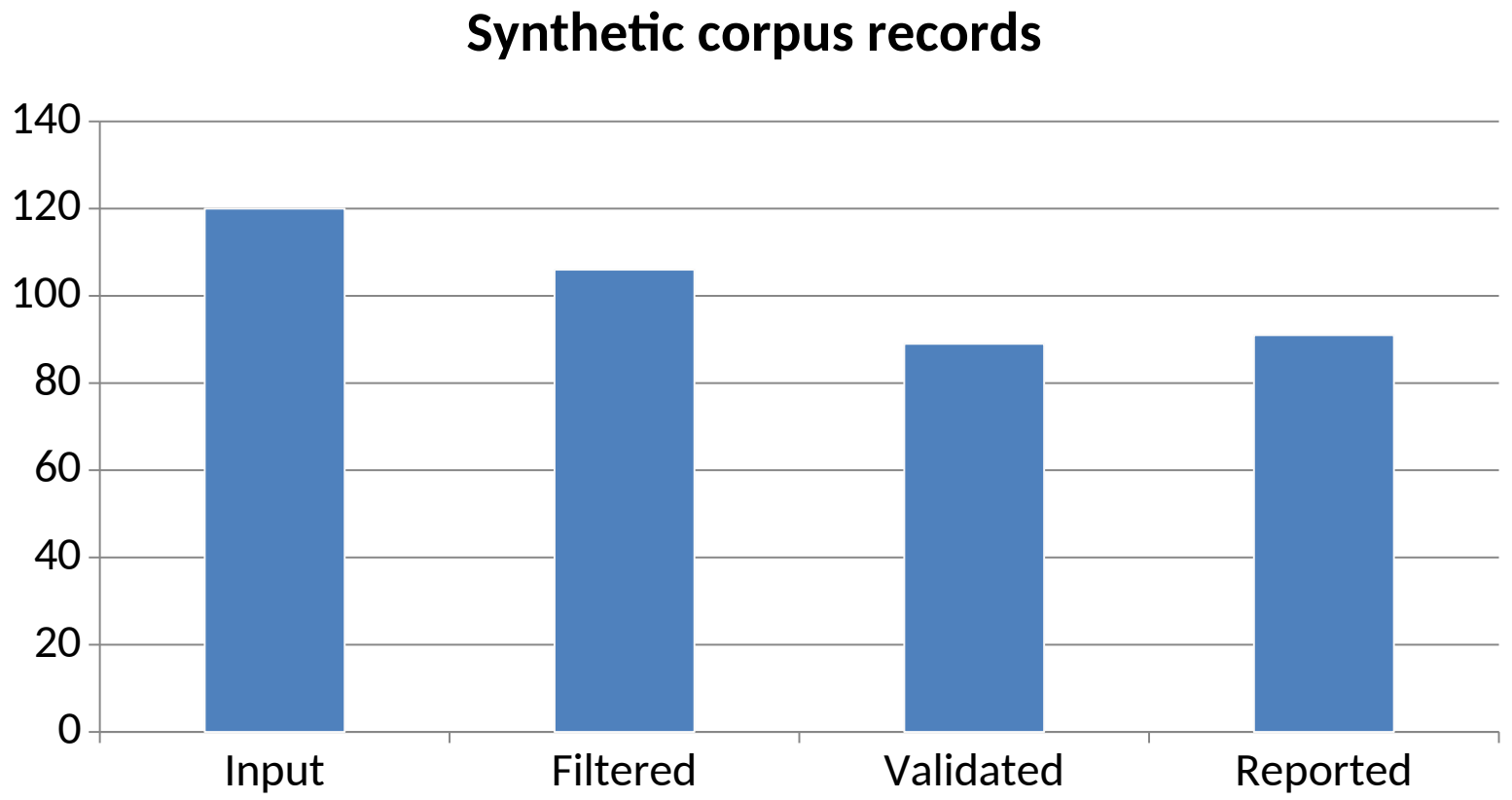
document-term matrix

PYTHON EXPRESSION OR ANALYTICAL RULE

```
CountVectorizer(min_df=2)
```

Evidence to inspect: inspect vocabulary dimensions.

Term frequency: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: inspect vocabulary dimensions.

Editable PowerPoint chart

Term frequency: failure and control

FAILURE MODE

rare-term noise

CONTROL / EVIDENCE

Check document-term matrix; confirm inspect vocabulary dimensions.

TF-IDF and n-grams

INPUT / ARTIFACT

weighted sparse matrix

ANALYTICAL DECISION

compare bigram signal

OBSERVE

weighted sparse matrix

TRANSFORM

compare bigram signal

CHECK

high-dimensional sparsity

TF-IDF and n-grams: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

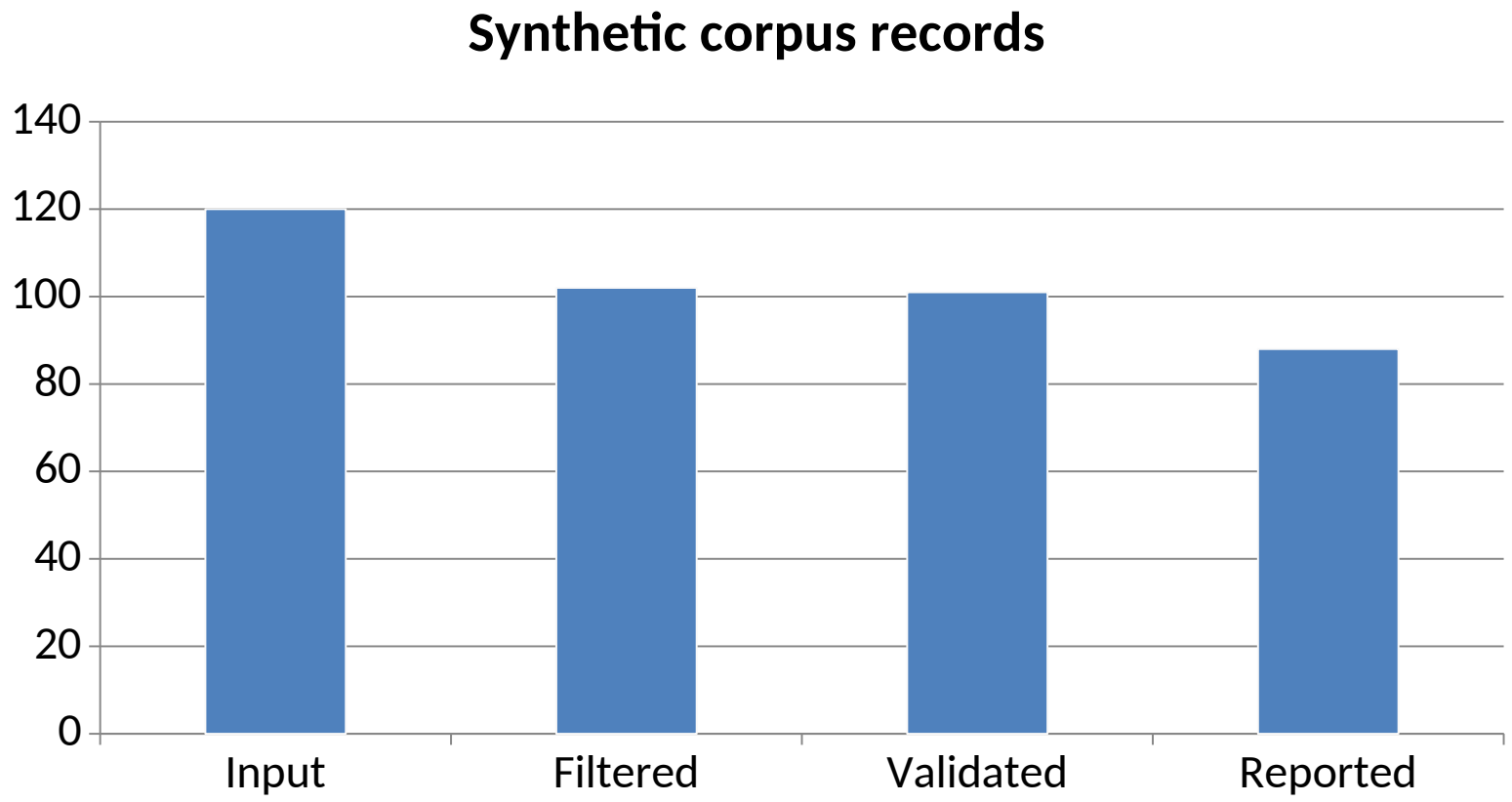
weighted sparse matrix

PYTHON EXPRESSION OR ANALYTICAL RULE

```
TfidfVectorizer(ngram_range=(1,2))
```

Evidence to inspect: compare bigram signal.

TF-IDF and n-grams: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: compare bigram signal.

Editable PowerPoint chart

TF-IDF and n-grams: failure and control

FAILURE MODE

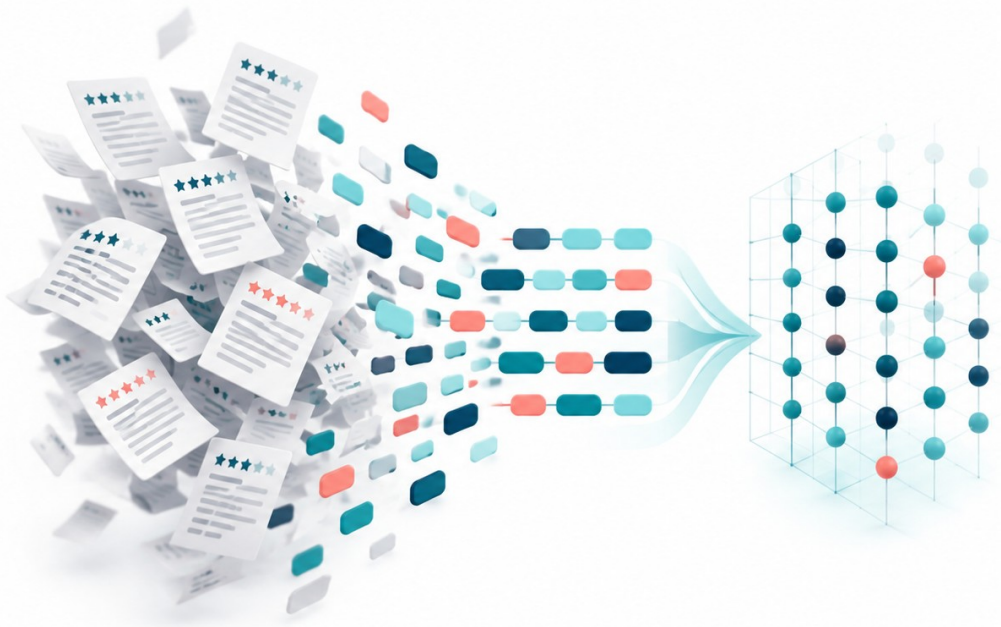
high-dimensional sparsity

CONTROL / EVIDENCE

Check weighted sparse matrix; confirm compare bigram signal.

Topic 3: Text analytics and feature engineering

Technical evidence from a synthetic Singapore hotel-review corpus



DATA → MODEL → INSIGHT

POS tags

INPUT / ARTIFACT

tokens with grammatical roles

ANALYTICAL DECISION

identify issue noun phrases

OBSERVE

tokens with grammatical roles

TRANSFORM

identify issue noun phrases

CHECK

tagger domain drift

POS tags: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

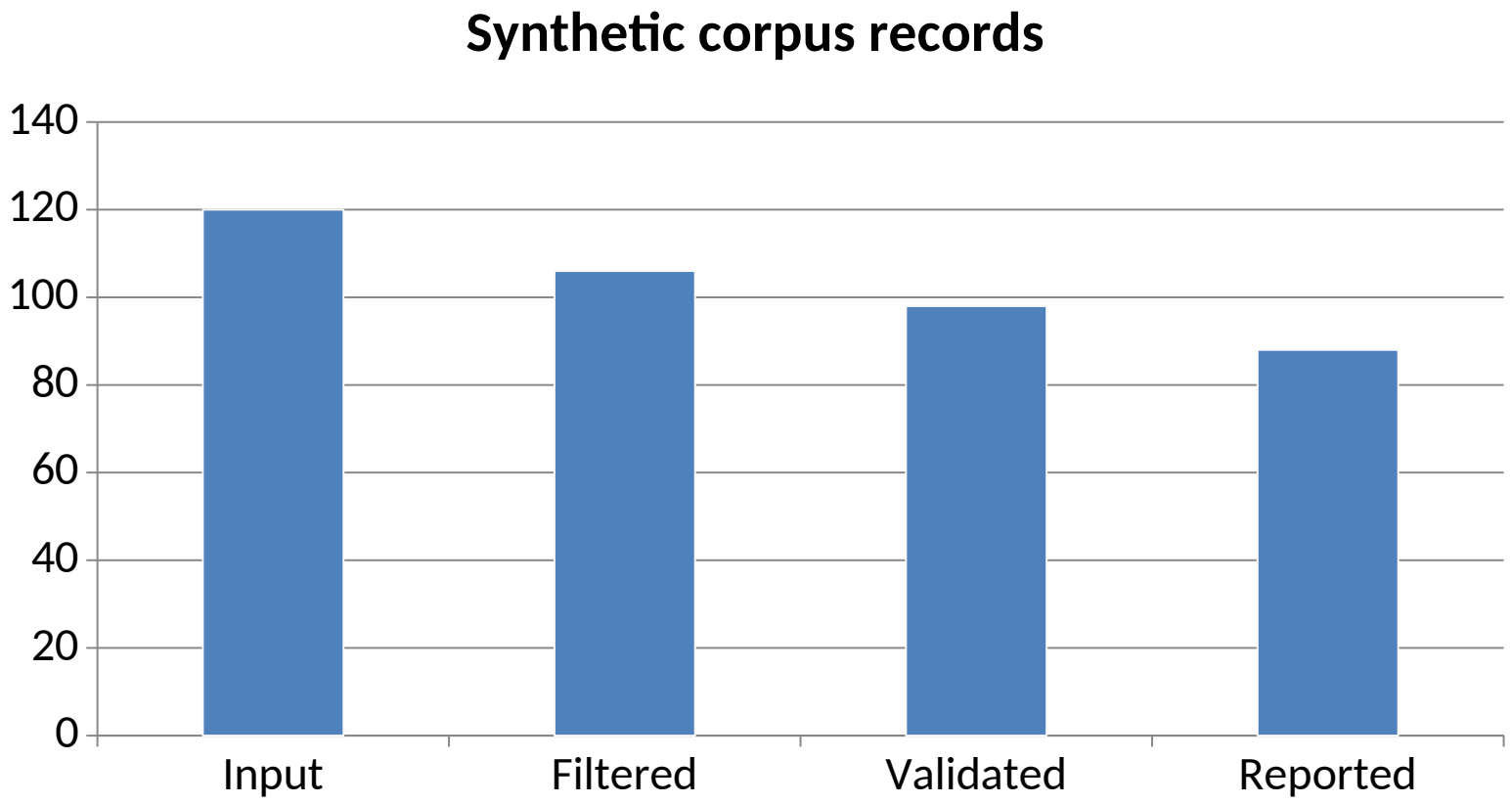
tokens with grammatical roles

PYTHON EXPRESSION OR ANALYTICAL RULE

```
nltk.pos_tag(tokens)
```

Evidence to inspect: identify issue noun phrases.

POS tags: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: identify issue noun phrases.

Editable PowerPoint chart

POS tags: failure and control

FAILURE MODE

tagger domain drift

CONTROL / EVIDENCE

Check tokens with grammatical roles; confirm identify issue noun phrases.

Named entities

INPUT / ARTIFACT

hotel and location spans

ANALYTICAL DECISION

measure entity precision

OBSERVE

hotel and location spans

TRANSFORM

measure entity precision

CHECK

brand aliases missed

Named entities: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

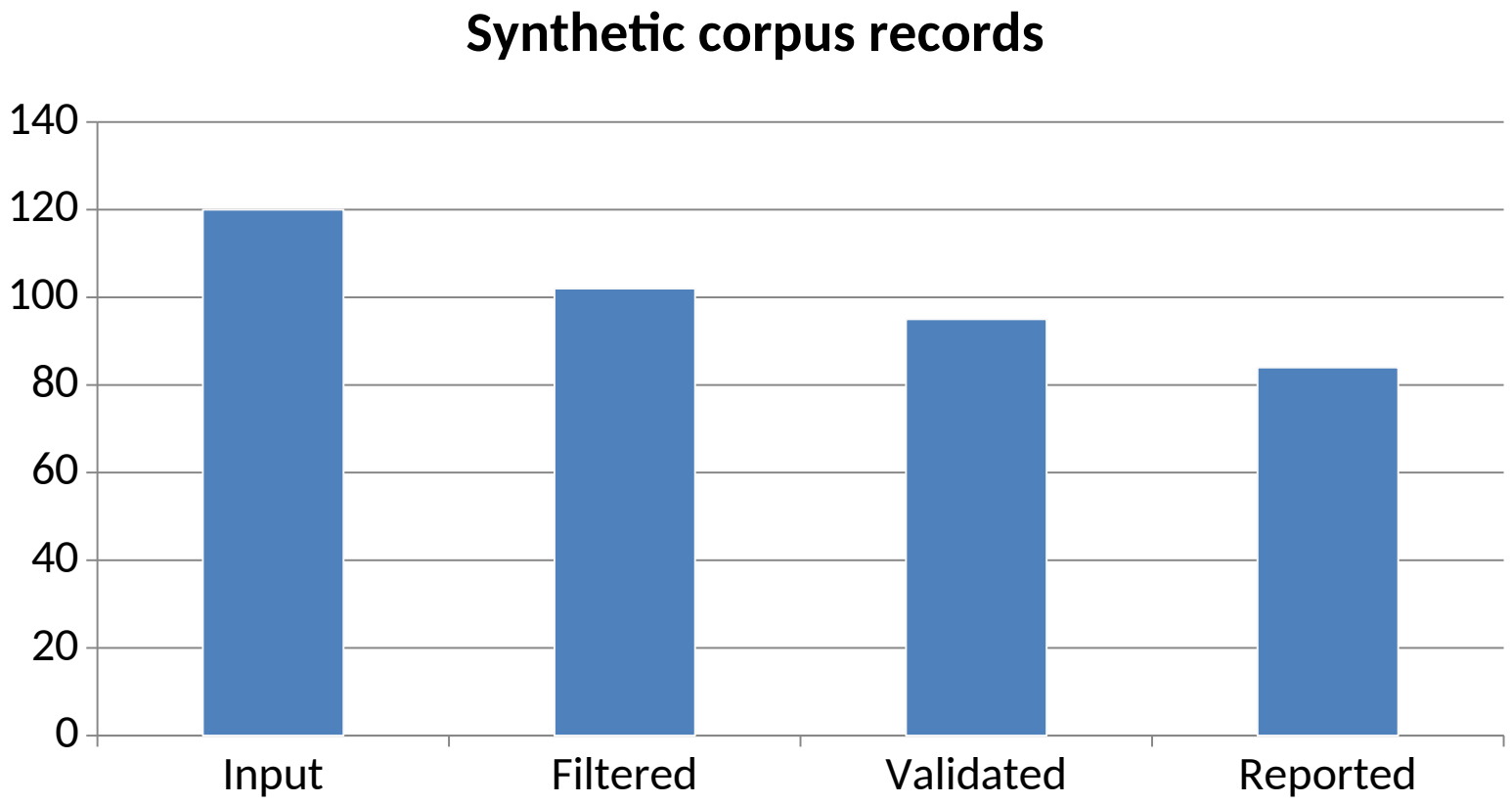
hotel and location spans

PYTHON EXPRESSION OR ANALYTICAL RULE

entity = hotel_name

Evidence to inspect: measure entity precision.

Named entities: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: measure entity precision.

Editable PowerPoint chart

Named entities: failure and control

FAILURE MODE

brand aliases missed

CONTROL / EVIDENCE

Check hotel and location spans; confirm measure entity precision.

Entity normalization

INPUT / ARTIFACT

canonical hotel names

ANALYTICAL DECISION

merge spelling variants

OBSERVE

canonical hotel names

TRANSFORM

merge spelling variants

CHECK

false merges

Entity normalization: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

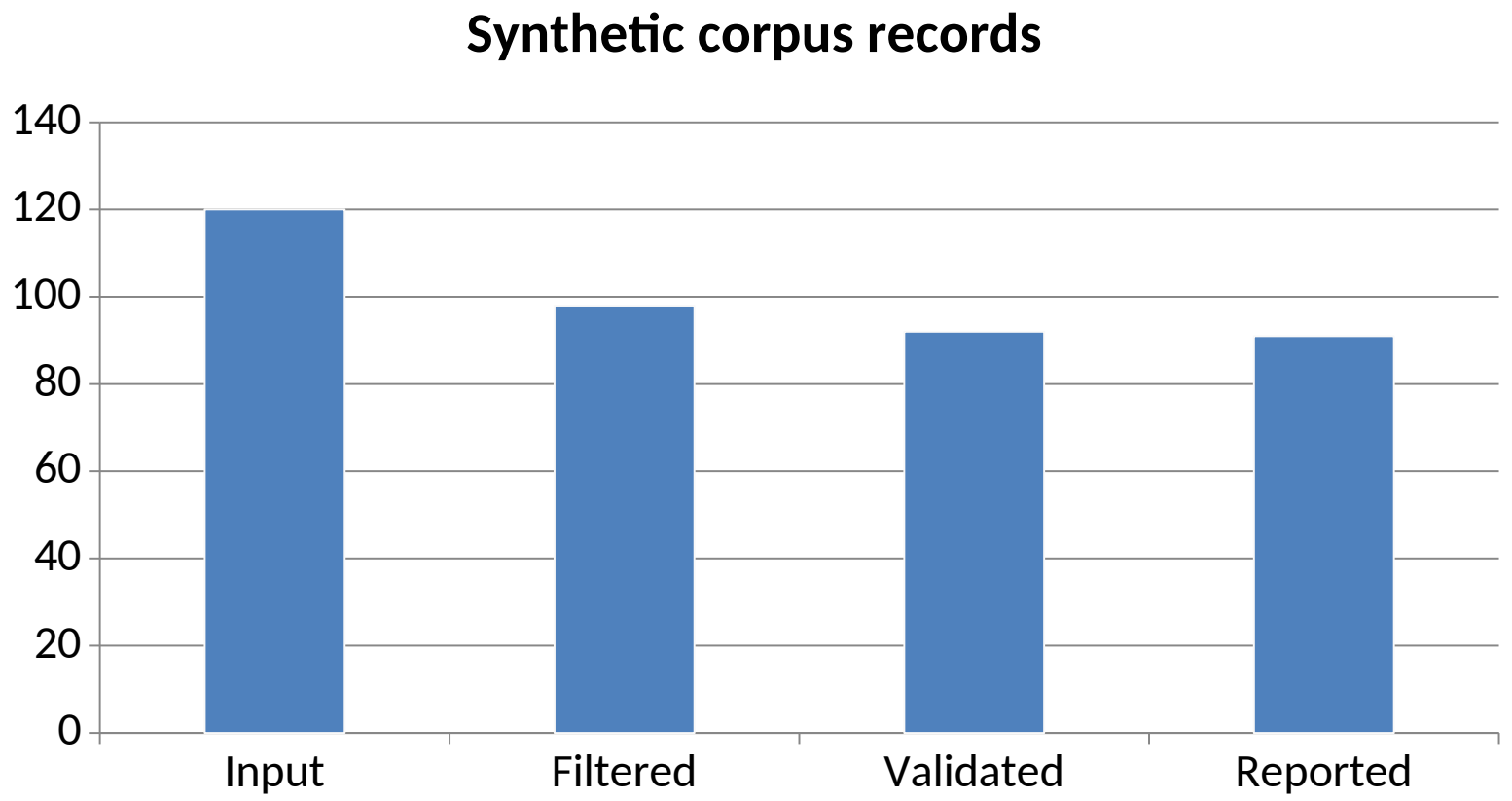
canonical hotel names

PYTHON EXPRESSION OR ANALYTICAL RULE

alias_map.get(name,name)

Evidence to inspect: merge spelling variants.

Entity normalization: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: merge spelling variants.

Editable PowerPoint chart

Entity normalization: failure and control

FAILURE MODE

false merges

CONTROL / EVIDENCE

Check canonical hotel names; confirm merge spelling variants.

Co-occurrence links

INPUT / ARTIFACT

hotel × aspect edges

ANALYTICAL DECISION

count review-level links

OBSERVE

hotel × aspect edges

TRANSFORM

count review-level links

CHECK

long documents inflate links

Co-occurrence links: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

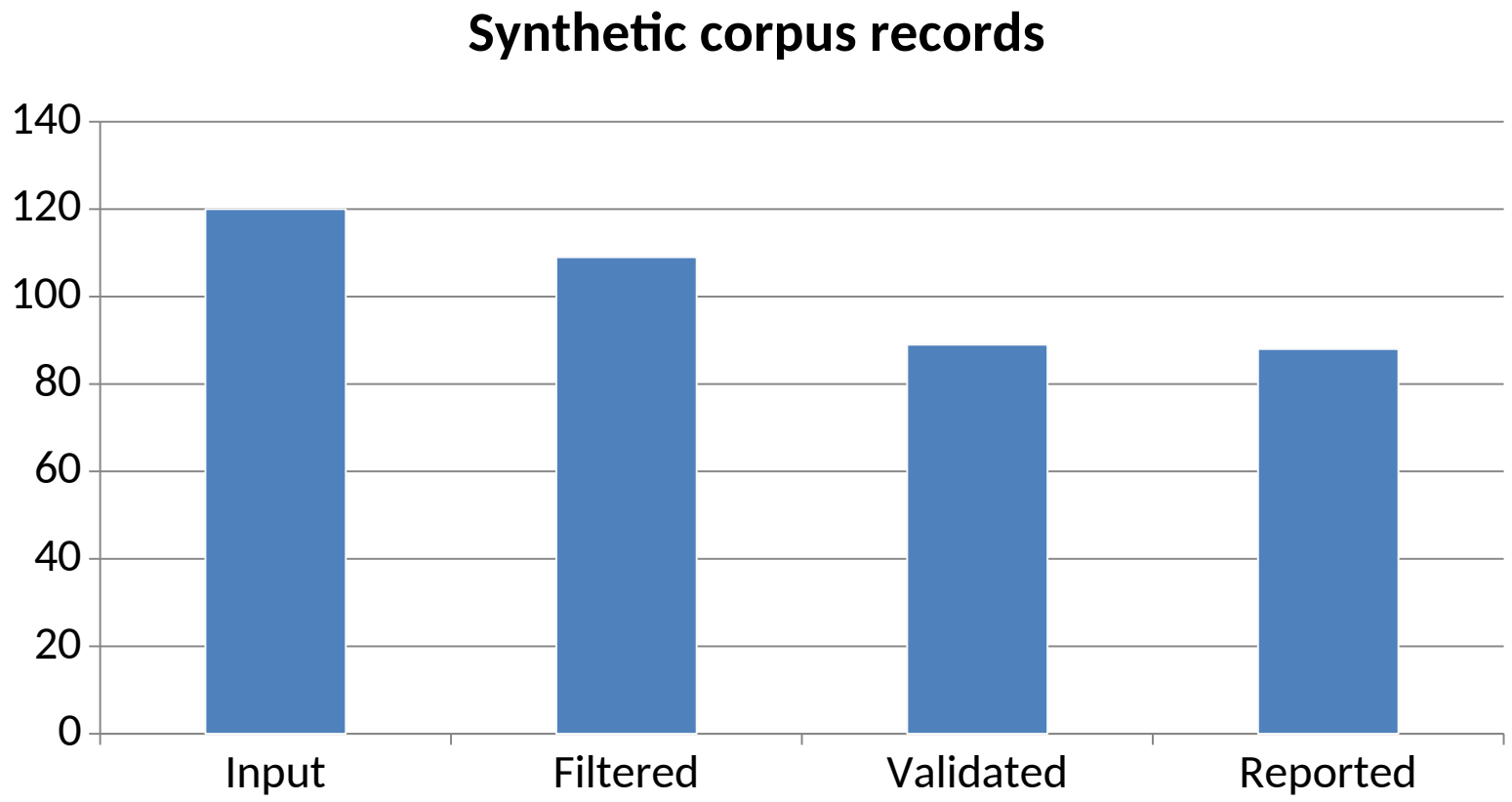
hotel × aspect edges

PYTHON EXPRESSION OR ANALYTICAL RULE

```
links[(hotel, aspect)] += 1
```

Evidence to inspect: count review-level links.

Co-occurrence links: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: count review-level links.

Editable PowerPoint chart

Co-occurrence links: failure and control

FAILURE MODE

long documents inflate links

CONTROL / EVIDENCE

Check hotel × aspect edges; confirm count review-level links.

Aspect extraction

INPUT / ARTIFACT

booking / room / support

ANALYTICAL DECISION

inspect phrase context

OBSERVE

booking / room / support

TRANSFORM

inspect phrase context

CHECK

ambiguous terms

Aspect extraction: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

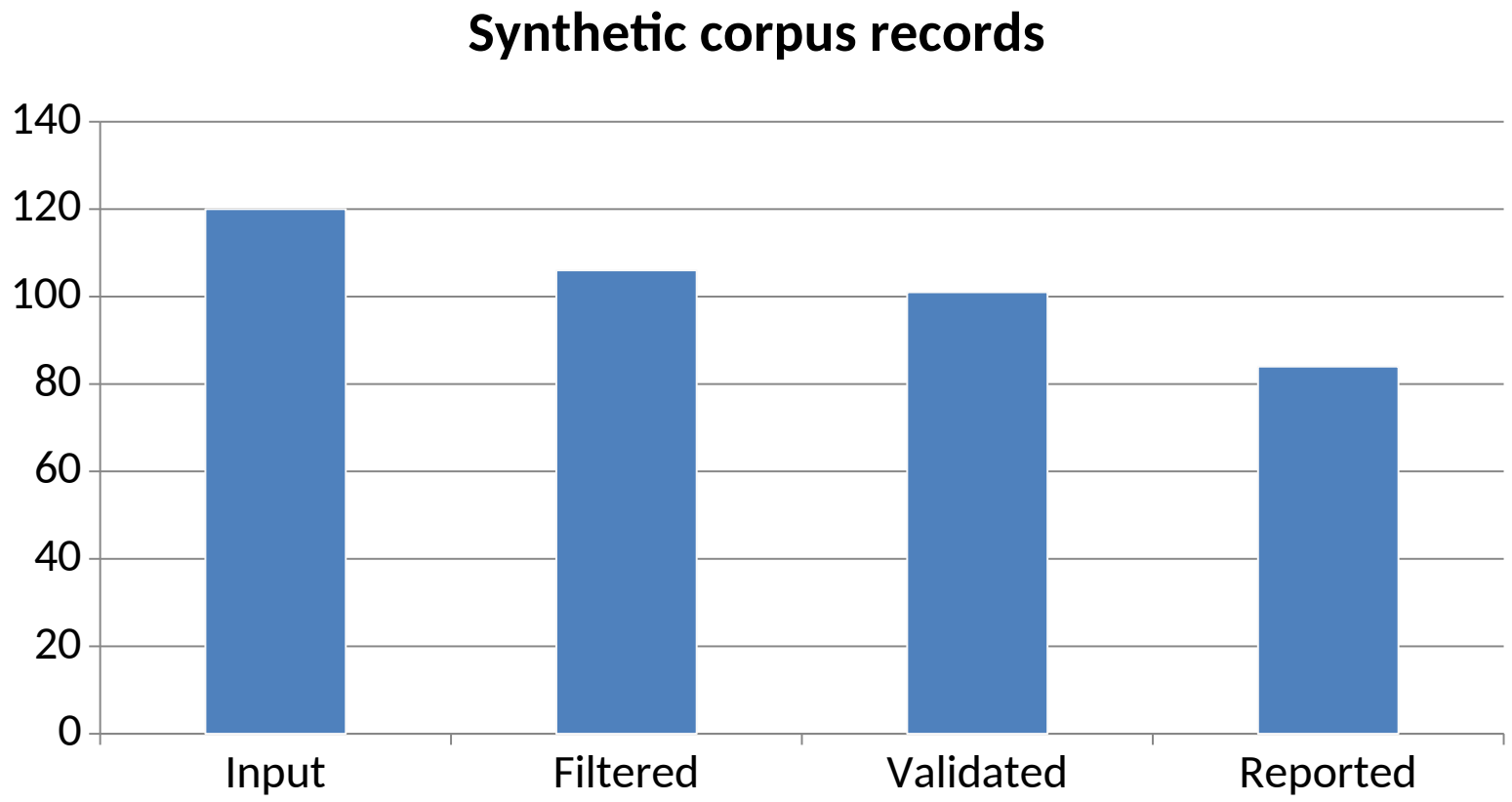
booking / room / support

PYTHON EXPRESSION OR ANALYTICAL RULE

```
text.str.contains("booking")
```

Evidence to inspect: inspect phrase context.

Aspect extraction: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: inspect phrase context.

Editable PowerPoint chart

Aspect extraction: failure and control

FAILURE MODE

ambiguous terms

CONTROL / EVIDENCE

Check booking / room / support; confirm inspect phrase context.

Feature engineering

INPUT / ARTIFACT

rating, channel, length

ANALYTICAL DECISION

compare incremental value

OBSERVE

rating, channel, length

TRANSFORM

compare incremental value

CHECK

label leakage via rating

Feature engineering: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

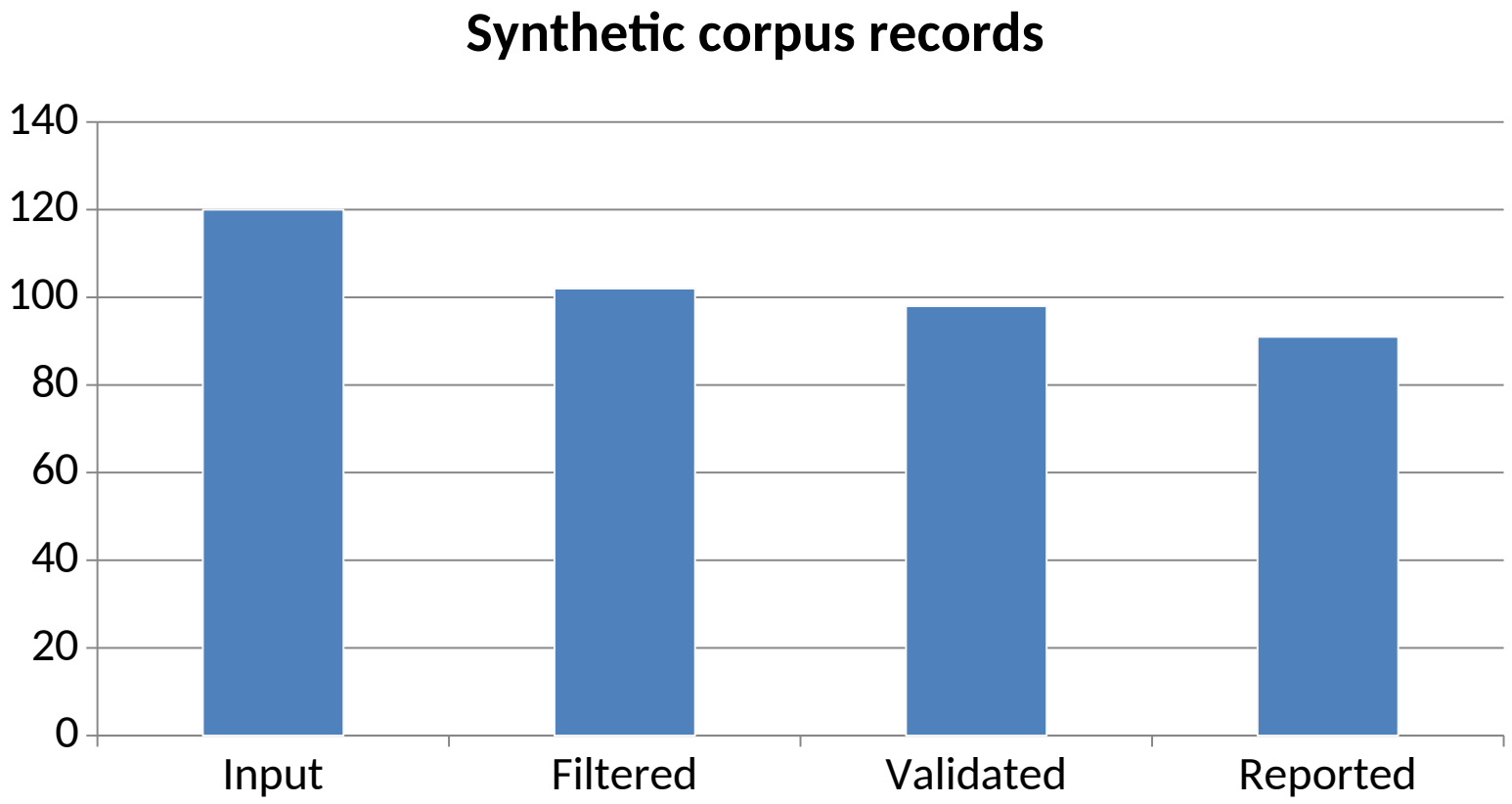
rating, channel, length

PYTHON EXPRESSION OR ANALYTICAL RULE

ColumnTransformer(...)

Evidence to inspect: compare incremental value.

Feature engineering: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: compare incremental value.

Editable PowerPoint chart

Feature engineering: failure and control

FAILURE MODE

label leakage via rating

CONTROL / EVIDENCE

Check rating, channel, length; confirm compare incremental value.

K-means clustering

INPUT / ARTIFACT

TF-IDF document vectors

ANALYTICAL DECISION

inspect inertia curve

OBSERVE

TF-IDF document vectors

TRANSFORM

inspect inertia curve

CHECK

cluster labels mistaken for truth

K-means clustering: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

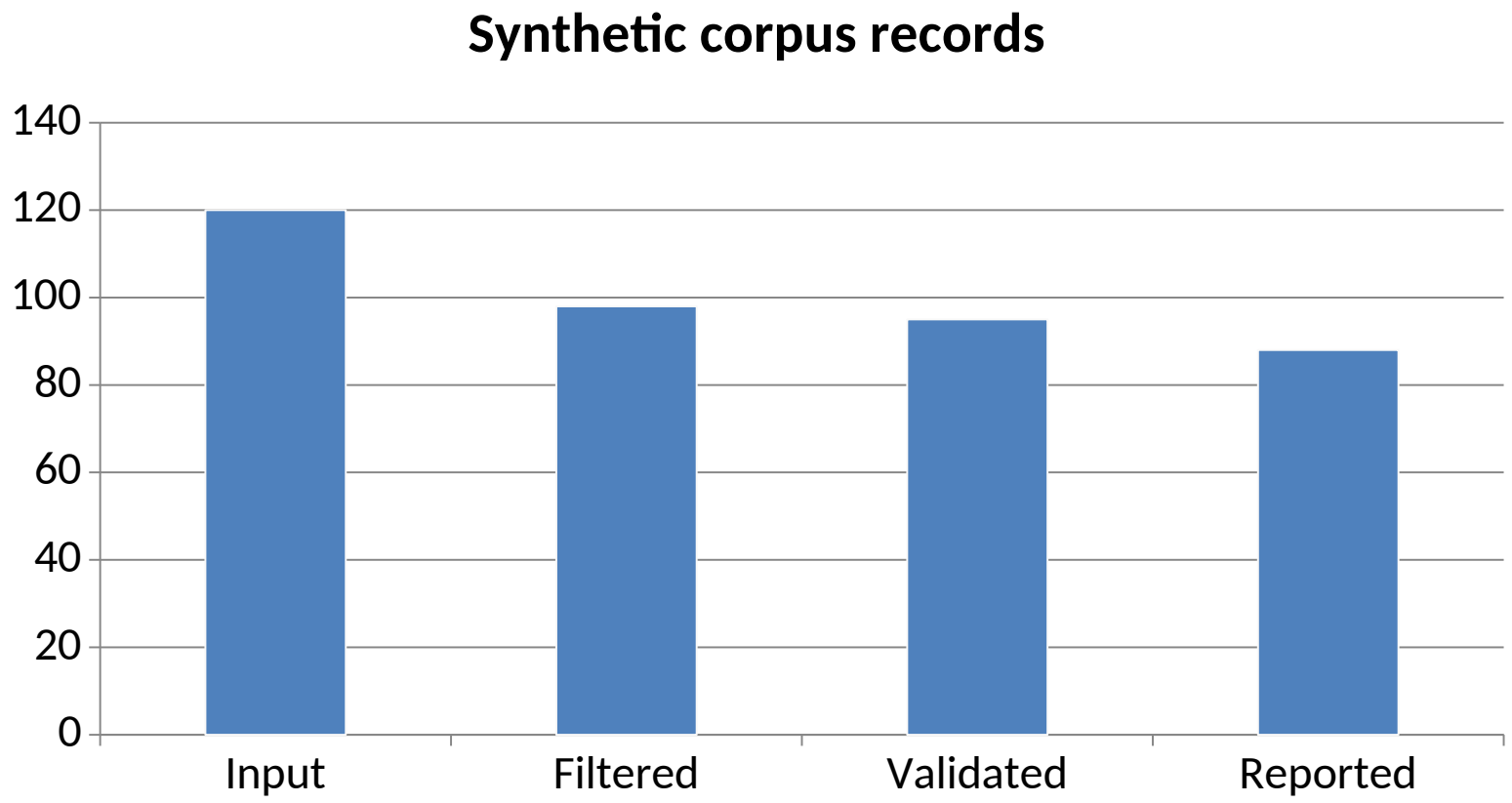
TF-IDF document vectors

PYTHON EXPRESSION OR ANALYTICAL RULE

```
KMeans(n_clusters=3)
```

Evidence to inspect: inspect inertia curve.

K-means clustering: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: inspect inertia curve.

Editable PowerPoint chart

K-means clustering: failure and control

FAILURE MODE

cluster labels mistaken for truth

CONTROL / EVIDENCE

Check TF-IDF document vectors; confirm inspect inertia curve.

Elbow diagnostic

INPUT / ARTIFACT

inertia for k=2..5

ANALYTICAL DECISION

look for diminishing returns

OBSERVE

inertia for k=2..5

TRANSFORM

look for diminishing returns

CHECK

no clear elbow

Elbow diagnostic: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

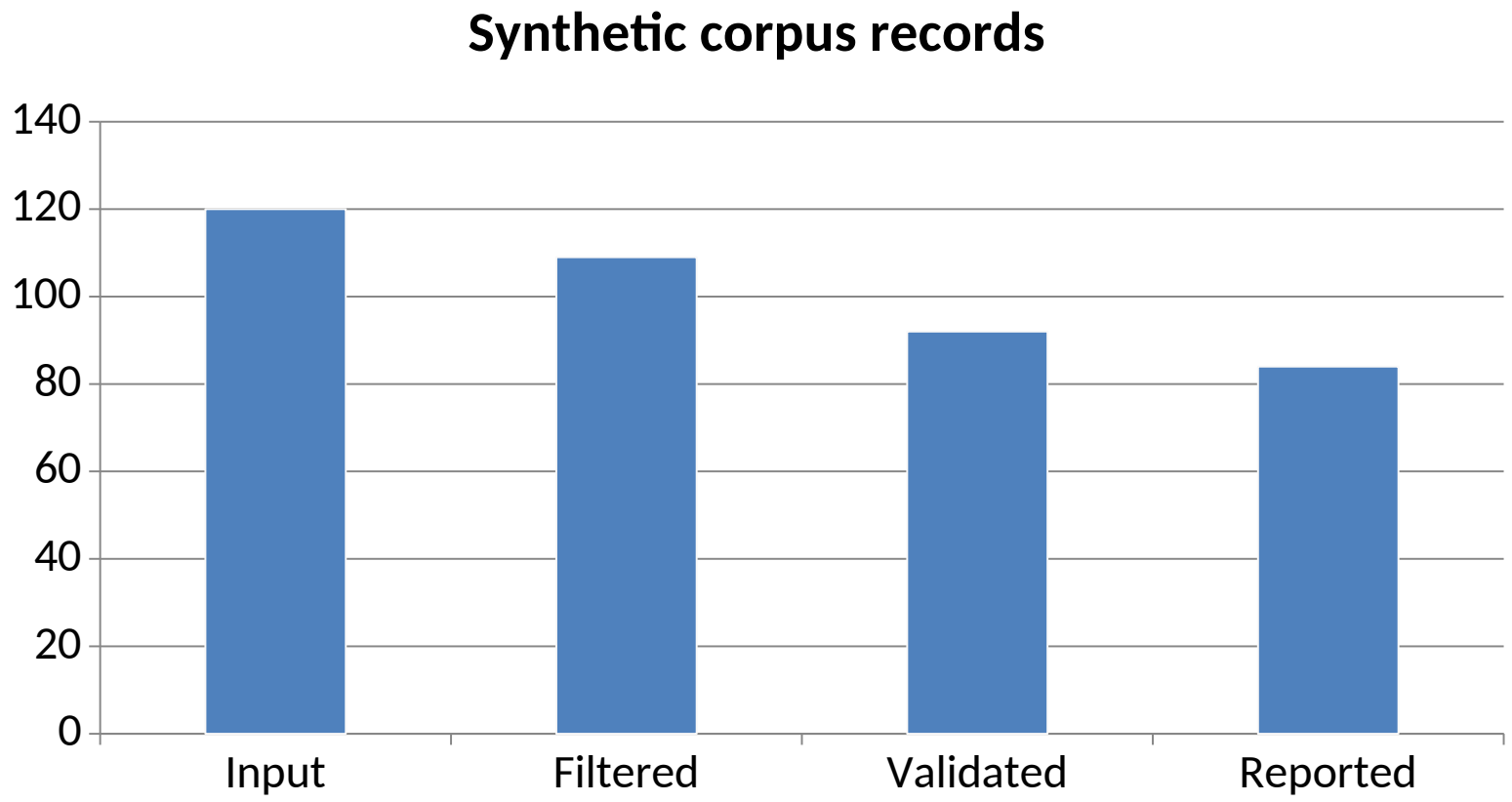
inertia for k=2..5

PYTHON EXPRESSION OR ANALYTICAL RULE

model.inertia_

Evidence to inspect: look for diminishing returns.

Elbow diagnostic: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: look for diminishing returns.

Editable PowerPoint chart

Elbow diagnostic: failure and control

FAILURE MODE

no clear elbow

CONTROL / EVIDENCE

Check inertia for k=2..5; confirm look for diminishing returns.

Supervised classification

INPUT / ARTIFACT

labeled review corpus

ANALYTICAL DECISION

compare macro F1

OBSERVE

labeled review corpus

TRANSFORM

compare macro F1

CHECK

class imbalance

Supervised classification: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

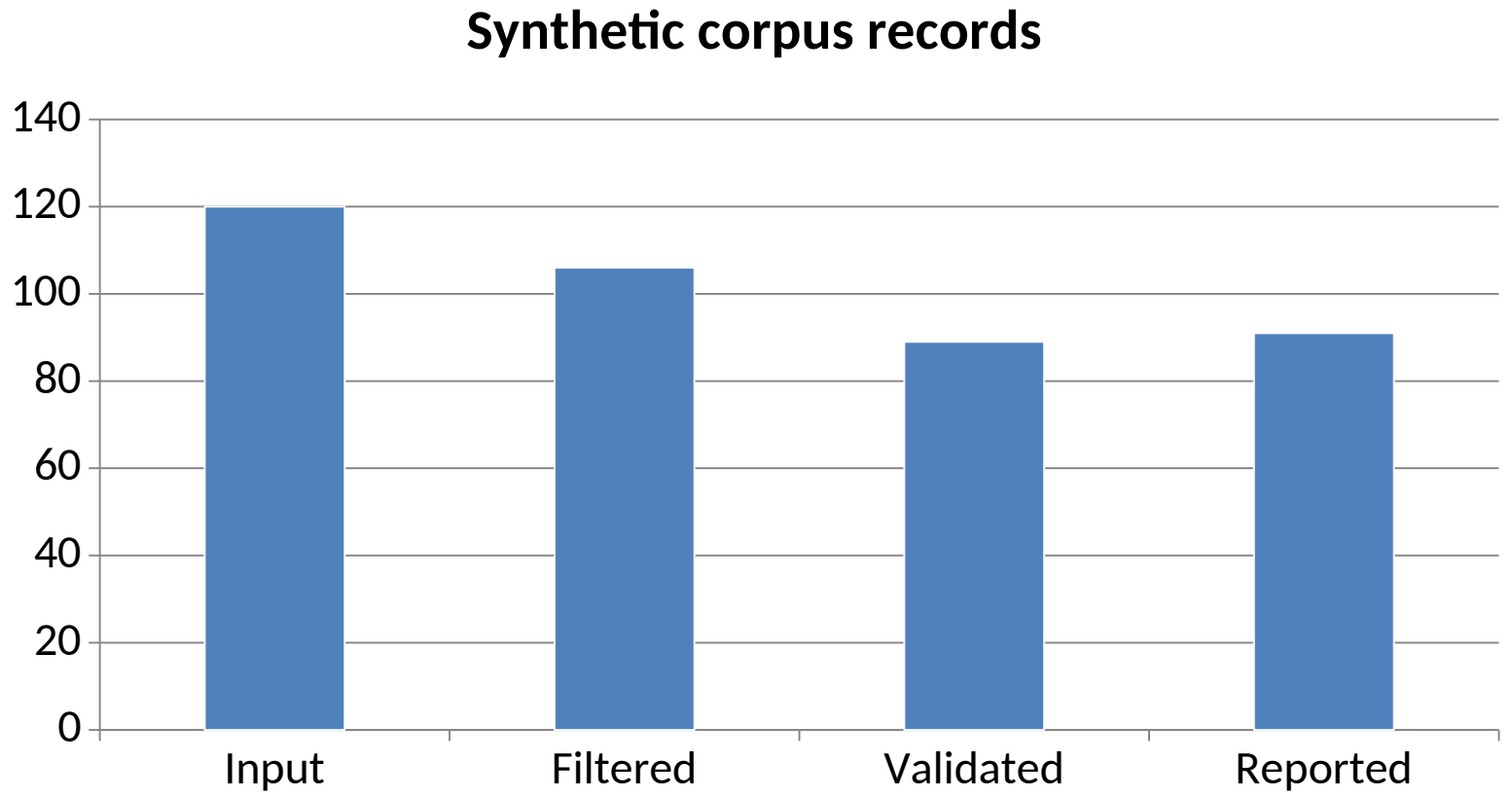
labeled review corpus

PYTHON EXPRESSION OR ANALYTICAL RULE

```
LogisticRegression(max_iter=1000)
```

Evidence to inspect: compare macro F1.

Supervised classification: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: compare macro F1.

Editable PowerPoint chart

Supervised classification: failure and control

FAILURE MODE

class imbalance

CONTROL / EVIDENCE

Check labeled review corpus; confirm compare macro F1.

Error analysis

INPUT / ARTIFACT

false negative examples

ANALYTICAL DECISION

review mistakes by aspect

OBSERVE

false negative examples

TRANSFORM

review mistakes by aspect

CHECK

metric-only evaluation

Error analysis: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

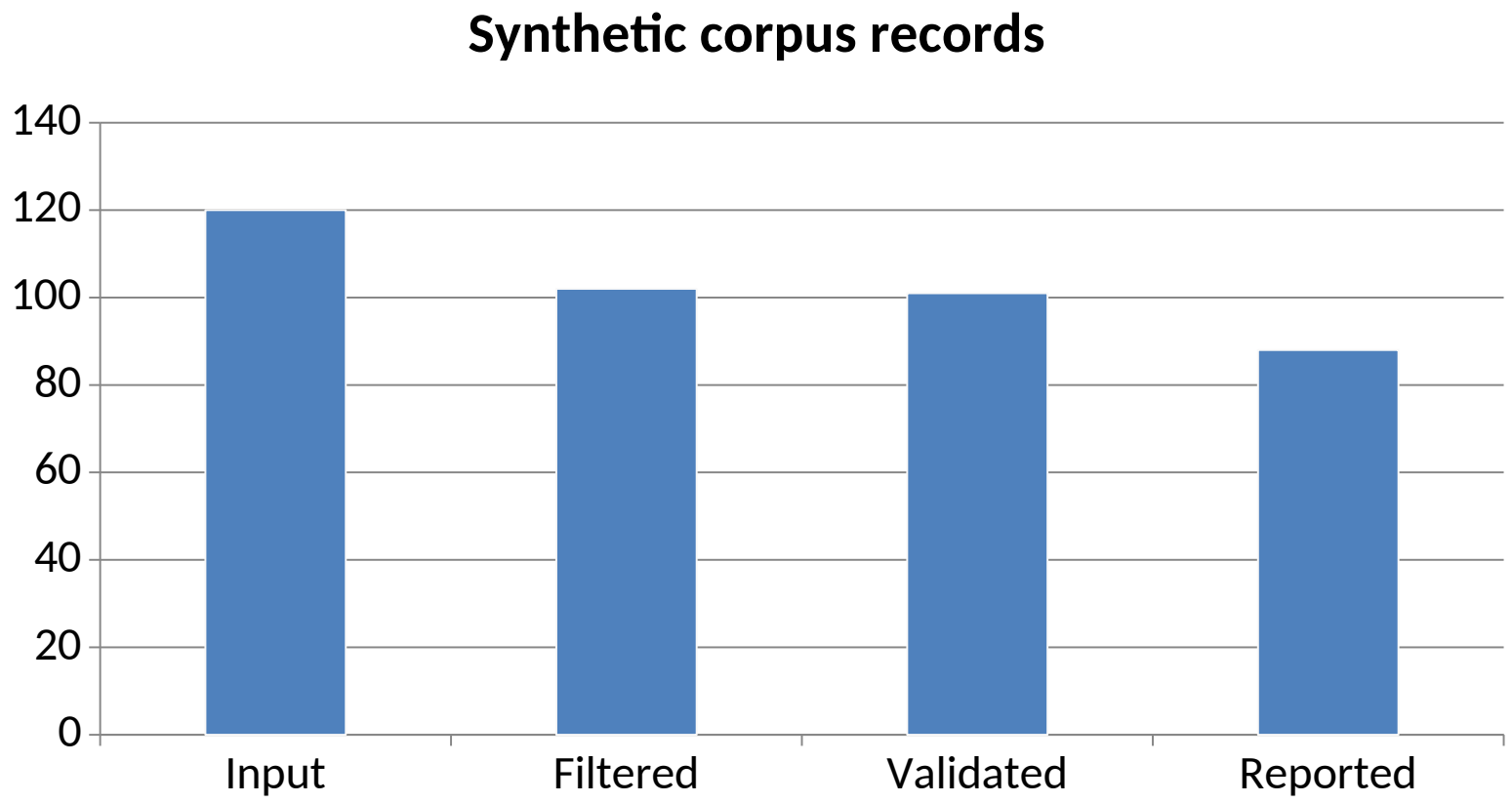
false negative examples

PYTHON EXPRESSION OR ANALYTICAL RULE

```
confusion_matrix(y_true,y_pred)
```

Evidence to inspect: review mistakes by aspect.

Error analysis: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: review mistakes by aspect.

Editable PowerPoint chart

Error analysis: failure and control

FAILURE MODE

metric-only evaluation

CONTROL / EVIDENCE

Check false negative examples; confirm review mistakes by aspect.

Topic 4: Sentiment modelling and evaluation

Technical evidence from a synthetic Singapore hotel-review corpus

DATA → MODEL → INSIGHT



Lexicon baseline

INPUT / ARTIFACT

positive/negative term scores

ANALYTICAL DECISION

inspect negation examples

OBSERVE

positive/negative term scores

TRANSFORM

inspect negation examples

CHECK

domain jargon

Lexicon baseline: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

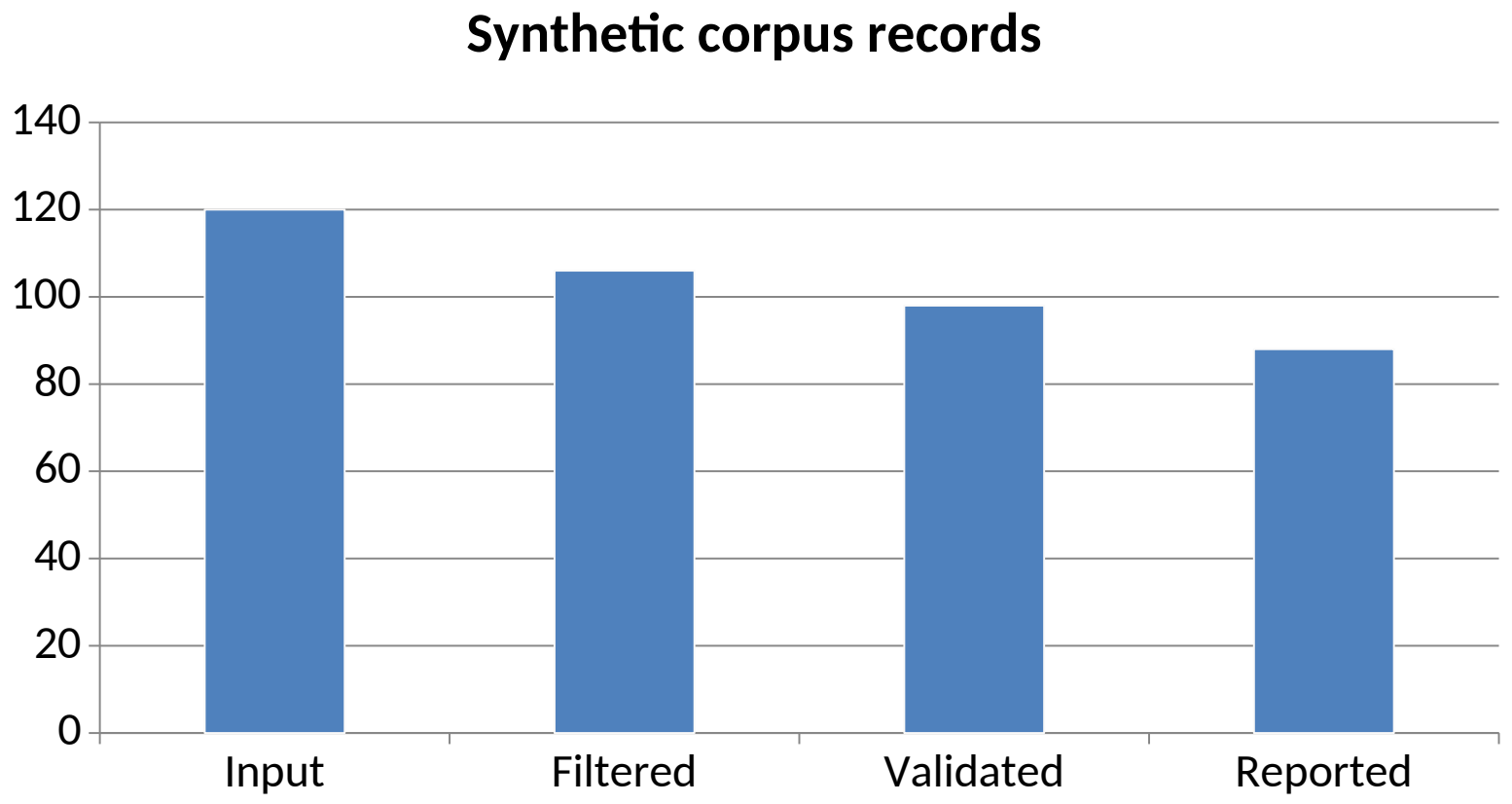
positive/negative term scores

PYTHON EXPRESSION OR ANALYTICAL RULE

score = pos_hits - neg_hits

Evidence to inspect: inspect negation examples.

Lexicon baseline: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: inspect negation examples.

Editable PowerPoint chart

Lexicon baseline: failure and control

FAILURE MODE

domain jargon

CONTROL / EVIDENCE

Check positive/negative term scores; confirm inspect negation examples.

Sentiment labels

INPUT / ARTIFACT

three-class annotation

ANALYTICAL DECISION

audit rating mismatch

OBSERVE

three-class annotation

TRANSFORM

audit rating mismatch

CHECK

weak labels

Sentiment labels: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

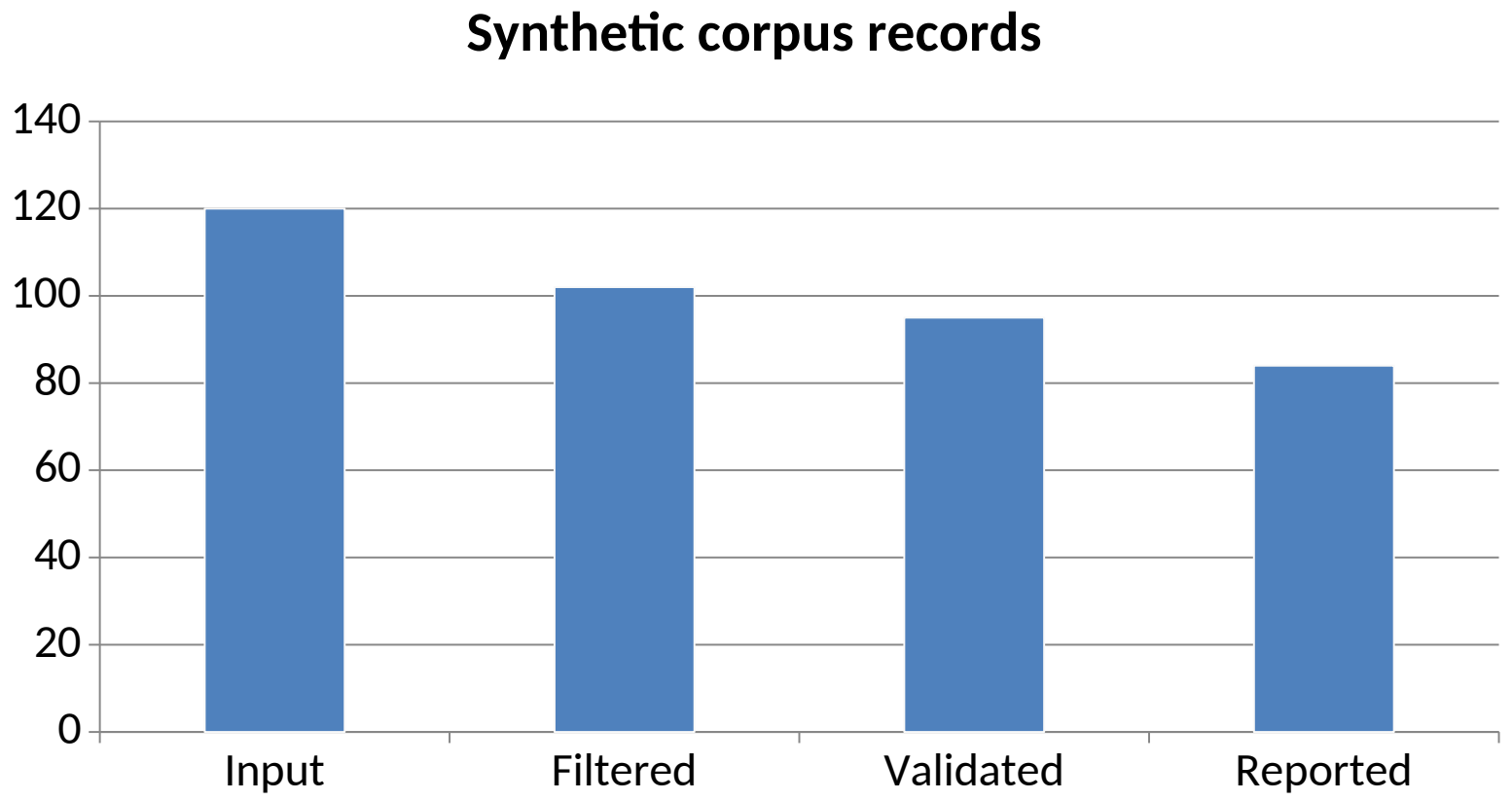
three-class annotation

PYTHON EXPRESSION OR ANALYTICAL RULE

```
pd.crosstab(rating,sentiment)
```

Evidence to inspect: audit rating mismatch.

Sentiment labels: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: audit rating mismatch.

Editable PowerPoint chart

Sentiment labels: failure and control

FAILURE MODE

weak labels

CONTROL / EVIDENCE

Check three-class annotation; confirm audit rating mismatch.

Stratified split

INPUT / ARTIFACT

75/25 train/holdout

ANALYTICAL DECISION

preserve class proportions

OBSERVE

75/25 train/holdout

TRANSFORM

preserve class proportions

CHECK

tiny holdout

Stratified split: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

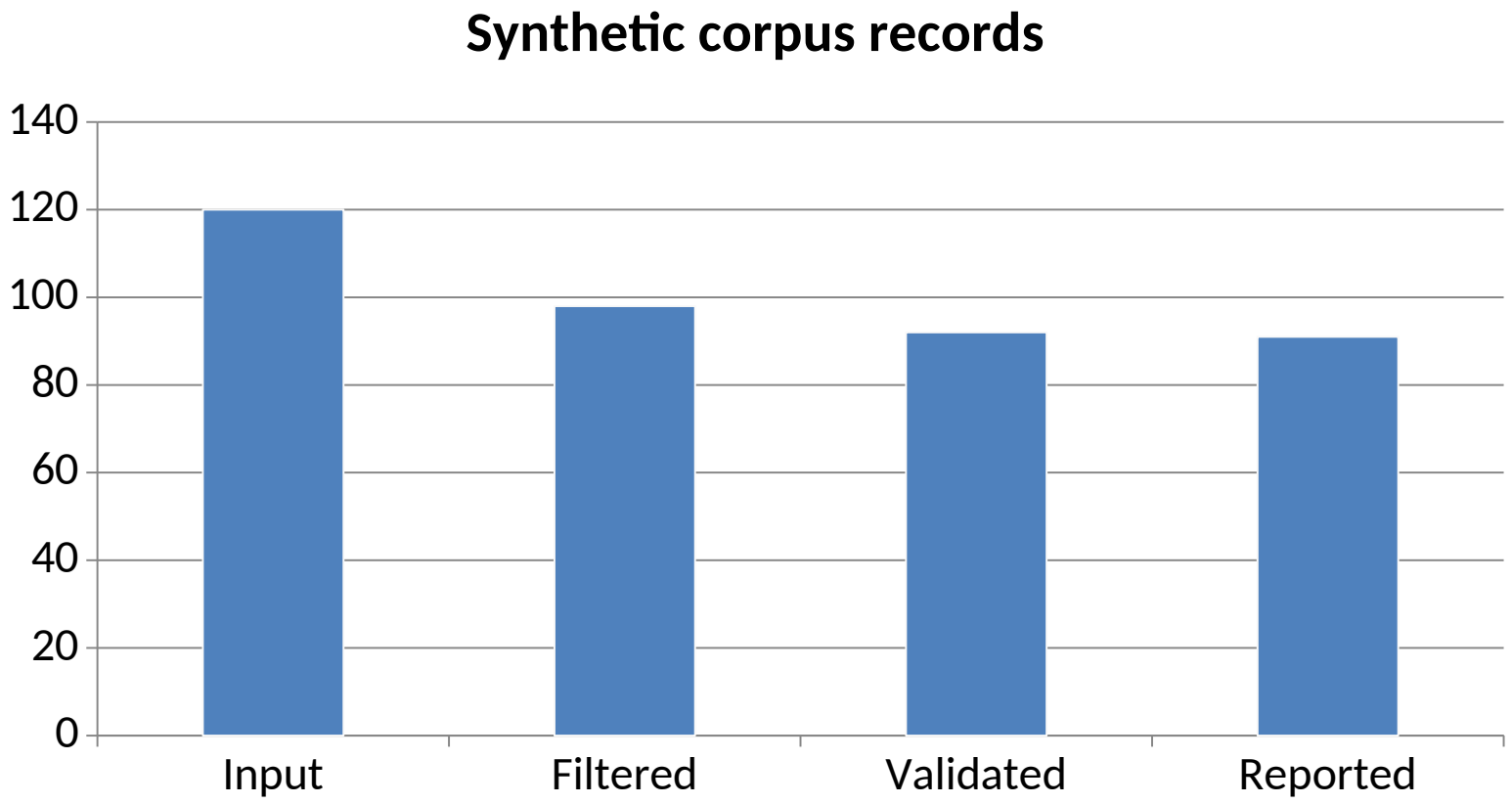
75/25 train/holdout

PYTHON EXPRESSION OR ANALYTICAL RULE

train_test_split(..., stratify=y)

Evidence to inspect: preserve class proportions.

Stratified split: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: preserve class proportions.

Editable PowerPoint chart

Stratified split: failure and control

FAILURE MODE

tiny holdout

CONTROL / EVIDENCE

Check 75/25 train/holdout; confirm preserve class proportions.

Pipeline isolation

INPUT / ARTIFACT

TF-IDF then logistic model

ANALYTICAL DECISION

fit on training fold only

OBSERVE

TF-IDF then logistic model

TRANSFORM

fit on training fold only

CHECK

pre-fit vectorizer leakage

Pipeline isolation: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

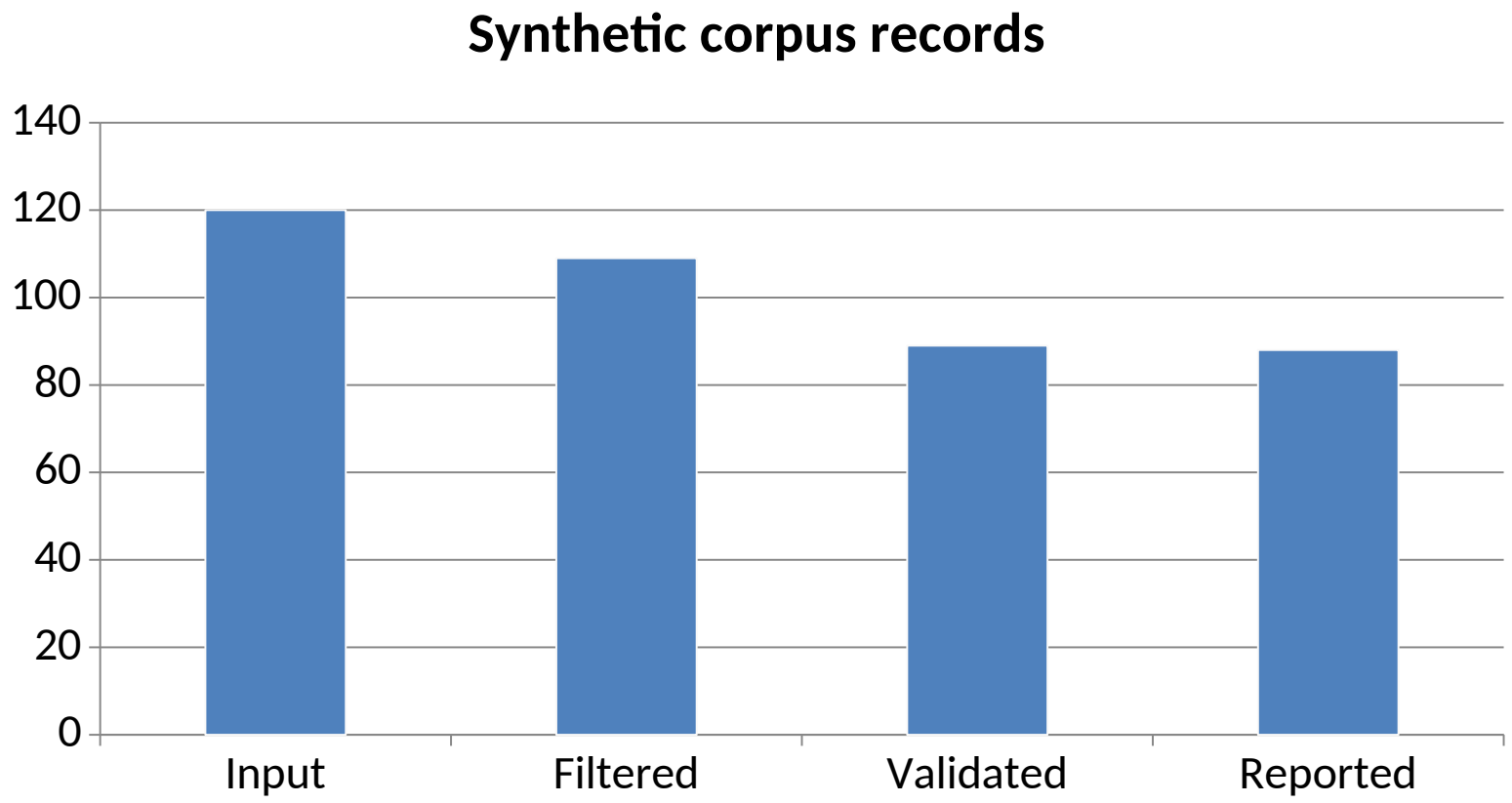
TF-IDF then logistic model

PYTHON EXPRESSION OR ANALYTICAL RULE

```
make_pipeline(TfidfVectorizer(), LogisticRegression())
```

Evidence to inspect: fit on training fold only.

Pipeline isolation: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: fit on training fold only.

Editable PowerPoint chart

Pipeline isolation: failure and control

FAILURE MODE

pre-fit vectorizer leakage

CONTROL / EVIDENCE

Check TF-IDF then logistic model; confirm fit on training fold only.

Model fit

INPUT / ARTIFACT

regularized logistic regression

ANALYTICAL DECISION

inspect coefficients

OBSERVE

regularized logistic regression

TRANSFORM

inspect coefficients

CHECK

overfitting

Model fit: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

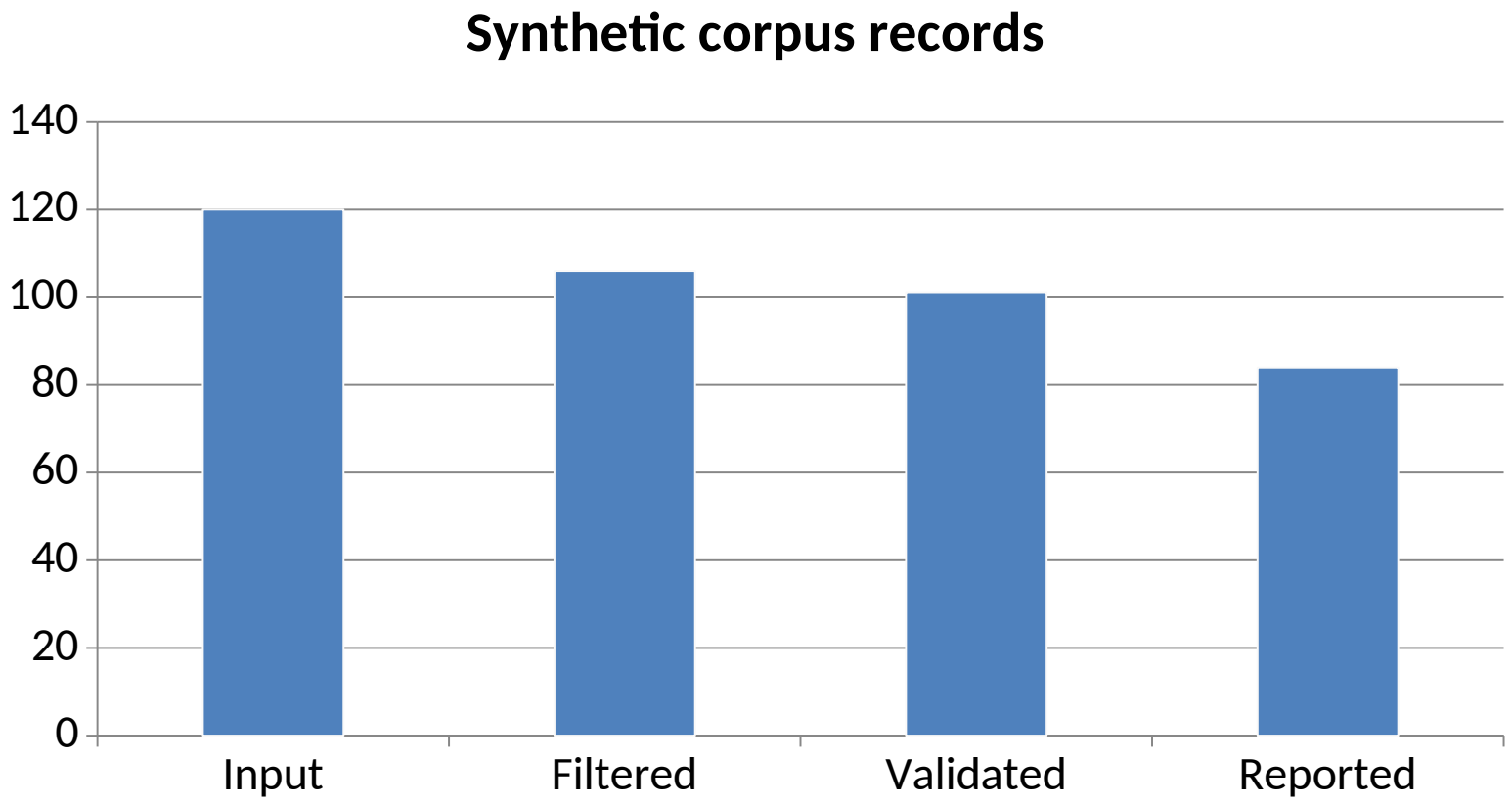
regularized logistic regression

PYTHON EXPRESSION OR ANALYTICAL RULE

```
pipe.fit(X_train,y_train)
```

Evidence to inspect: inspect coefficients.

Model fit: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: inspect coefficients.

Editable PowerPoint chart

Model fit: failure and control

FAILURE MODE

overfitting

CONTROL / EVIDENCE

Check regularized logistic regression; confirm inspect coefficients.

Predictions

INPUT / ARTIFACT

class and confidence

ANALYTICAL DECISION

sample uncertain cases

OBSERVE

class and confidence

TRANSFORM

sample uncertain cases

CHECK

confidence calibration

Predictions: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

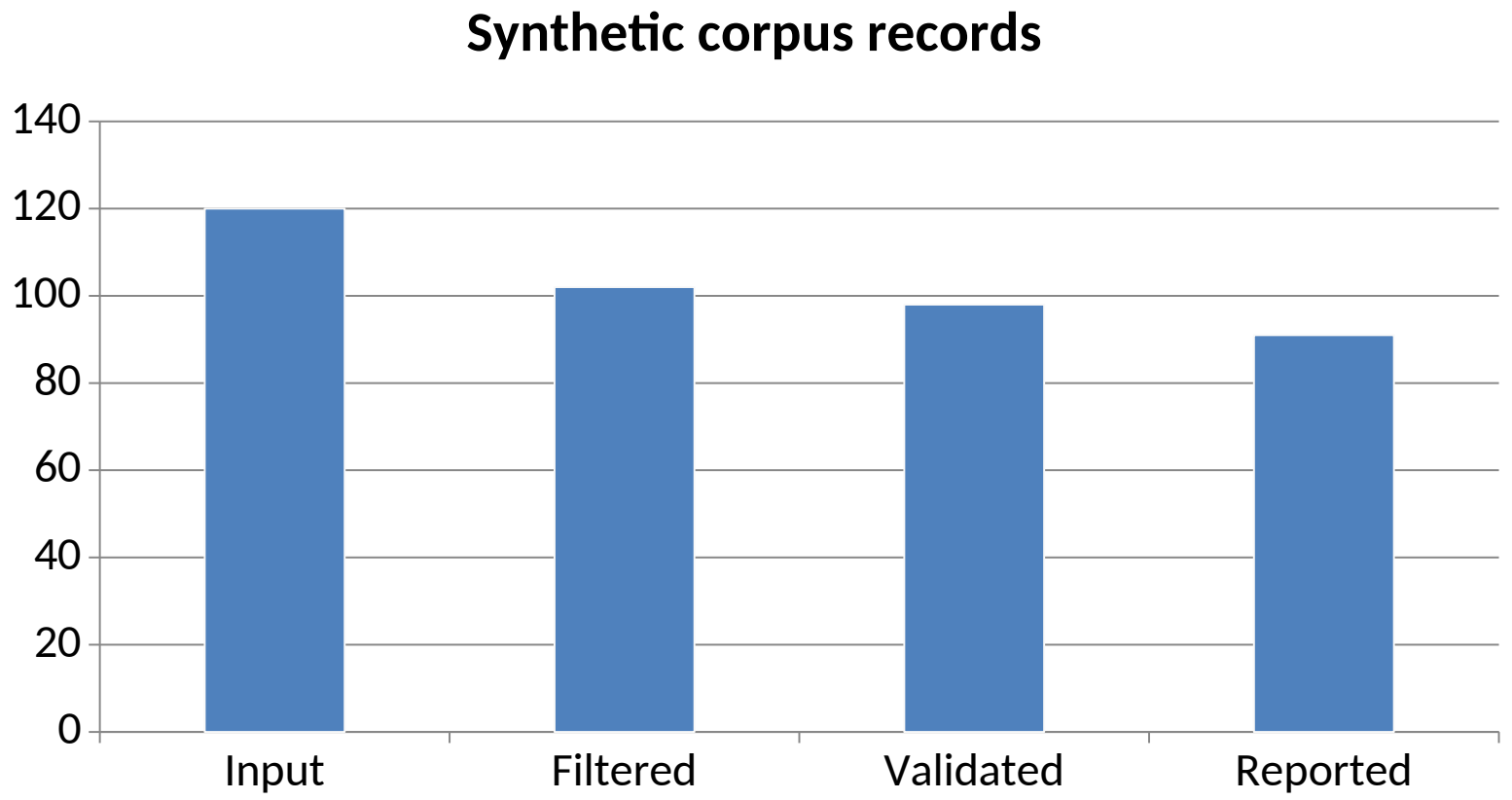
class and confidence

PYTHON EXPRESSION OR ANALYTICAL RULE

```
pipe.predict(X_test)
```

Evidence to inspect: sample uncertain cases.

Predictions: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: sample uncertain cases.

Editable PowerPoint chart

Predictions: failure and control

FAILURE MODE

confidence calibration

CONTROL / EVIDENCE

Check class and confidence; confirm sample uncertain cases.

Confusion matrix

INPUT / ARTIFACT

true × predicted counts

ANALYTICAL DECISION

inspect negative misses

OBSERVE

true × predicted counts

TRANSFORM

inspect negative misses

CHECK

axis swap

Confusion matrix: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

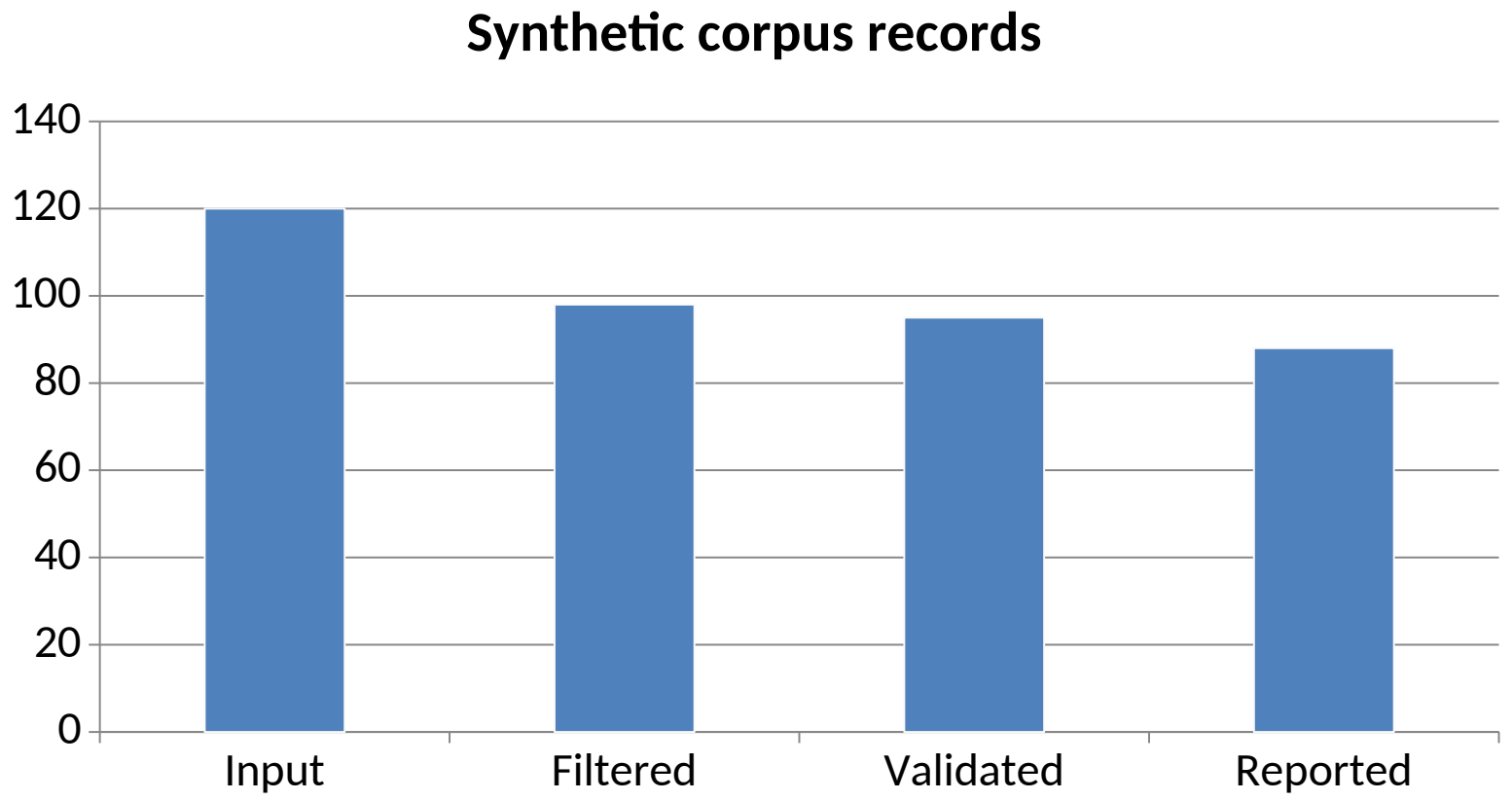
true × predicted counts

PYTHON EXPRESSION OR ANALYTICAL RULE

```
confusion_matrix(y_test,pred)
```

Evidence to inspect: inspect negative misses.

Confusion matrix: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: inspect negative misses.

Editable PowerPoint chart

Confusion matrix: failure and control

FAILURE MODE

axis swap

CONTROL / EVIDENCE

Check true × predicted counts; confirm inspect negative misses.

Precision and recall

INPUT / ARTIFACT

per-class metrics

ANALYTICAL DECISION

trade triage capacity vs misses

OBSERVE

per-class metrics

TRANSFORM

trade triage capacity vs misses

CHECK

accuracy-only reporting

Precision and recall: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

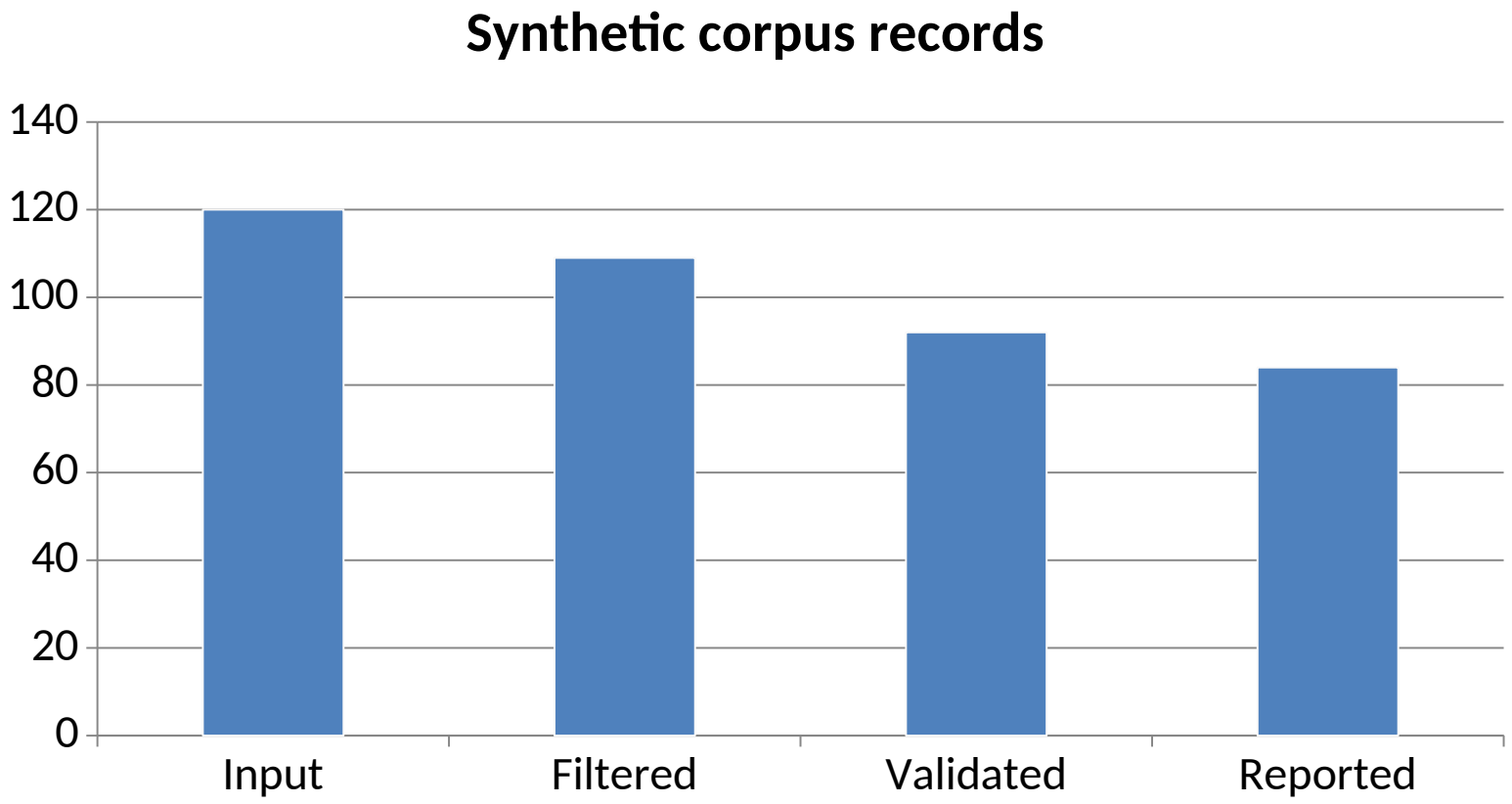
per-class metrics

PYTHON EXPRESSION OR ANALYTICAL RULE

```
classification_report(y_test,pred)
```

Evidence to inspect: trade triage capacity vs misses.

Precision and recall: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: trade triage capacity vs misses.

Editable PowerPoint chart

Precision and recall: failure and control

FAILURE MODE

accuracy-only reporting

CONTROL / EVIDENCE

Check per-class metrics; confirm trade triage capacity vs misses.

Macro F1

INPUT / ARTIFACT

unweighted class mean

ANALYTICAL DECISION

compare to majority baseline

OBSERVE

unweighted class mean

TRANSFORM

compare to majority baseline

CHECK

support ignored

Macro F1: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

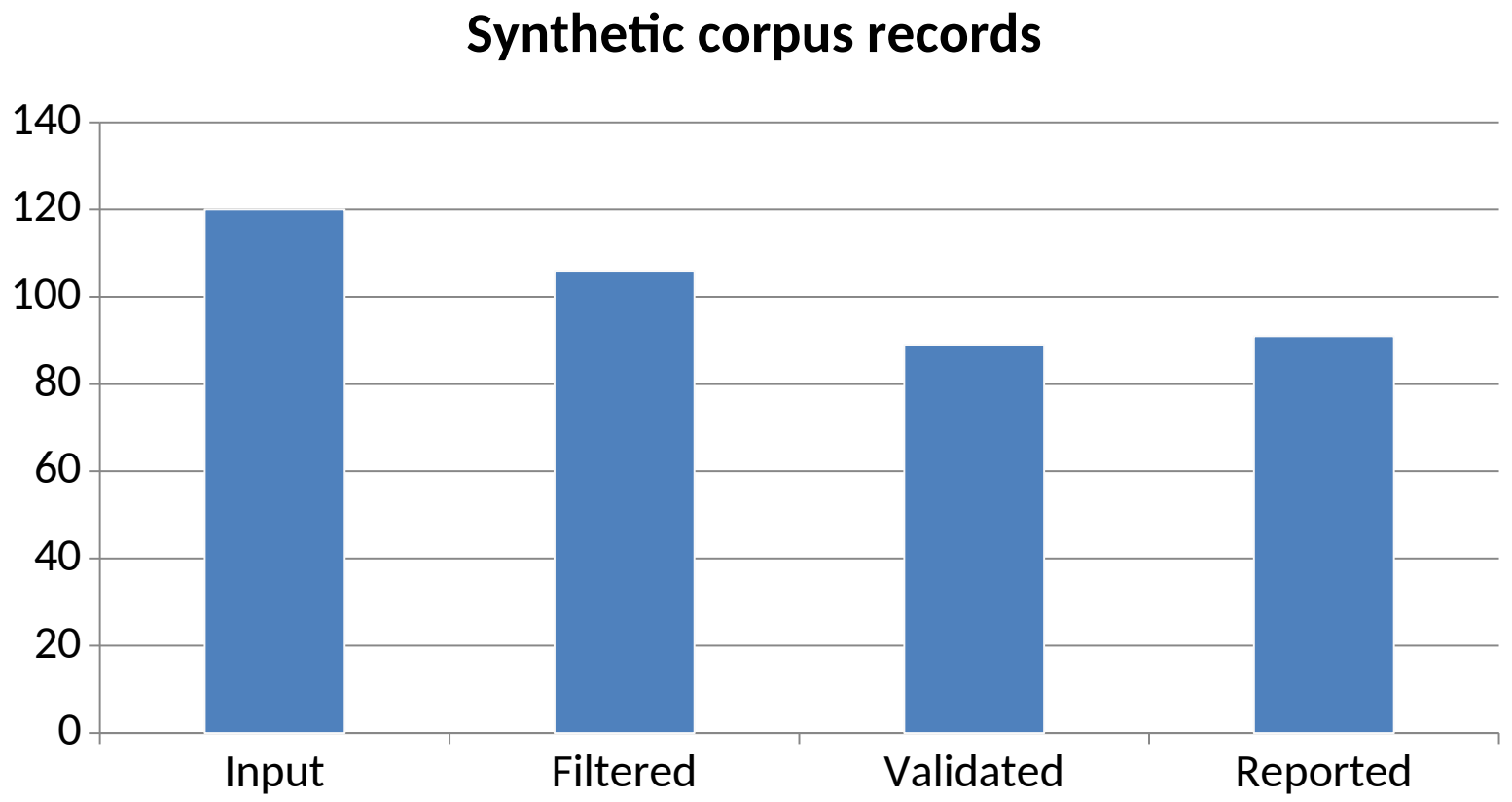
unweighted class mean

PYTHON EXPRESSION OR ANALYTICAL RULE

```
f1_score(...,average="macro")
```

Evidence to inspect: compare to majority baseline.

Macro F1: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: compare to majority baseline.

Editable PowerPoint chart

Macro F1: failure and control

FAILURE MODE

support ignored

CONTROL / EVIDENCE

Check unweighted class mean; confirm compare to majority baseline.

Model card

INPUT / ARTIFACT

data, metrics, limits

ANALYTICAL DECISION

document deployment threshold

OBSERVE

data, metrics, limits

TRANSFORM

document deployment threshold

CHECK

unsupported claims

Model card: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

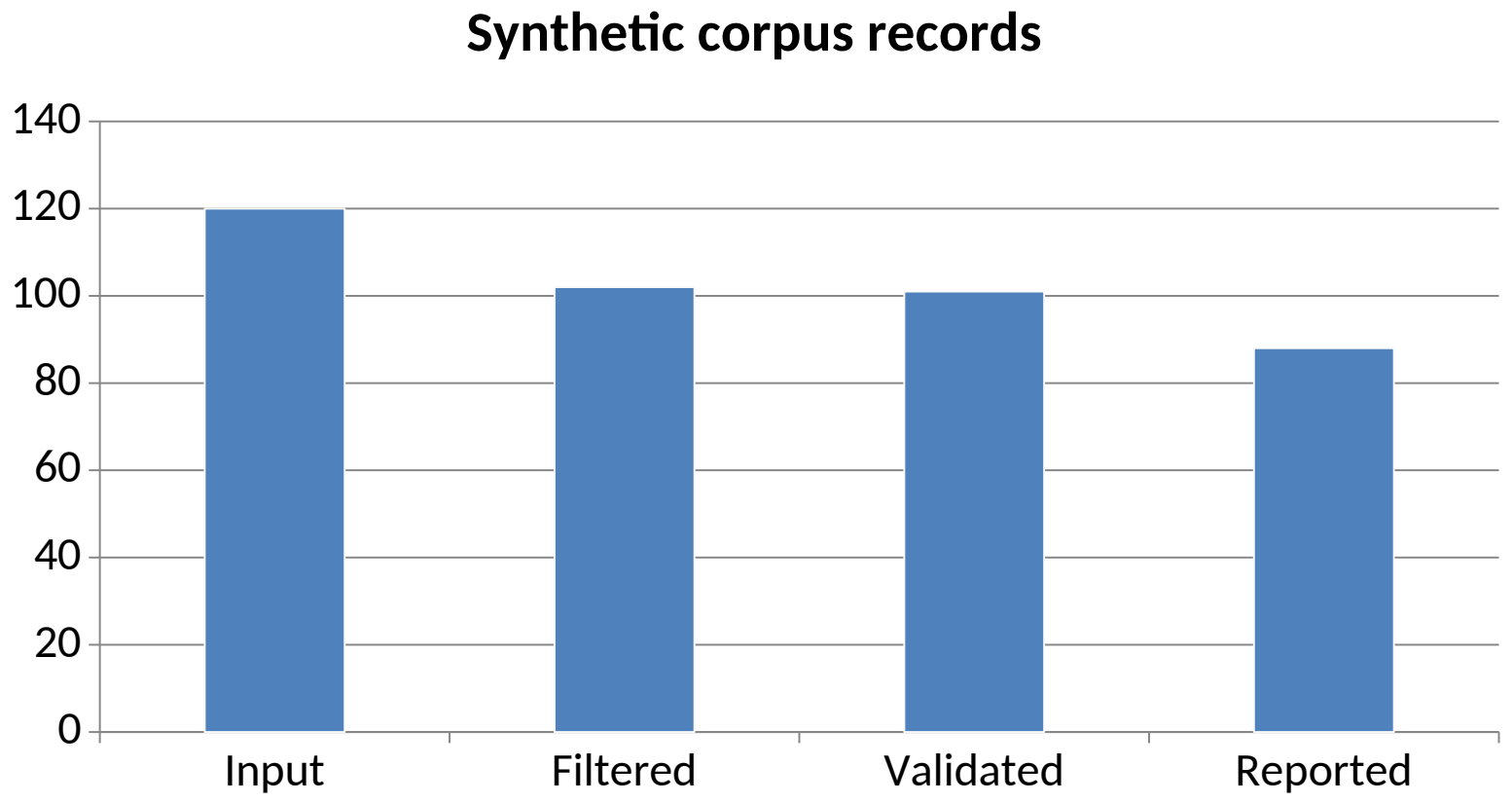
data, metrics, limits

PYTHON EXPRESSION OR ANALYTICAL RULE

model_version + training_date

Evidence to inspect: document deployment threshold.

Model card: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: document deployment threshold.

Editable PowerPoint chart

Model card: failure and control

FAILURE MODE

unsupported claims

CONTROL / EVIDENCE

Check data, metrics, limits; confirm document deployment threshold.

TextBlob polarity

INPUT / ARTIFACT

lexicon-derived polarity score

ANALYTICAL DECISION

compare to supervised holdout

OBSERVE

lexicon-derived polarity score

TRANSFORM

compare to supervised holdout

CHECK

domain-specific wording

TextBlob polarity: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

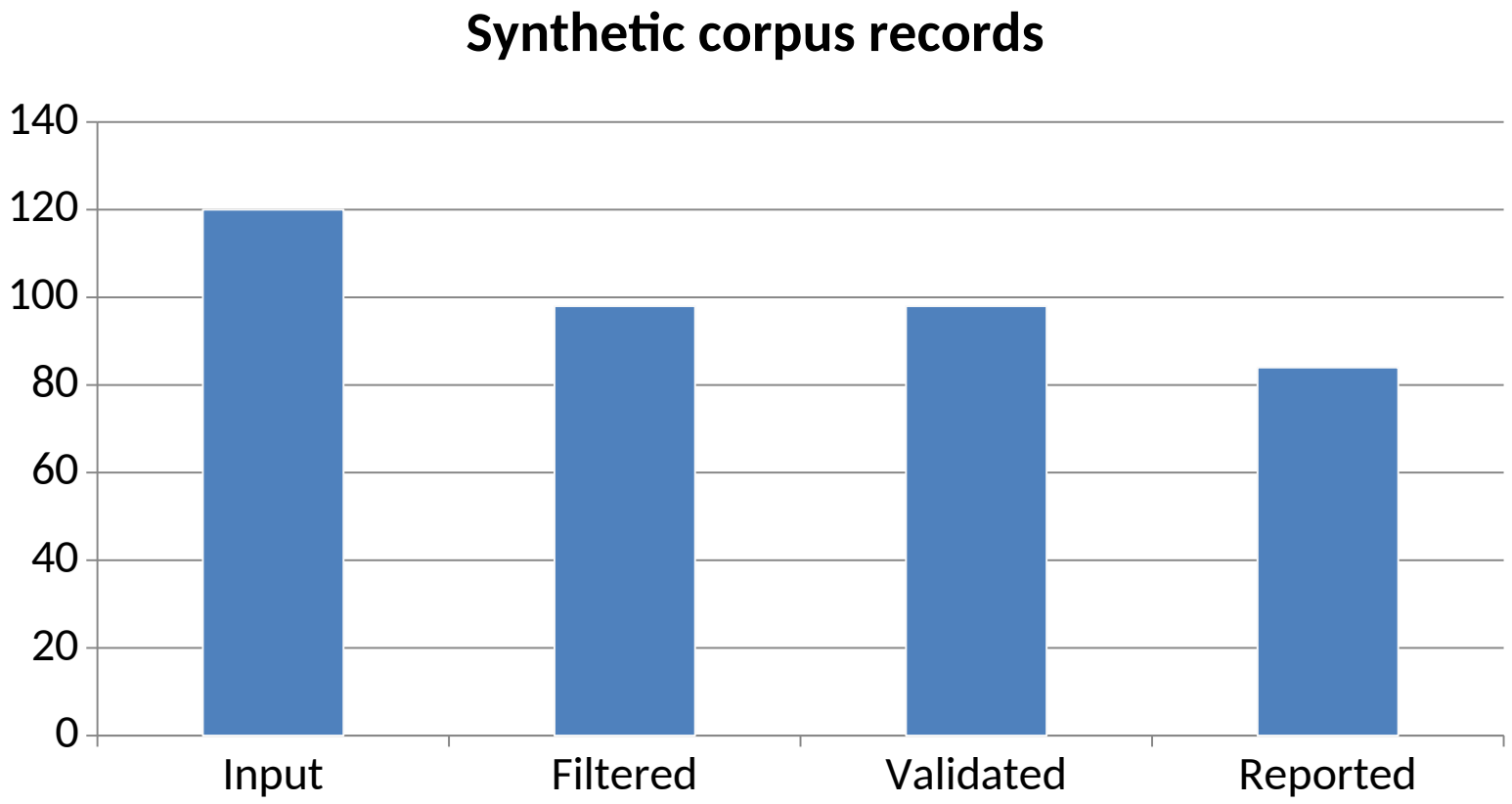
lexicon-derived polarity score

PYTHON EXPRESSION OR ANALYTICAL RULE

```
TextBlob(text).sentiment.polarity
```

Evidence to inspect: compare to supervised holdout.

TextBlob polarity: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: compare to supervised holdout.

Editable PowerPoint chart

TextBlob polarity: failure and control

FAILURE MODE

domain-specific wording

CONTROL / EVIDENCE

Check lexicon-derived polarity score; confirm compare to supervised holdout.

Topic 5: Summarization and visualization

Technical evidence from a synthetic Singapore hotel-review corpus

DATA → MODEL → INSIGHT



Aspect sentiment

INPUT / ARTIFACT

aspect × polarity table

ANALYTICAL DECISION

rank negative issues

OBSERVE

aspect × polarity table

TRANSFORM

rank negative issues

CHECK

mention overlap

Aspect sentiment: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

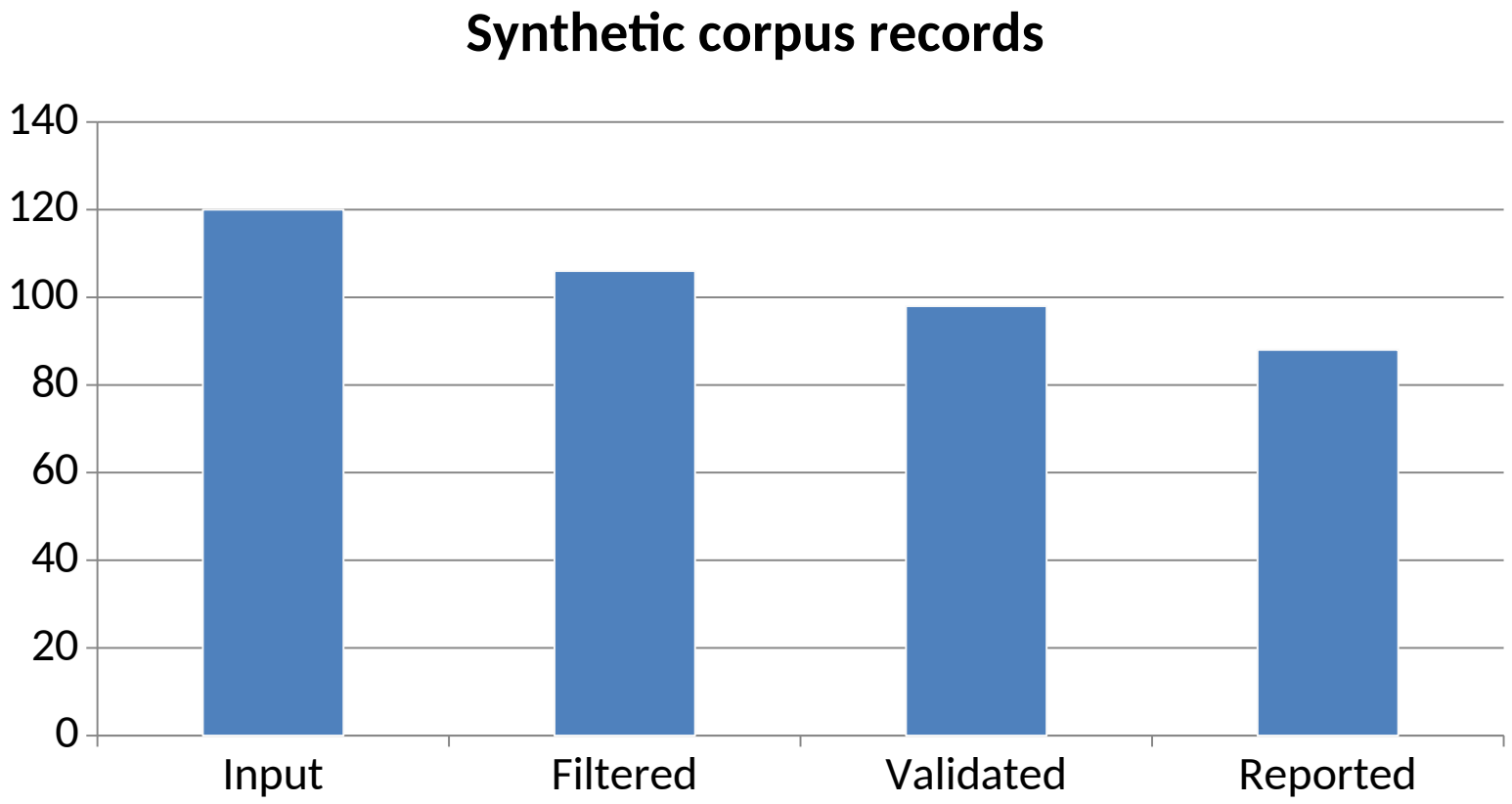
aspect × polarity table

PYTHON EXPRESSION OR ANALYTICAL RULE

```
pd.crosstab(aspect,sentiment)
```

Evidence to inspect: rank negative issues.

Aspect sentiment: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: rank negative issues.

Editable PowerPoint chart

Aspect sentiment: failure and control

FAILURE MODE

mention overlap

CONTROL / EVIDENCE

Check aspect × polarity table; confirm rank negative issues.

Hotel comparison

INPUT / ARTIFACT

normalized sentiment shares

ANALYTICAL DECISION

compare property mix

OBSERVE

normalized sentiment shares

TRANSFORM

compare property mix

CHECK

different sample sizes

Hotel comparison: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

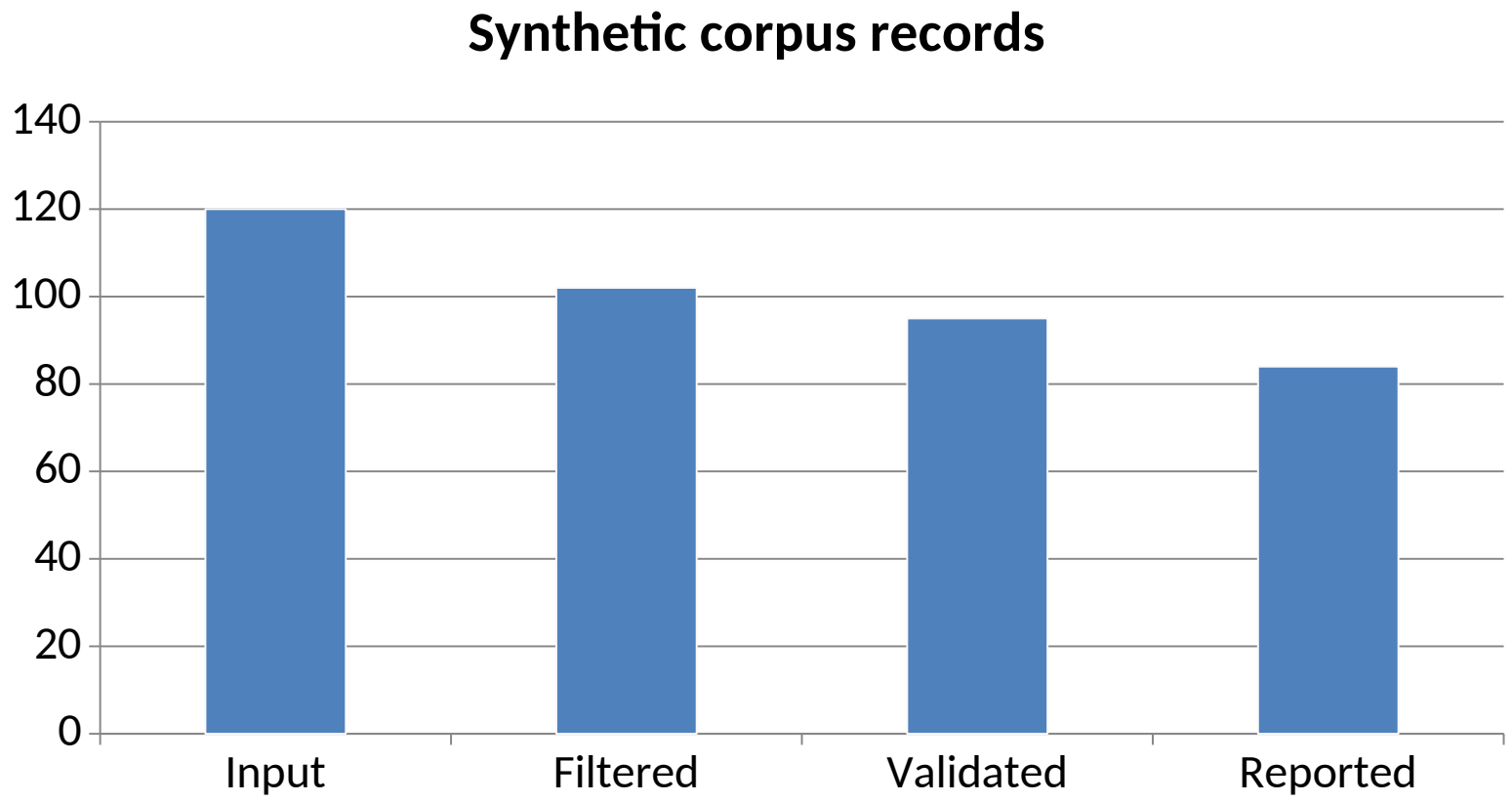
normalized sentiment shares

PYTHON EXPRESSION OR ANALYTICAL RULE

```
pd.crosstab(hotel,sentiment,normalize="index")
```

Evidence to inspect: compare property mix.

Hotel comparison: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: compare property mix.

Editable PowerPoint chart

Hotel comparison: failure and control

FAILURE MODE

different sample sizes

CONTROL / EVIDENCE

Check normalized sentiment shares; confirm compare property mix.

Trend analysis

INPUT / ARTIFACT

monthly negative share

ANALYTICAL DECISION

detect sustained change

OBSERVE

monthly negative share

TRANSFORM

detect sustained change

CHECK

small cell counts

Trend analysis: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

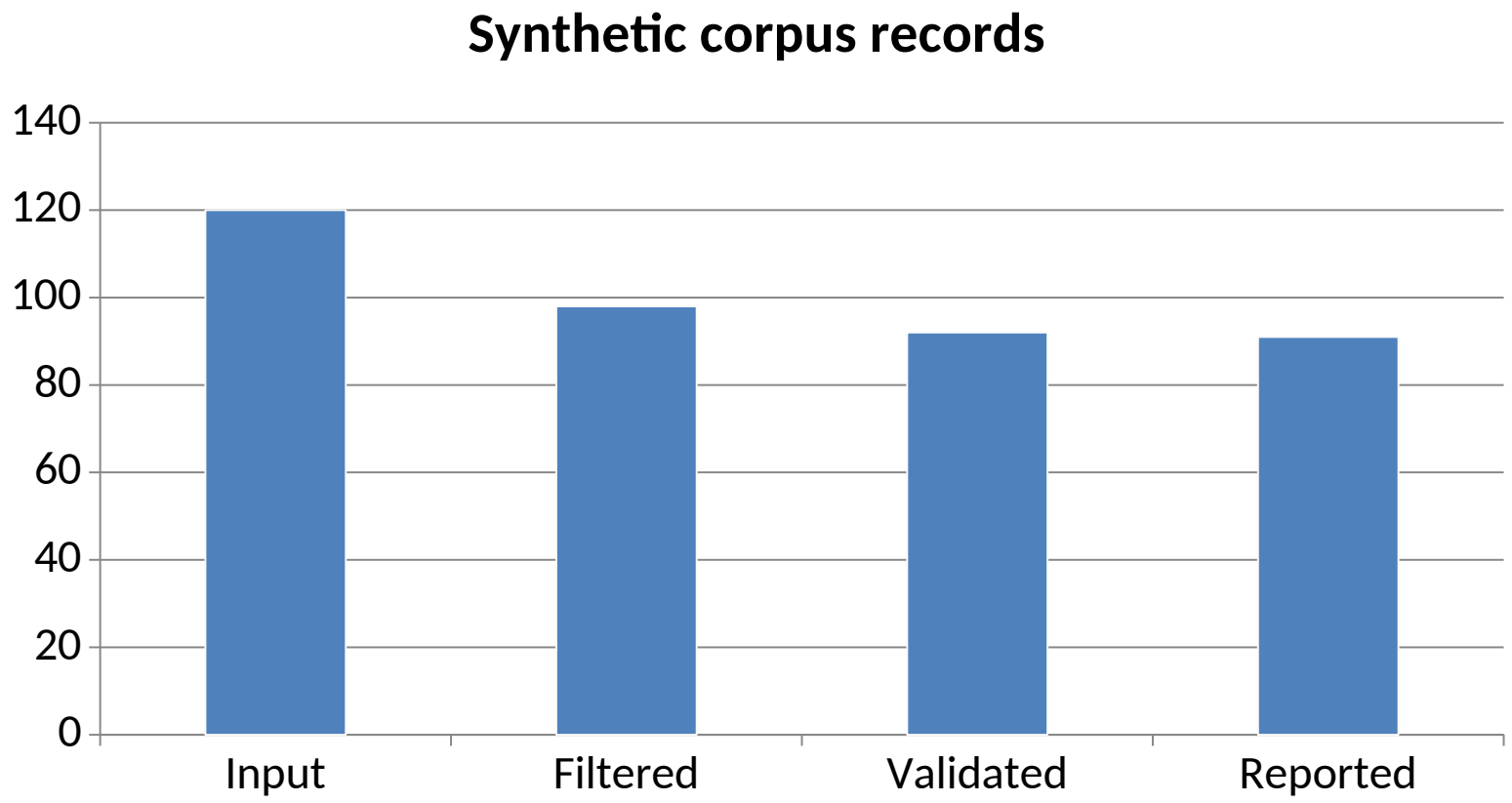
monthly negative share

PYTHON EXPRESSION OR ANALYTICAL RULE

groupby(month).mean()

Evidence to inspect: detect sustained change.

Trend analysis: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: detect sustained change.

Editable PowerPoint chart

Trend analysis: failure and control

FAILURE MODE

small cell counts

CONTROL / EVIDENCE

Check monthly negative share; confirm detect sustained change.

Extractive summary

INPUT / ARTIFACT

representative review sentences

ANALYTICAL DECISION

preserve source ID

OBSERVE

representative review sentences

TRANSFORM

preserve source ID

CHECK

misleading cherry-pick

Extractive summary: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

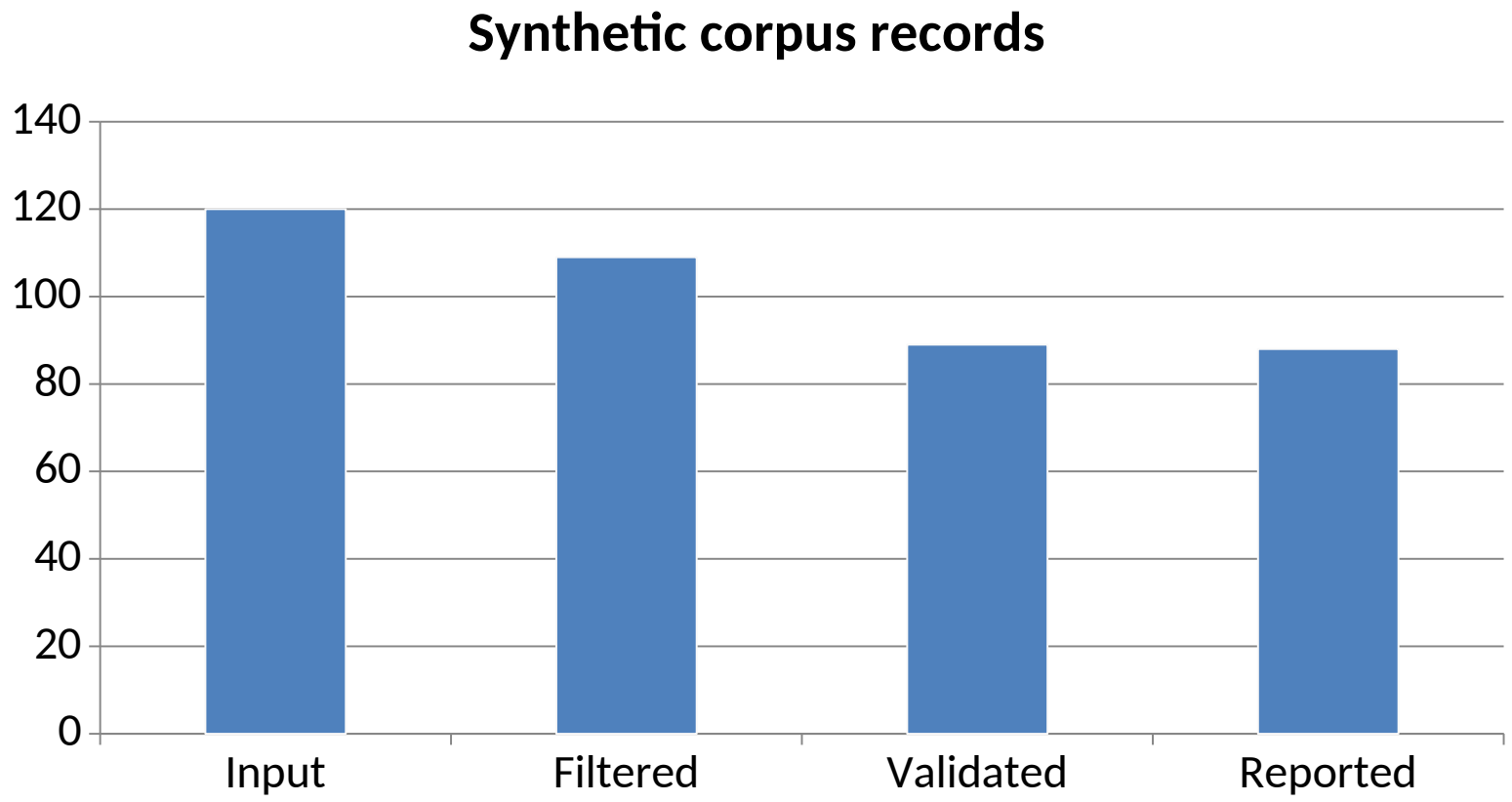
representative review sentences

PYTHON EXPRESSION OR ANALYTICAL RULE

rank by cosine similarity

Evidence to inspect: preserve source ID.

Extractive summary: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: preserve source ID.

Editable PowerPoint chart

Extractive summary: failure and control

FAILURE MODE

misleading cherry-pick

CONTROL / EVIDENCE

Check representative review sentences; confirm preserve source ID.

Topic modelling

INPUT / ARTIFACT

document-topic distribution

ANALYTICAL DECISION

inspect top terms

OBSERVE

document-topic distribution

TRANSFORM

inspect top terms

CHECK

unstable topics

Topic modelling: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

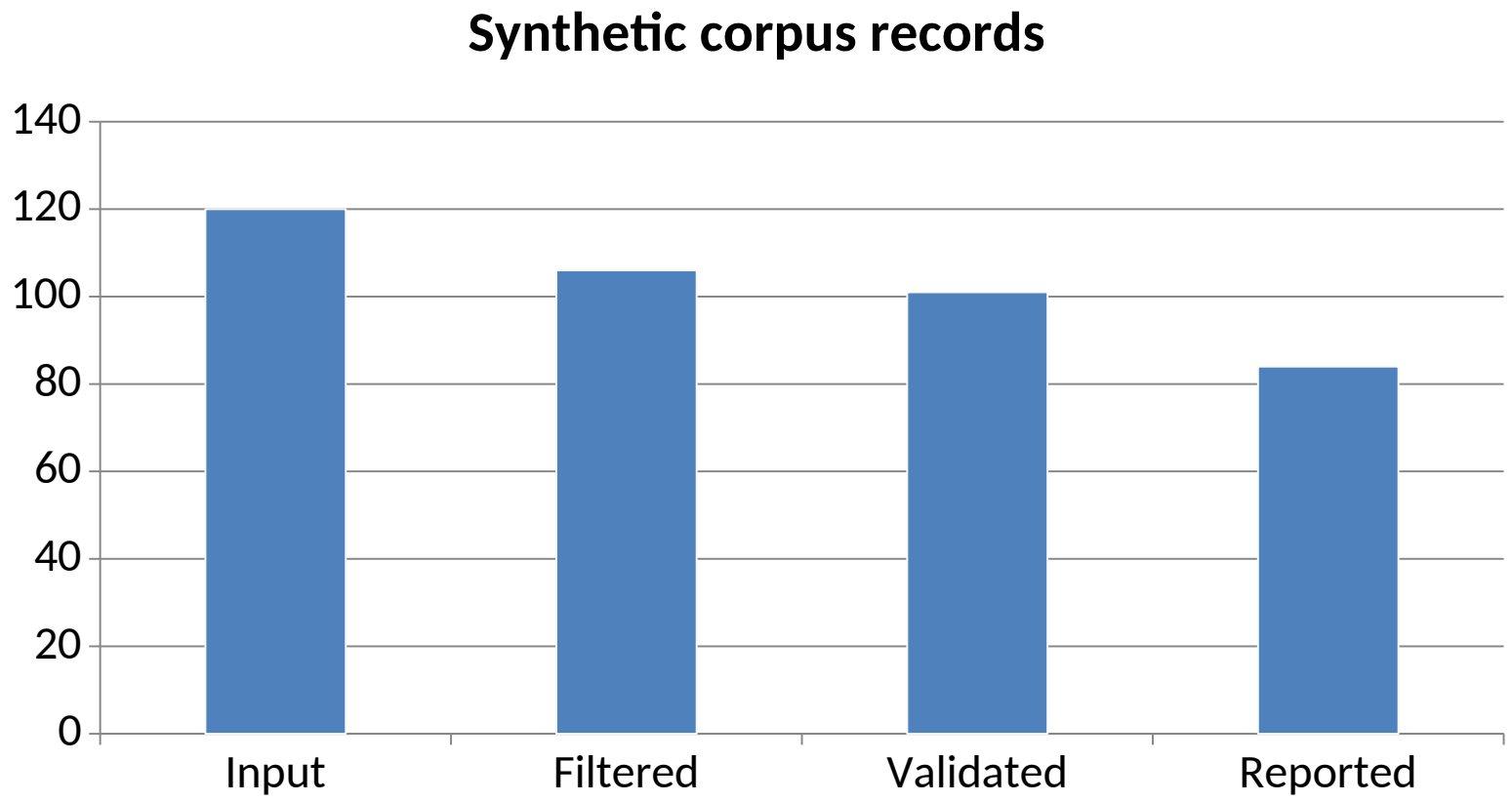
document-topic distribution

PYTHON EXPRESSION OR ANALYTICAL RULE

```
LatentDirichletAllocation(...)
```

Evidence to inspect: inspect top terms.

Topic modelling: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: inspect top terms.

Editable PowerPoint chart

Topic modelling: failure and control

FAILURE MODE

unstable topics

CONTROL / EVIDENCE

Check document-topic distribution; confirm inspect top terms.

LDA vs NMF

INPUT / ARTIFACT

probabilistic vs matrix factorization

ANALYTICAL DECISION

compare topic coherence

OBSERVE

probabilistic vs matrix factorization

TRANSFORM

compare topic coherence

CHECK

topic count arbitrary

LDA vs NMF: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

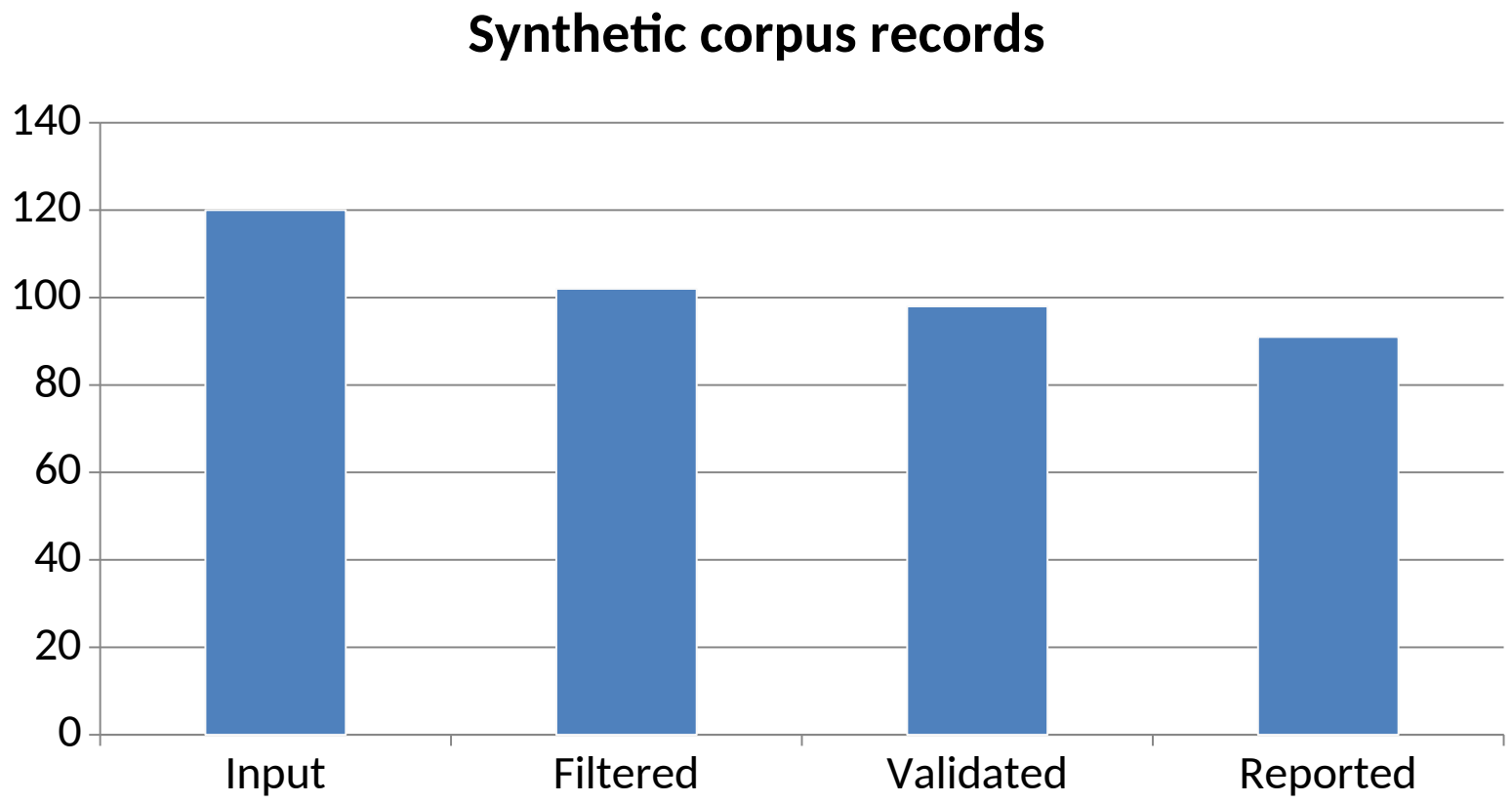
probabilistic vs matrix factorization

PYTHON EXPRESSION OR ANALYTICAL RULE

```
NMF(n_components=3)
```

Evidence to inspect: compare topic coherence.

LDA vs NMF: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: compare topic coherence.

Editable PowerPoint chart

LDA vs NMF: failure and control

FAILURE MODE

topic count arbitrary

CONTROL / EVIDENCE

Check probabilistic vs matrix factorization; confirm compare topic coherence.

Visual encoding

INPUT / ARTIFACT

stacked sentiment bars

ANALYTICAL DECISION

show denominators

OBSERVE

stacked sentiment bars

TRANSFORM

show denominators

CHECK

colour-only meaning

Visual encoding: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

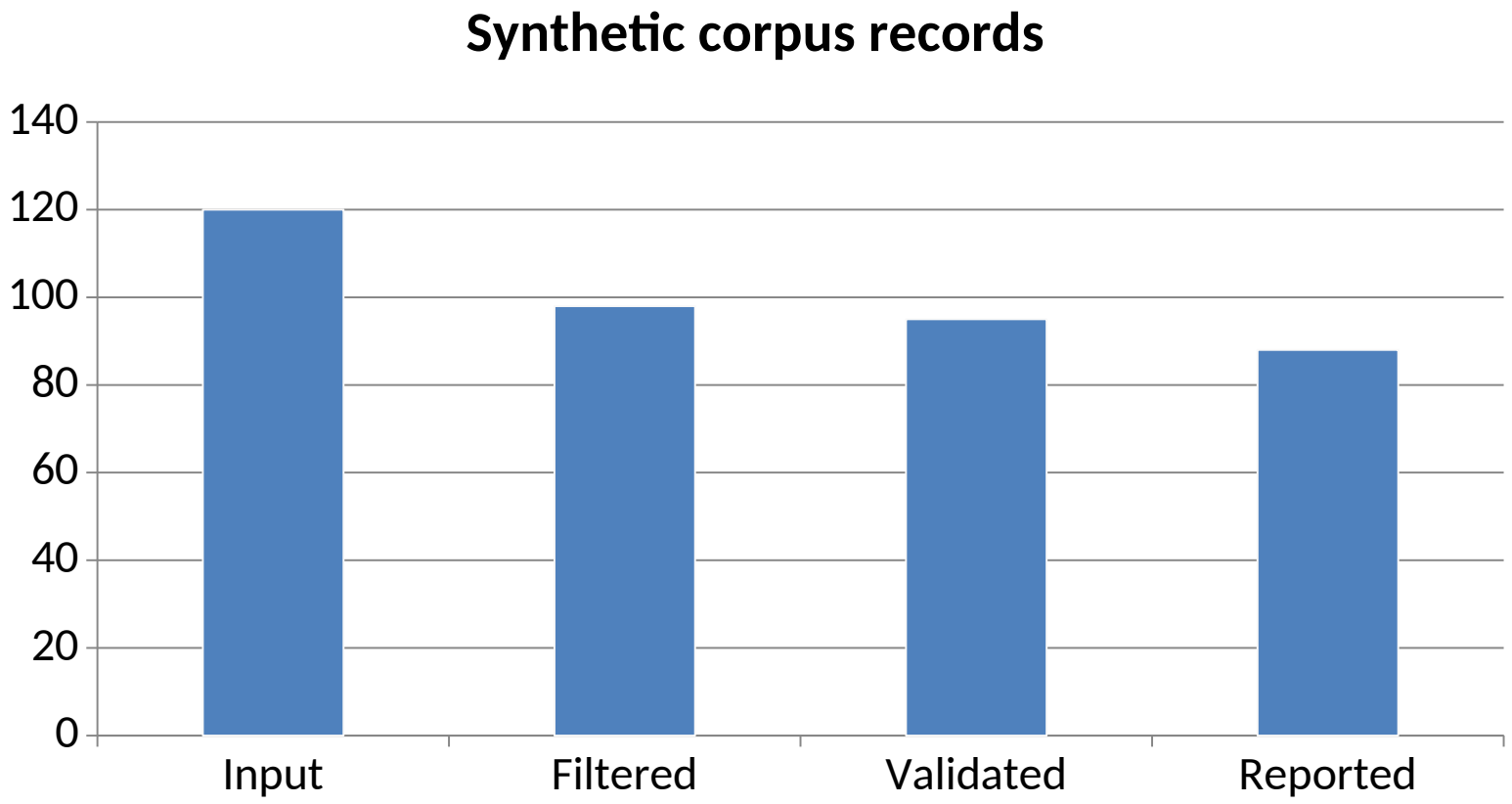
stacked sentiment bars

PYTHON EXPRESSION OR ANALYTICAL RULE

ax.bar(..., label=...)

Evidence to inspect: show denominators.

Visual encoding: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: show denominators.

Editable PowerPoint chart

Visual encoding: failure and control

FAILURE MODE

colour-only meaning

CONTROL / EVIDENCE

Check stacked sentiment bars; confirm show denominators.

Insight statement

INPUT / ARTIFACT

observation + evidence + action

ANALYTICAL DECISION

tie finding to hotel owner

OBSERVE

observation + evidence + action

TRANSFORM

tie finding to hotel owner

CHECK

causality overclaim

Insight statement: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

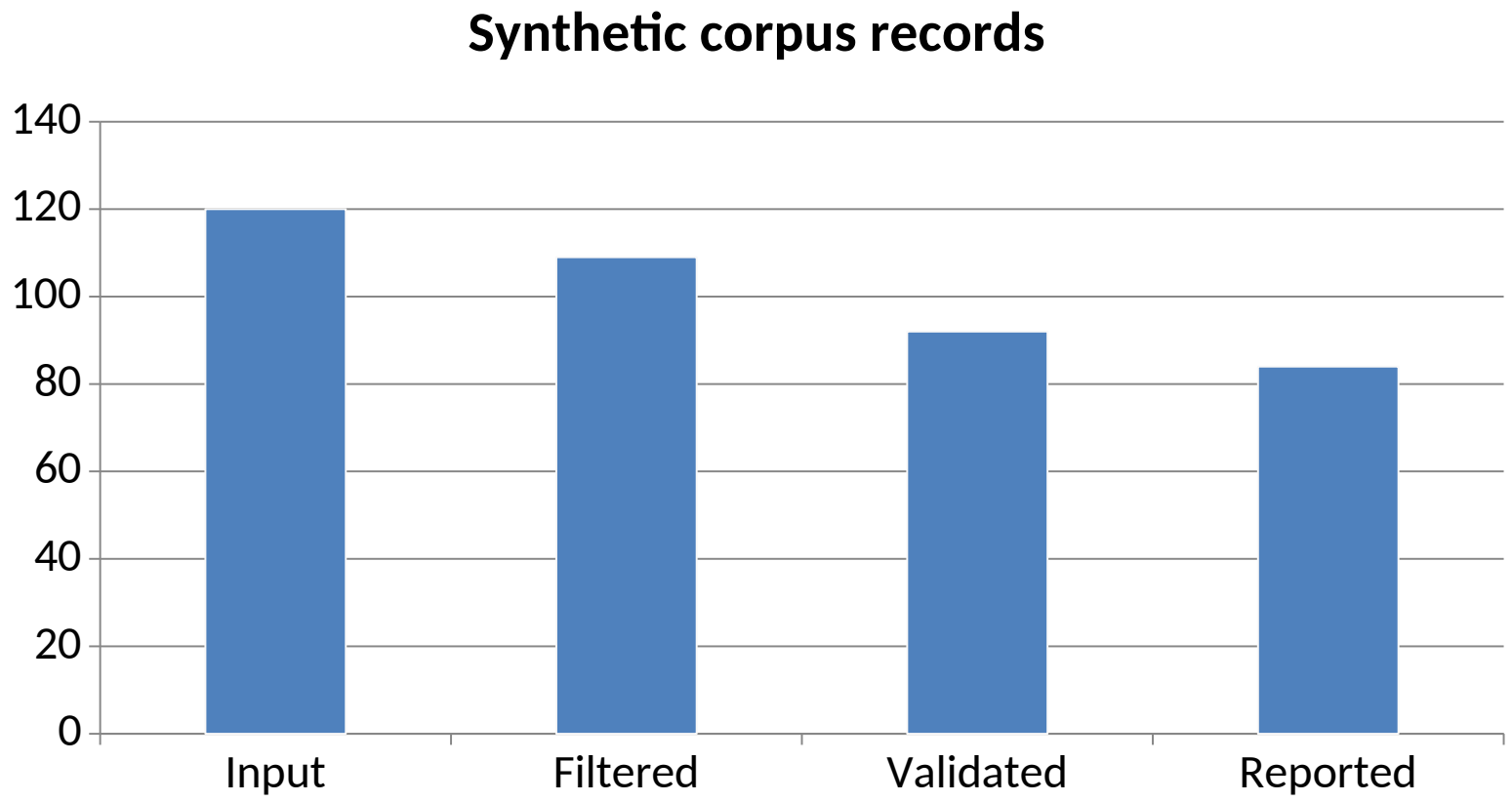
observation + evidence + action

PYTHON EXPRESSION OR ANALYTICAL RULE

negative share and count

Evidence to inspect: tie finding to hotel owner.

Insight statement: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: tie finding to hotel owner.

Editable PowerPoint chart

Insight statement: failure and control

FAILURE MODE

causality overclaim

CONTROL / EVIDENCE

Check observation + evidence + action; confirm tie finding to hotel owner.

Quality check

INPUT / ARTIFACT

spot-check source reviews

ANALYTICAL DECISION

confirm summary fidelity

OBSERVE

spot-check source reviews

TRANSFORM

confirm summary fidelity

CHECK

hallucinated summary

Quality check: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

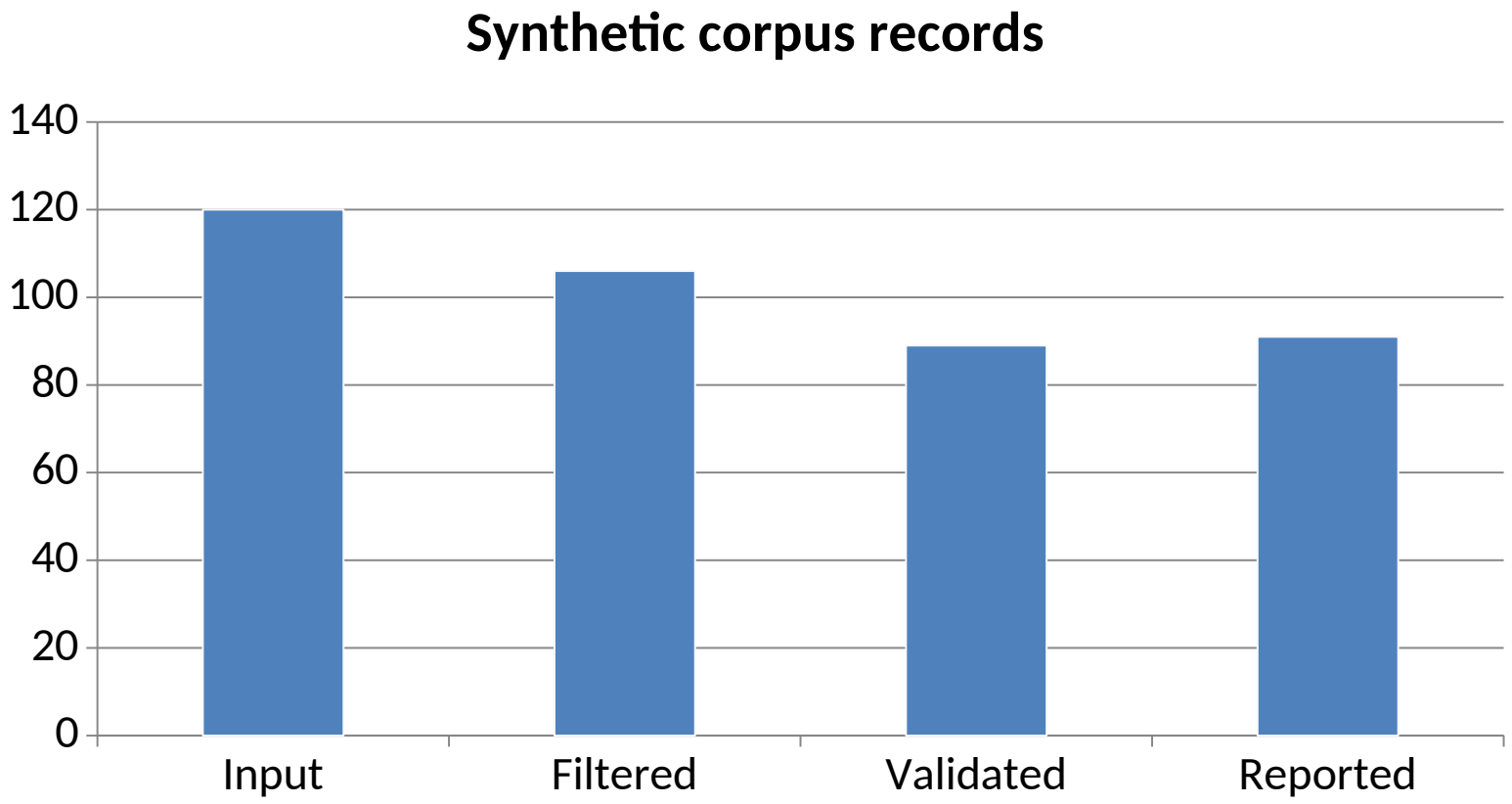
spot-check source reviews

PYTHON EXPRESSION OR ANALYTICAL RULE

review_id trace

Evidence to inspect: confirm summary fidelity.

Quality check: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: confirm summary fidelity.

Editable PowerPoint chart

Quality check: failure and control

FAILURE MODE

hallucinated summary

CONTROL / EVIDENCE

Check spot-check source reviews; confirm confirm summary fidelity.

Delivery pack

INPUT / ARTIFACT

chart, table, methods

ANALYTICAL DECISION

reproduce with script and CSV

OBSERVE

chart, table, methods

TRANSFORM

reproduce with script and CSV

CHECK

unreproducible dashboard

Delivery pack: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

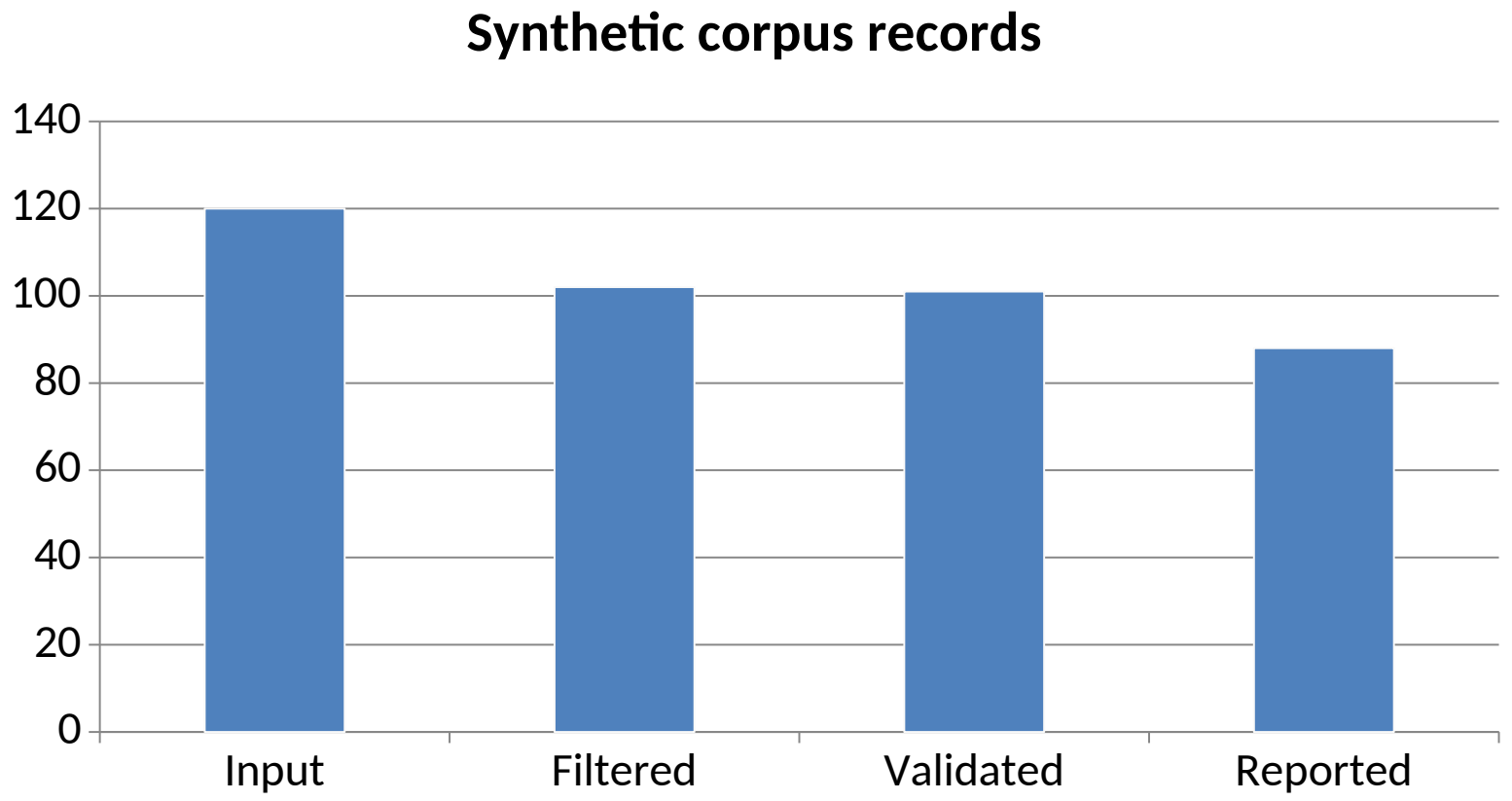
chart, table, methods

PYTHON EXPRESSION OR ANALYTICAL RULE

output/sentiment_by_hotel.csv

Evidence to inspect: reproduce with script and CSV.

Delivery pack: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: reproduce with script and CSV.

Editable PowerPoint chart

Delivery pack: failure and control

FAILURE MODE

unreproducible dashboard

CONTROL / EVIDENCE

Check chart, table, methods; confirm reproduce with script and CSV.

Transformer summarization

INPUT / ARTIFACT

source text to condensed output

ANALYTICAL DECISION

compare summary with review IDs

OBSERVE

source text to condensed output

TRANSFORM

compare summary with review IDs

CHECK

unsupported generated claim

Transformer summarization: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

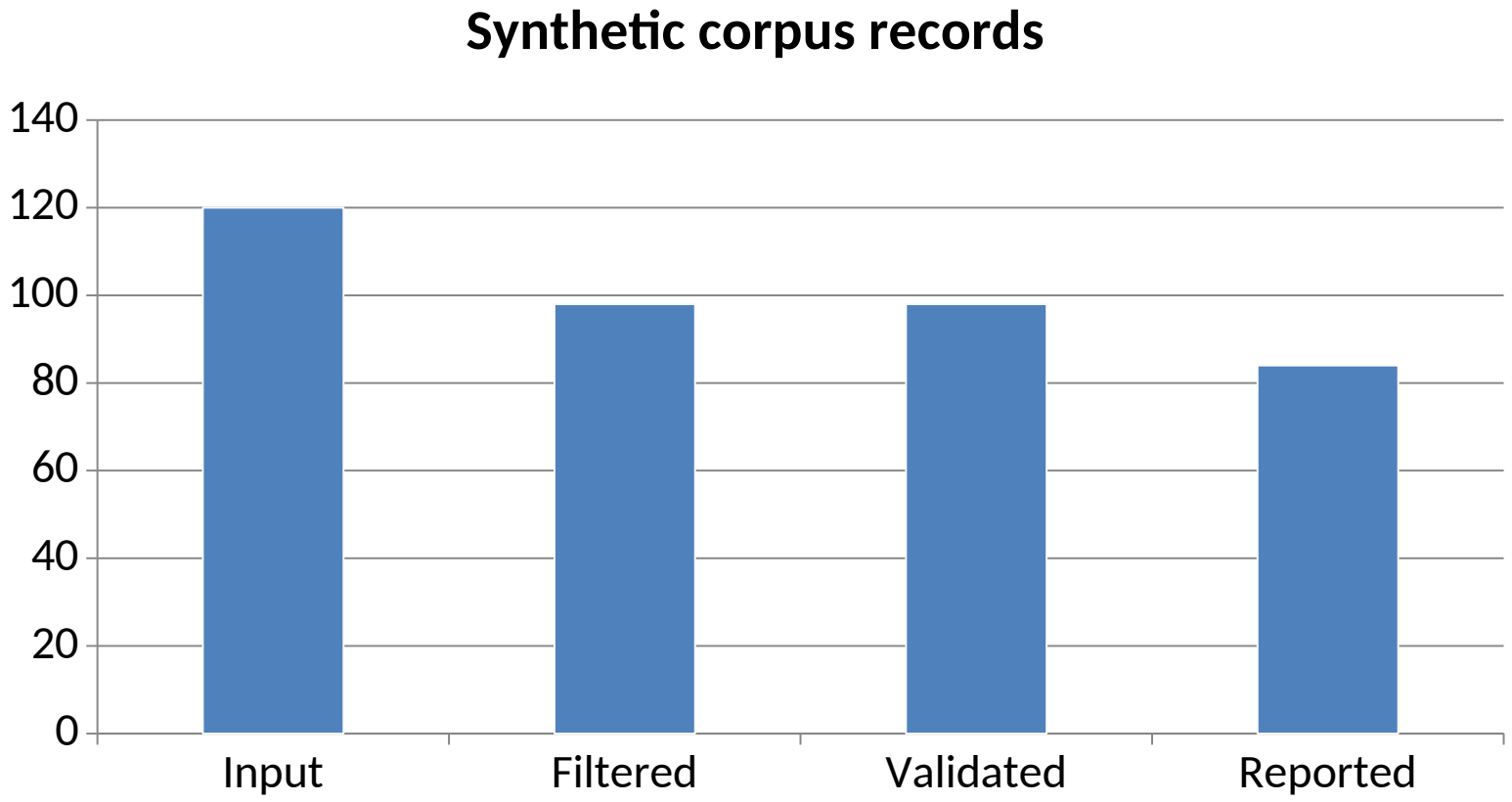
source text to condensed output

PYTHON EXPRESSION OR ANALYTICAL RULE

```
pipeline("summarization")(text)
```

Evidence to inspect: compare summary with review IDs.

Transformer summarization: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: compare summary with review IDs.

Editable PowerPoint chart

Transformer summarization: failure and control

FAILURE MODE

unsupported generated claim

CONTROL / EVIDENCE

Check source text to condensed output; confirm compare summary with review IDs.

Extractive vs abstractive

INPUT / ARTIFACT

selected sentences vs generated paraphrase

ANALYTICAL DECISION

verify factual consistency

OBSERVE

selected sentences vs generated paraphrase

TRANSFORM

verify factual consistency

CHECK

missing complaint qualifier

Extractive vs abstractive: reproducible Python artifact

EXAMPLE INPUT / SCHEMA

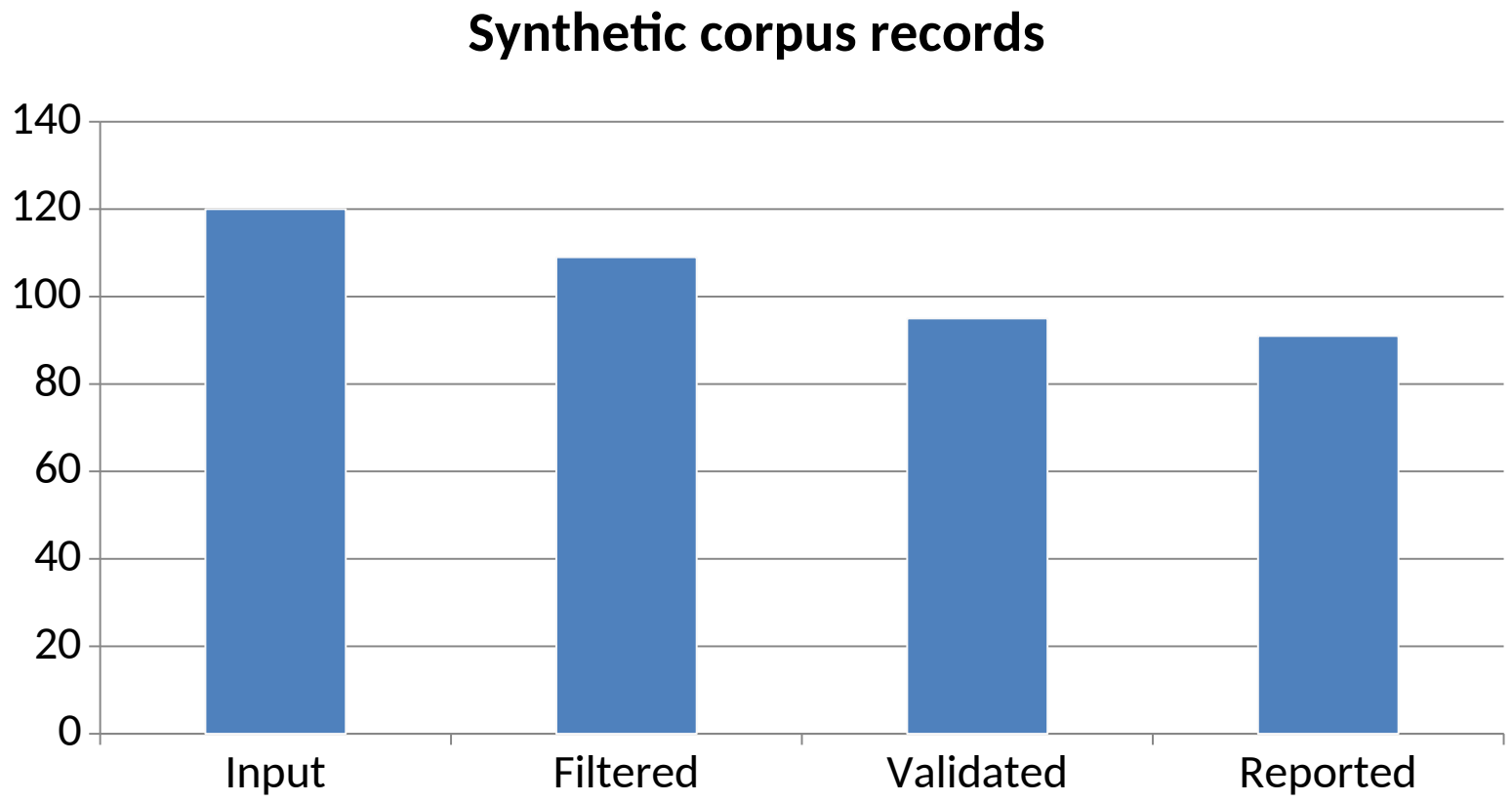
selected sentences vs generated paraphrase

PYTHON EXPRESSION OR ANALYTICAL RULE

source_sentence_id + summary

Evidence to inspect: verify factual consistency.

Extractive vs abstractive: evidence trace



READ THE TRACE

Illustrative counts from a 120-review synthetic corpus.

Interpret alongside: verify factual consistency.

Editable PowerPoint chart

Extractive vs abstractive: failure and control

FAILURE MODE

missing complaint qualifier

CONTROL / EVIDENCE

Check selected sentences vs generated paraphrase; confirm verify factual consistency.

Your trainer — profile template

TRAINER NAME

To be completed by the assigned trainer

Role / designation
Relevant qualifications
Text analytics experience
Contact

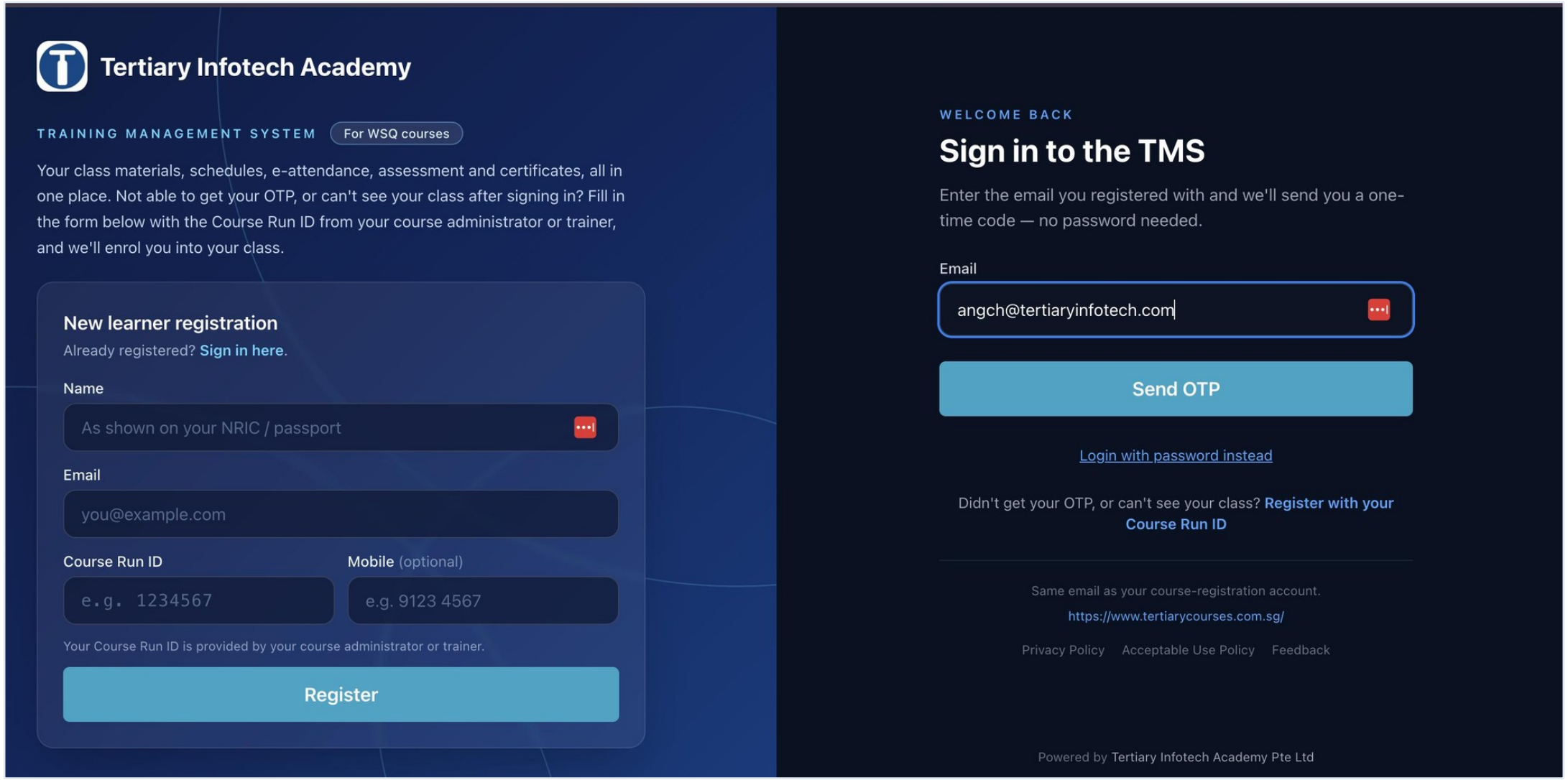
Dr Alfred Ang

Dr Alfred Ang

Trainer profile

Python and applied analytics instruction
Course developer and adult educator
Facilitates evidence-based practical learning

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Learner Guide

Lab package

lms-tms.tertiaryinfotech.com · open the course record and use its material links

Assessment briefing

Written Assessment: 60 min, open book.
Practical Performance: 90 min, open book.
Submit only your own work through the LMS.

Assessment flow

Attendance

WA (SAQ)

PP tasks

LMS submission

Assessor sign-off

Digital attendance

Complete AM, PM and assessment attendance using the provider QR process.

Thank you

Reproduce the analysis, inspect errors, and support every insight with a source review ID.